B.Sc. Physics

ELECTRICITY AND MAGNETISM

Notes (HRK)



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HOUSE OF PHYSICS PUBLICATIONS

B.Sc. Physics

Complete Notes of

ELECTRICITY AND MAGNETISM

Chapter # 1: ELECTRICAL FORCE AND COULOMB'S LAW

Chapter # 2: THE ELECTRIC FIELD

Chapter # 3: GAUSS'S LAW

Chapter # 4: ELECTRIC POTENTIAL

Chapter # 5: CAPACITANCE AND DIELECTRICS

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Chapter # 8: MAGNETIC FIELD EFFECTS

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B.Sc. Physics ELECTRICITY AND MAGNETISM in Past Papers of University of Sargodha

B.Sc. Physics, Paper B, Annual 2017

B.Sc. Physics, Paper B, Annual 2016

B.Sc. Physics, Paper B, Annual 2015

B.Sc. Physics, Paper B, Annual 2014

B.Sc. Physics, Paper B, Annual 2013

B.Sc. Physics, Paper B, Annual 2012

Electrical Charge and Coulomb's Law

Electromagnetic forces are responsible for the structure of atoms and for the binding of atoms in molecules and solids. Many properties of materials that we have studied so far are electromagnetic in their nature. Such as the elasticity of solids and the surface tension of liquids. The spring force, friction, and the normal force all originate with the electromagnetic force between atoms.

Among the examples of electromagnetism that we shall study are the force between electric charges, such as occurs between an electron and the nucleus in an atom; the motion of a charged body subject to an external electric force. such as an electron in an oscilloscope beam; the flow of electric charges through circuits and the behavior of circuit elements; the force between permanent magnets and the properties of magnetic materials; and electromagnetic radiation, which ultimately leads to the study of optics, the nature and propagation of light.

In this chapter, we begin with a discussion of electric charge, some properties of charged bodies, and the fundamental electric force between two charged bodies.

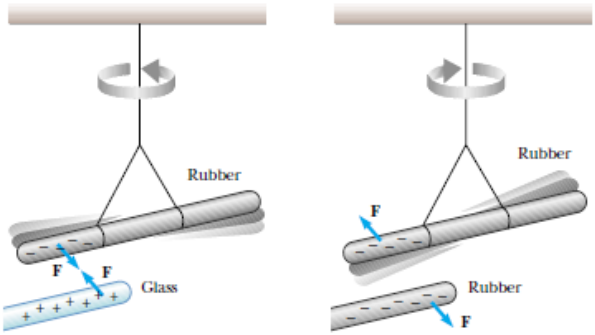
27.1 Electric Charge

27.1.1 Electric Charge and Electrical Forces

A body is said to be electrical neutral if it contains equal number of $+ve$ and $-ve$ charges. When two bodies are rubbed together, their neutrality is distributed due to transfer of electrons from one body to the other. The body which gives electrons becomes electrically $+ve$ and the body which gains electrons becomes negative.

"*Charges of the same signs repel each other and*

Charges of the oppositely sign attract each other."



These attractive and repulsive forces among the charges are called electrical forces.

27.1.2 Application of Electrical Forces

An electrical force between charged bodies has many industrial applications.

i. Photocopying or xerography
ii. Ink-jet printing
iii. Electrostatic paint spraying
iv. Powder coating etc.

27.1.3 Xerography

The photocopying process is called "Xerography". The main parts of photocopier are (1) Lamp, (2) Rotating drum, (3) Toner, (4) Sheet of paper and (5) Heated roller.

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The lamp forms a positively charged image of the document on a rotating drum (which is an aluminum cylinder coated with a layer of selenium). The negatively charged toner particles stick to the drum at the places of positively charged image. The toner particles are shifted from the drum to a charged sheet of paper, after which they are fused on the paper by heated rollers. This produces a powder print i.e. photocopy.

Q # 1. Explain what is meant by the term "a neutral atom." Explain what "a negatively charged atom" means.

Ans. A neutral atom is one that has no net charge. This means that it has the same number of electrons orbiting the nucleus as it has protons in the nucleus. A negatively charged atom has one or more excess electrons.

Q # 2. Explain from an atomic viewpoint why charge is usually transferred by electrons.

Ans. Electrons are less massive and more mobile than protons. Also, they are more easily detached from atoms than protons.

Q # 3. Would life be different if the electrons were positively charged and the protons were negatively charged? Does the choice of signs have any bearing on physical and chemical interactions? Explain.

Ans. No. Life would be no different if electrons were + charged and protons were – charged. Opposite charges would still attract, and like charges would repel. The naming of + and – charge is merely a convention.

27.2 Conductors and Insulators

Insulators

The materials through which the charges don't flow are called insulators. Glass, chemically pure water, and plastics are common examples of insulators. If the charge is placed on an insulator, the charges will stay where they are placed.

Conductors

The materials through which charge can flow easily are called conductors. Metals in general, tap water and the human body are common examples of conductors. The copper rod cannot be charged because any charges placed on it easily flow through the rod, through your body (which is also a conductor), and to the ground. The insulating handle, however, blocks the flow and allows charge to build up on the copper.

27.2.1 Distinction among Conductors, Insulators and Semi-Conductors on the basis of Free Electron

An experiment, called the Hall Effect, shows that it is the negative charges (electrons) that are free to move in metal. In metals, the atoms are so close to each other, their outermost shell is overlapped. The electrons in the outermost shells are already loosely bound; they are attracted by the neighboring nuclei and become free to move among the lattice atoms. These electrons are called free electrons or conduction electrons. The distinction between conductors and insulators can be made on the basis of number of conduction electrons.

- In conductors, each atom contributes one conduction electron. Therefore, there will be on the average about 10^{23} conduction electrons per cm^3.
- In insulators, at room temperature, it is very difficult to find even one conduction electrons per cm^3.

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- Intermediate between conductor and insulators are the semi-conductors e.g. Ge and Si, which might contain $10^{10} - 10^{12}$ conduction electrons per cm^3.The main characteristics of these materials is that the density of conduction electrons can be increased by
  i. Adding small quantity impurities (1 atom in $10^6 - 10^9$)
  ii. Increase the applied voltage
  iii. Increasing the temperature
  iv. Increasing the intensity of incident

($Definition$) Point Charges

The charge bodies whose sizes are much smaller than the distance between them are called point charges.

27.5 Coulomb's Law

Charles Augustin Coulomb (l 736- 1806) measured electrical attractions and repulsions quantitatively and deduced the law that governs them. His apparatus, shown in Fig. consist of spheres A and B.



If A and B are charged, the electric force on A tends to twist the suspension fiber. The angle θ is then a relative measure of the electric force acting on charge A. Experiments due to Coulomb and his contemporaries showed that

Statement

The magnitude of electrical force between two point charges is directly proportional to the product of magnitude of charges and inversely proportional to the square of the distance between their centers.

Mathematical Form

Suppose two point charges q_1 and q_2 separated by distance r. According to the Coulomb's law, the electrical force F between two charges is:

$$F \quad \alpha \quad q_1\, q_2 \qquad ---(1)$$

$$F \quad \alpha \quad \frac{1}{r^2} \qquad ---(2)$$

Combining (1) and (2), we have:

$$\Rightarrow F \quad \alpha \quad \frac{q_1\, q_2}{r^2}$$

$$\Rightarrow F \quad = constant \quad \frac{q_1\, q_2}{r^2}$$

$$\Rightarrow F \quad = k \quad \frac{q_1\, q_2}{r^2}$$

where k is called Coulomb's constant. Its value depends upon the system of units and medium between the charges. For free space and in system international 'k' is expressed as:

$$k = \frac{1}{4\pi\, \varepsilon_0}$$

where ε_0 is the permittivity of free space and its value in SI unit is $8.85 \times 10^{-12}\ C^2 N^{-1}\ m^{-2}$.

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Sample problem 5. Nucleus of an iron atom has radius $4 \times 10^{-15}\ m$ and contains 26 protons. What electric repulsive force act between them in such a way that they are separated by a distance of one radius.

Solution:

Radius $r = 4 \times 10^{-15}\ m$

Charge of proton $q_1 = q_2 = 1.6 \times 10^{-19}\ C$

From Coulomb's law:

$$F = k\frac{q_1 q_2}{r^2} = 9 \times 10^9 \times \frac{1.6\times10^{-19}\times1.6\times10^{-19}}{(4\times10^{-15})^2} = 14\ N$$

Problem 25. In a crystal of salt, an atom of Na transfer an electron to Cl formed an ionic bond. The resulting positive and negative ion can attract each other by force F. Find F if the ions are 282 pm apart.

Solution:

$q_1 = 1.6 \times 10^{-19} C$

$q_2 = -1.6 \times 10^{-19} C$

$r = 282\ \text{pm} = 282\ \times 10^{-12} m$

From Coulomb's law:

$$F = k\frac{q_1 q_2}{r^2} = 9 \times 10^9 \times \frac{(1.6 \times 10^{-19}) \times (-1.6 \times 10^{-19})}{(282\ \times 10^{-12})^2} = 2.89 \times 10^{-9}\ N$$

27.6 Coulomb's Law and Newton's Law of Gravitation (A Comparison)

The electrical force between two charges is directly proportional to the product of the magnitude of charges and inversely proportional to the square of the distance between them:

$F = k\frac{q_1 q_2}{r^2}$ ---------- **(1)**

The gravitational force between two masses is directly proportional to the product of their masses and inversely proportional to the square of the distance between them:

$F = G\frac{m_1 m_2}{r^2}$ ---------- **(2)**

Similarities among the Electrical and Gravitational Force

(i) Both forces are the conservative forces.

(ii) Both forces obey the inverse square law.

(iii) The charge 'q' plays the same role in the Coulomb's law that the mass 'm' plays in Newton's law of gravitation

Differences among the Electrical and Gravitational Force

(i) Electrical force is might be attractive as well as repulsive, while the gravitational force is only attractive.

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(ii) Electrostatic force is medium dependent and can be shielded while the gravitational force lacks this property.

(iii) The value of gravitational constant is very small while the electrical constant is very large. It is because of the fact that gravitational force is very weak as compared to electrical force.

(iv) In using the law of gravitation, we define mass 'm' from $F = ma$, then determine 'G' by applying gravitational law to known masses. While in case of coulomb's law, we define 'K' for a particular value and then determine 'q' by applying coulomb's law.

Problem 4. Two equally charged particles held $3.3\ mm$ apart when they are released from rest, the initial acceleration of 1st particle $7.22\ \frac{m}{s^2}$ and 2nd is $9.16\ \frac{m}{s^2}$. The mass of the 1st particle is $6.31 \times 10^{-7}\ kg$. Find the mass of 2nd particle and common charge?

Solution:

$$a_1 = 7.22\frac{m}{s^2}$$

$$a_2 = 9.16\ \frac{m}{s^2}$$

$$r = 3.3\ mm = 3.3 \times 10^{-3}\ m$$

$$m_1 = 6.31 \times 10^{-7}\ kg$$

$$m_2 = ?$$

$$q = ?$$

As both particles exerts equal force on each other, therefore:

$$m_1 a_1 = m_2 a_2$$

$$\Longrightarrow m_2 = \frac{m_1 a_1}{a_2} = \frac{6.31 \times 10^{-7} \times 7.22}{9.16} = 4.97 \times 10^{-7}\ kg$$

Now for present case:

$$F = m_1 a_1 \Longrightarrow k\frac{q^2}{r^2} = m_1 a_1 \Longrightarrow q^2 = \frac{m_1 a_1 r^2}{k} = \frac{6.31 \times 10^{-7} \times 7.22 \times (3.3 \times 10^{-3})^2}{9 \times 10^9}$$

$$= 55.055 \times 10^{-4}$$

$$\Longrightarrow q = 7.41 \times 10^{-2} C$$

Problem 36. In the radioactive decay of U^{238}, the center of the emerging He^4 particel is at a certain distance $12 \times 10^{-15}\ m$ from the center of residual Th^{234} nucleus at that instant. (a) What is the force on helium atom and (b) what is its acceleration?

Solution:

$$U^{238} \longrightarrow He^4 + Th^{234}$$

$$r = 12 \times 10^{-15}\ m$$

$F = ?$

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a =?

$$F = k\frac{q_1 q_2}{r^2} = 9\times10^9 \times \frac{(2e)\times(90e)}{(12\times10^{-15})^2}$$

$$= 9\times10^9 \times \frac{180\times1.6\times10^{-19}\times1.6\times10^{-19}}{(12\times10^{-15})^2} = 288\ \text{N}$$

Mass of helium atom m = 4 amu = $4\times1.67\times10^{-27}$kg = 6.67×10^{-27}kg

From Newton's 2nd law of motion:

$$\text{F} = \text{ma} \Longrightarrow \text{a} = \frac{\text{F}}{\text{m}} = \frac{288}{6.67\times10^{-27}} = 43.18\times10^{27}\ \frac{\text{m}}{\text{s}^2}$$

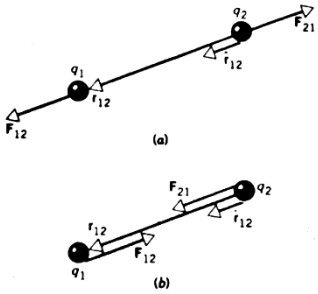
Q # 4. A free electron and a free proton are released in identical electric fields. Compare the electric forces on the two particles. Compare their accelerations.

Ans. The electric forces on the particles have the same magnitude, but are in opposite directions. The electron will have a much larger acceleration (by a factor of about 2 000) than the proton, due to its much smaller mass.

Vector Form of Coulomb's Law

So far we have considered only the magnitude of the force between two charges determined according to Coulomb's law. Force, being a vector, has directional properties as well. In case of coulomb's law, the direction of the force is determined by the relative sign of the two electric charges.

Suppose we have the two point charges 'q_1' and 'q_2' separated by a distance r_{12}. For the moment, we assume the two charges to have the same sign, so that they repel each other. Let us consider the force exerted on q_1 by q_2 is denoted by $\vec{\text{F}}_{12}$. The position vector that locates particle 1 relative to particle 2 is $\vec{\text{r}}_{12}$; that is, if we were to define the origin of our coordinate system at the location of q_2, then $\vec{\text{r}}_{12}$ would be the position vector of q_1.



(*a*) Two point charges q_1 and q_2 of the same sign exert equal and opposite repulsive forces on one another. The vector $\mathbf{r}_{12}$ locates q_1 relative to q_2, and the unit vector $\hat{\mathbf{r}}_{12}$ points in the direction of $\mathbf{r}_{12}$. Note that $\mathbf{F}_{12}$ is parallel to $\mathbf{r}_{12}$. (*b*) The two charges now have opposite signs, and the force is attractive. Note that $\mathbf{F}_{12}$ is antiparallel to $\mathbf{r}_{12}$.

If the charges have the same sign, then the force is repulsive and $\vec{\text{F}}_{12}$ must be parallel to $\vec{\text{r}}_{12}$, as shown in figure a. If the charges are opposite sign, as in figure b, then $\vec{\text{F}}_{12}$ is attractive and anti-parallel to $\vec{\text{r}}_{12}$. In either case we can represent the force as:

$$\vec{\text{F}}_{12} = k\frac{q_1 q_2}{r_{12}^2}\hat{\mathbf{r}}_{12}$$

Question. Show that Coulomb force is a mutual force

Ans. Coulomb's force is a mutual force, it means that if a charge 'q_1' exerts a force on charge 'q_2', then 'q_2' also exerts an equal and opposite force on 'q_1'.

If charge 'q_2' exerts an electrostatic force '$\vec{\text{F}}_{12}$' on charge 'q_1' and 'q_1' exerts electrical force '$\vec{\text{F}}_{21}$' on charge 'q_2' and, then $\vec{\text{F}}_{12} = -\vec{\text{F}}_{21}$

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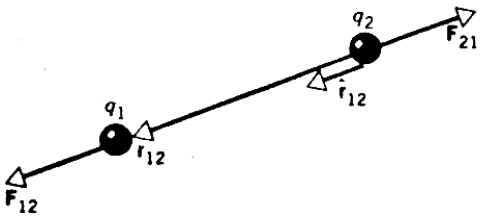
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Proof.

If $\hat{\mathbf{r}}_{12}$ represents the direction of force exerted on charge 'q_1' by 'q_2' and $\hat{\mathbf{r}}_{21}$ is the unit vector which represent the direction of force on charge 'q_2' by 'q_1', then

$$\vec{F}_{21} = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r^2}\hat{r}_{21} \quad \text{---------- (1)}$$

$$\vec{F}_{12} = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r^2}\hat{r}_{12} \quad \text{---------- (2)}$$



As $\hat{r}_{21} = -\hat{r}_{12}$, so the eq. (1) becomes

$$\vec{F}_{21} = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r^2}(-\hat{r}_{12})$$

$$\vec{F}_{21} = -\frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r^2}\hat{r}_{12}$$

By eq. (2)

$$\vec{F}_{21} = -\vec{F}_{12}$$

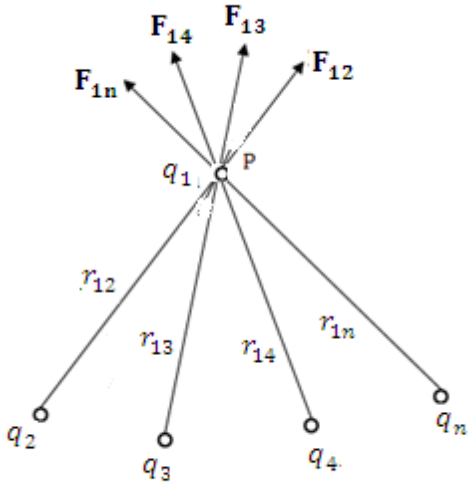
This expression shows that Coulomb force is a mutual force.

27.5.1 Significance of Vector Form of Coulomb's Law

Vector form of coulomb's law has the critical importance, when there is an assembly of point charges. In this case, the resultant force on any one of the charges is the vector sum of the forces due to each of the other forces. This is called principle of superposition.

27.5.2 Coulomb force due to many point charges

Let $q_1, q_2, q_3, \ldots\ldots\ldots, q_n$ are the 'n' point charges as shown in the figure. We want to find out electrical force on charge q_1 by the assembly of n point charges. The point charges $q_2, q_3, q_4, \ldots\ldots\ldots, q_n$ are at the distances r_{12}, $r_{13}, r_{14}\ldots\ldots\ldots, r_{1n}$ from charge q_1, respectively.



Now if $\vec{F}_{12}, \vec{F}_{13}, \vec{F}_{14}\ldots\ldots\ldots, \vec{F}_{1n}$ be the electrostatic force on charge q_1 due to the point charges $q_2, q_3, q_4, \ldots\ldots\ldots, q_n$ respectively. Thus, the total electrostatic force $\vec{F}_1$ on charge q_1 is given by;

$$\vec{F}_1 = \vec{F}_{12} + \vec{F}_{13} + \vec{F}_{14} + \ldots\ldots\ldots + \vec{F}_{1n} \quad \text{--------(1)}$$

where

$\vec{F}_{12}$= Force on charge 'q_1' exerted by 'q_2' = $\frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{(r_{12})^2}\hat{r}_{12}$

$\vec{F}_{13}$ = Force on charge 'q_1' exerted by 'q_3' = $\frac{1}{4\pi\varepsilon_0}\frac{q_1 q_3}{(r_{13})^2}\hat{r}_{13}$

$\vec{F}_{14}$ = Force on charge 'q_1' exerted by 'q_4' = $\frac{1}{4\pi\varepsilon_0}\frac{q_1 q_4}{(r_{14})^2}\hat{r}_{14}$

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$\vec{F}_{1n}$ = Force on charge 'q_1' exerted by 'q_n' = $\frac{1}{4\pi\varepsilon_0}\frac{q_1 q_n}{(r_{1n})^2}\hat{r}_{1n}$

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Putting values in equation (1), we get

$$\vec{F}_1 = \frac{1}{4\pi\varepsilon_0}\frac{q_1q_2}{(r_{12})^2}\hat{r}_{12} + \frac{1}{4\pi\varepsilon_0}\frac{q_1q_3}{(r_{13})^2}\hat{r}_{13} + \frac{1}{4\pi\varepsilon_0}\frac{q_1q_4}{(r_{14})^2}\hat{r}_{14} + \cdots\cdots + \frac{1}{4\pi\varepsilon_0}\frac{q_1q_n}{(r_{1n})^2}\hat{r}_{1n}$$

$$= \frac{q_1}{4\pi\varepsilon_0}\left(\frac{q_2}{(r_{12})^2}\hat{r}_{12} + \frac{q_3}{(r_{13})^2}\hat{r}_{13} + \frac{q_4}{(r_{14})^2}\hat{r}_{14} + \cdots\cdots + \frac{q_n}{(r_{1n})^2}\hat{r}_{1n}\right)$$

$$= \frac{q_1}{4\pi\varepsilon_0}\sum_{i=1}^{n}\frac{q_i}{(r_{1i})^2}\hat{r}_{1i}$$

This expression gives the electrical force between on a point charge due to many point charges.

27.3 Quantization of Charges

When the two bodies are rubbed together, transfer of electrons from one body to the other takes place and they are said to be electrified. The magnitude of charge q that can be detected and measured on any object is given by

$$q = ne \qquad \text{---------- (1)}$$

where $n = 0, \pm1, \pm2, \ldots\ldots\ldots$ and e is the elementary unit of charge called on unit charge, has the experimentally determined value

$$e = 1.6 \times 10^{-19}\ C$$

When a physical quantity is discrete values, it is called quantized quantity.

Equation (1) shows that charge is also a quantized quantity like matter, energy, angular momentum etc. It means that we can find a body that can have a charge of $10e$ or $-5e$ but it is not possible to find a body with fractional charge such as $+3.57e$ or $-2.35e$.

According to the theory of elementary particles, protons and neutrons are not the elementary particles like electrons. They are consider to be composite particles made up of more fundamental particles called "QUARKS", which have fractional charges of magnitude $+\frac{2}{3}e$ and $-\frac{1}{3}e$.

Protons and neutrons each is made up of three quarks. Proton with the positive charge composed of two $+\frac{2}{3}e$ quarks and one $-\frac{1}{3}e$ quark i.e.,

$$\frac{2}{3}e + \frac{2}{3}e - \frac{1}{3}e = \frac{2e + 2e - e}{3} = \frac{3e}{3} = e$$

Which is the charge of a proton.

Neutron with the zero charge is made up of two $-\frac{1}{3}e$ quarks and one $+\frac{2}{3}e$ i.e.,

$$-\frac{1}{3}e - \frac{1}{3}e + \frac{2}{3}e = \frac{-e - e + 2e}{3} = 0$$

Although there is a strong evidence of existence of quarks within the proton and neutrons, but yet it is impossible to create free quark.

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Problem 24. Find total charge in coulombs of 75 kg of electron.

Solution:

Total mass of electrons $m = 75\ kg$

Mass of one electrons $m_e = 9.1 \times 10^{-31} kg$

Number of electron $n = \frac{m}{m_e} = \frac{75}{9.1 \times 10^{-31}} = 8.23 \times 10^{31}$

Total Charge $q = ne = 8.23 \times 10^{31} \times 1.6 \times 10^{-19} = 13.17 \times 10^{12} C$

Problem 26. The electrostatic force between identical ions that are separated by a distance of $5 \times 10^{-10} m$ is $3.7 \times 10^{-9}\ N$. (a) Find the charge on each ion? (b) How many electrons are missing from each ion?

Solution: $F = 3.7 \times 10^{-9}\ N$

$r = 5 \times 10^{-10} m$

(a) $q_1 = q_2 = q = ?$

$$F = k\frac{q_1 q_2}{r^2} = k\frac{q^2}{r^2}$$

$$\Longrightarrow q^2 = \frac{Fr^2}{k} = \frac{3.7 \times 10^{-9} \times (5 \times 10^{-10})^2}{9 \times 10^9} = 10.27 \times 10^{-38}$$

$$\Longrightarrow q = 3.20 \times 10^{-19} C$$

(b) $n = ?$

$$q = ne \Longrightarrow n = \frac{q}{e} = \frac{3.20 \times 10^{-19}}{1.6 \times 10^{-19}} = 2$$

27.4 Conservation of Charges

When the two bodies are rubbed together, they are electrified. The process of rubbing does not create charge but only transfer it from one body to the other. Thus the charges can neither be created nor destroyed. This hypothesis is called conservation of charge.

Examples of Charge Conservation

i. **Pair Production**

When a high energy γ-ray photon strike the heavy nucleus, the energy of the photon is converted into an electron-positron pair. This process is called pair production. This process is written as;

$$\gamma \longrightarrow e^- + e^+$$

The net charge is zero on the both sides i.e., the charge is conserved.

ii. **Pair Annihilation**

When an electron of charge $-e$ comes close to a positron of charge $+e$, they disappear and their rest mass energy is converted into radiant energy which appear in the form of two oppositely directed γ-ray photons.

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$$e^- + e^+ \longrightarrow \gamma + \gamma$$

The net charge is zero on the both sides i.e., the charge is conserved.

iii. **Decay of π^0 −Meson**

The neutral π^0 −Meson decays into two γ-ray photons as

$$\pi^0 \longrightarrow \gamma + \gamma$$

The net charge is zero on the both sides.

In another example of charge conservation, a neutron decays into a proton of charge $+e$ and an electron of charge $-e$ and a neutrino:

$${}^1_0n \longrightarrow {}^1_1p + {}^0_{-1}e + \nu$$

The net charge is zero on the both sides i.e., the charge is conserved.

iv. **D-D Reactions**

Two deuterium nuclei fused to tritium and helium as

$${}^2_1H + {}^2_1H \longrightarrow {}^3_1H + {}^1_1H + Energy$$

$${}^2_1H + {}^2_1H \longrightarrow {}^3_2He + {}^1_0n + Energy$$

In these nuclear reactions, charge is constant on both sides.

Problem 35. Identify the elements in the following nuclear reactions:

a) $\boldsymbol{H_1^1 + Be_4^9 \longrightarrow X + n_0^1}$

b) $\boldsymbol{C_6^{12} + H_1^1 \longrightarrow X}$

c) $\boldsymbol{N_7^{15} + H_1^1 \longrightarrow X}$

Solution:

By using law of conservation of charge number and mass number, we have

a) $H_1^1 + Be_4^9 \longrightarrow B_5^9 + n_0^1$

b) $C_6^{12} + H_1^1 \longrightarrow N_7^{13}$

c) $N_7^{15} + H_1^1 \longrightarrow O_8^{16}$

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THE ELECTRIC FIELD

Electric charges interact with each other over vast distances. Electrons or ionized atoms at the furthest reaches of the known universe can exert forces that cause electrons to move on the Earth.

How can we explain these interactions? We do so in terms of electric field- the distant charge set up an electric field, which exist throughout the space between the Earth and is the origin of the field.

In this chapter we consider only the static electric field due to charges at rest.

28.1 Electric Field Intensity

To describe the mechanism by which one charge particle exert the force the force on other charge particles, Michael Faraday introduced the concept of electric field.

Electric Field

The region or space around a charge in which it can exert the force of attraction or repulsion on other charged bodies is called electric field.

Electric Field Intensity

The electrostatic force on unit positive charge at a specific field point is called the electric field intensity. In order to find out electric field intensity, a test charge 'q_0' is placed in the electric field at a field point. The electric field intensity $\vec{E}$ is expressed as,

$$\vec{E} = \frac{\vec{F}}{q_0}$$

where $\vec{F}$ is the electrostatic force on test charge 'q_0'.

The test charge 'q_0' should be very small, so that it cannot disturb the field produced by source charge 'q'. Therefore the electric field intensity can be written as,

$$\vec{E} = \lim_{q_0 \to 0} \frac{\vec{F}}{q_0}$$

Analogy among Gravitation Field Intensity and Electric Field Intensity

The gravitational field strength $\vec{g}$ is described as the gravitational force $\vec{F}_g$ per unit test mass m_0:

$$\vec{g} = \frac{\vec{F}_g}{m_0}$$

In analogy with gravitational field intensity, the electric field intensity $\vec{E}$ is defined as electrostatic force $\vec{F}_e$ per unit test charge q_0:

$$\vec{E} = \frac{\vec{F}}{q_0}$$

Electric Field Lines or Lines of Force

A visual representation of the electric field can be obtained in terms of electric field lines. Electric field lines can be thought of a map that provides information about the direction and strength of the electric field at various places. As electric field line provides the information about the electric force exerted on a charge, the lines are commonly called "Lines of Force".

Properties of Electric Field Lines

i) Electric field lines originate from positive charges and end on negative charges.

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ii) The tangent to a field line at any point gives the direction of the electric field intensity at that point.

iii) The lines are closer where the field is strong, the lines are farther apart where the field is weak.

iv) No two lines cross each other.

Sample Problem 1. A proton is placed in a uniform electric field. What must be the magnitude and direction of this field f the electrostatic force acting on proton is just to balance its weight.

Solution: Charge on proton $q = 1.6 \times 10^{-19}\ C$

Mass of proton $m = 1.67 \times 10^{-27}\ kg$

For present case:

$$F_e = F_g$$

$$qE = mg \Longrightarrow E = \frac{mg}{q} = \frac{1.67 \times 10^{-27} \times 9.8}{1.6 \times 10^{-19}} = 10.02 \times 10^{-8} \frac{N}{C}$$

The direction of electric field will be opposite to that of weight, i.e., electric field will be directed upward.

Problem 41. In Millikan experiment a drop of radius $1.64\ \mu m$ and density $0.851 \frac{g}{cm^3}$ is balanced when $1.92 \times 10^5\ N/C$ is applied. Find the charge on drop in terms of e.

Solution: Radius of drop $r = 1.64\ \mu m = 1.64 \times 10^{-6} m$

Density $\rho = 0.851 \frac{g}{cm^3} = 0.851 \times 10^{-6} \frac{g}{m^3}$

Electric field $E = 1.92 \times 10^5\ N/C$

Charge q =?

For present case: $F_e = F_g$

$$qE = mg \Longrightarrow q = \frac{mg}{E} \Longrightarrow q = \frac{\rho V g}{E} = \frac{\rho \left(\frac{4}{3}\pi r^3\right) g}{E}$$

$$\Longrightarrow q = \frac{0.851 \times 10^{-6} \times \left(\frac{4}{3} \times 3.14 \times [1.64 \times 10^{-6}]^3\right) \times 9.8}{1.92 \times 10^5} = 8 \times 10^{-19} C$$

$$\Longrightarrow q = \frac{8 \times 10^{-19}}{1.6 \times 10^{-19}} e = 5e$$

Conceptual Q # 1. When defining the electric field, why is it necessary to specify that the magnitude of the test charge be very small?

Ans. So the electric field created by the test charge does not distort the electric field you are trying to measure, by moving the charges that create it.

Conceptual Q # 2. An object with negative charge is placed in a region of space where the electric field is directed vertically upward. What is the direction of the electric force exerted on this charge?

Ans. Vertically downward.

Conceptual Q # 3. Is it possible for an electric field to exist in empty space? Explain.

Ans. An electric field once established by a positive or negative charge extends in all directions from the charge. Thus, it can exist in empty space if that is what surrounds the charge.

Conceptual Q # 4. Explain why electric field lines never cross.

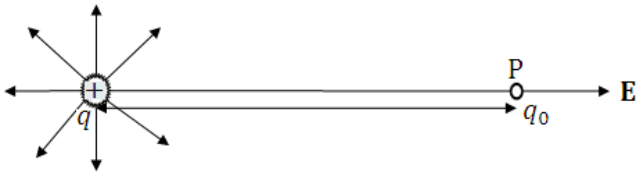
Ans. The direction of the electric field is the direction in which a positive test charge would feel a force when placed in the field. A charge will not experience two electrical forces at the same time, but the vector sum of the two. If electric field lines crossed, then a test charge placed at the point at which they cross would feel a force in two directions, which is not possible.

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28.2 Electric Field Intensity Due To a Point Charge

Consider a test charge 'q_0' placed at point P in the electric field of a point charge 'q' at a distance 'r' apart.



We want to find out electric field intensity at point 'P' due to a point charge 'q'.

The electrostatic force 'F' between 'q' and 'q_0' can be find out by using expression,

$$F = \frac{1}{4\pi\varepsilon_0}\frac{qq_0}{r^2}$$

The electric field intensity 'E' due to a point charge 'q' can be obtained by putting the value of electrostatic force in expression of electric field intensity:

$$E = \frac{\left(\frac{1}{4\pi\varepsilon_0}\frac{qq_0}{r^2}\right)}{q_0}$$

$$E = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}$$

This expression gives the magnitude of electric field intensity due to a point charge 'q'. In vector form, the electric field intensity '**E**' will be:

$$\vec{E} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\hat{r}$$

where $\hat{r}$ is the unit vector which gives the direction of electric field intensity.

Conceptual Q # 5. Explain what happens to the magnitude of the electric field created by a point charge as *r* approaches zero.

Ans. The electric field around a point charge approaches infinity as *r* approaches zero.

Sample Problem 2. In an ionized helium atom (a helium atom in which one of the two electrons has been removed) the electron and nucleus are separated by a distance of 26.5 pm. What electric field due to the nucleus at the location of the electron.

Solution:

Total charge of helium nucleus $q = 2e = 2 \times 1.6 \times 10^{-19} = 3.2 \times 10^{-19}\ C$

Distance $r = 26.5\ pm = 26.5 \times 10^{-12} m$

E =?

$$E = k\frac{q}{r^2} = 9 \times 10^9 \times \frac{3.2\times10^{-19}}{(26.5\times10^{-12})^2} = 4.1 \times 10^{12}\ \frac{N}{C}$$

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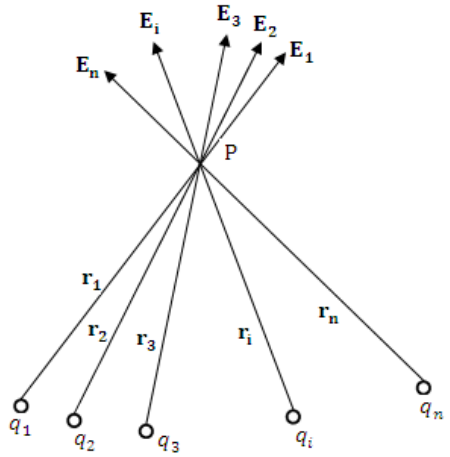
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28.3 Electric Field due to Many (*n*) Point Charges

Let $q_1, q_2, q_3, \ldots\ldots\ldots, q_n$ are the 'n' point charges which is at distances $r_1, r_2, r_3, \ldots\ldots\ldots, r_n$ respectively.

We want to find out the expression of electric field intensity due to assembly of n point charges at field point P.



Now if $\vec{E}_1, \vec{E}_2, \vec{E}_3, \ldots\ldots\ldots, \vec{E}_n$ be the electric field intensities at a field point due to the point charges $q_1, q_2, q_3, \ldots\ldots\ldots, q_n$ respectively. Then, the total electric field intensity due to assembly of 'n' point charges will be;

$$\vec{E} = \vec{E}_1 + \vec{E}_2 + \vec{E}_3 + \ldots\ldots\ldots + \vec{E}_n \quad ----(1)$$

where

$\vec{E}_1$ = Electric Field Intensity at a Field Point due to Point Charge 'q_1' $= \frac{1}{4\pi\varepsilon_0}\frac{q_1}{r_1^2}\hat{r}_1$

$\vec{E}_2$ = Electric Field Intensity at a Field Point due to Point Charge 'q_2' $= \frac{1}{4\pi\varepsilon_0}\frac{q_2}{r_2^2}\hat{r}_2$

$\vec{E}_3$ = Electric Field Intensity at a Field Point due to Point Charge 'q_3' $= \frac{1}{4\pi\varepsilon_0}\frac{q_3}{r_3^2}\hat{r}_3$

$\vec{E}_n$ = Electric Field Intensity at a Field Point due to Point Charge 'q_n' $= \frac{1}{4\pi\varepsilon_0}\frac{q_n}{r_n^2}\hat{r}_n$

Putting values in equation (1), we get:

$$\vec{E} = \frac{1}{4\pi\varepsilon_0}\frac{q_1}{r_1^2}\hat{r}_1 + \frac{1}{4\pi\varepsilon_0}\frac{q_2}{r_2^2}\hat{r}_2 + \frac{1}{4\pi\varepsilon_0}\frac{q_3}{r_3^2}\hat{r}_3 + \ldots\ldots\ldots + \frac{1}{4\pi\varepsilon_0}\frac{q_n}{r_n^2}\hat{r}_n$$

$$= \frac{1}{4\pi\varepsilon_0}\left(\frac{q_1}{r_1^2}\hat{r}_1 + \frac{q_2}{r_2^2}\hat{r}_2 + \frac{q_3}{r_3^2}\hat{r}_3 + \ldots\ldots\ldots + \frac{q_n}{r_n^2}\hat{r}_n\right)$$

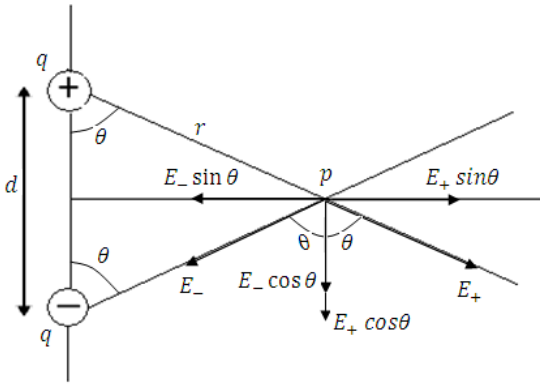
$$= \frac{1}{4\pi\varepsilon_0}\sum_{i=1}^{n}\frac{q_i}{r_i^2}\hat{r}_i$$

This equation gives the total electric field intensity due to assembly of 'n' point charges at a specific field point.

28.4 Electric Field due to a Dipole

Two equal and opposite charges separated by a small distance form an electrical dipole. Consider two point charges $+q$ and $-q$ of equal magnitude lying distance '*d*' apart as shown in the figure below.

We want to determine the electric field intensity **E** due to a dipole at point 'P'. The point 'P' is at a distance 'x' along the perpendicular bisector of the line joining the charges. Let the electric field intensities at point P due to the charges $+q$ and $-q$ be $\vec{E}_+$ and $\vec{E}_-$ respectively.



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The total electric field intensity at point P due to the charges $+q$ and $-q$ is given by the expression.

$$\vec{E} = \vec{E}_+ + \vec{E}_- \quad ---- \quad (1)$$

For the present case, $|E_+| = |E_-|$ because the point P is equidistant for the charges $+q$ and $-q$. Therefore,

$$|E_+| = |E_-| = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2} = \frac{1}{4\pi\varepsilon_0}\frac{q}{\left[x^2+\left(\frac{d}{2}\right)^2\right]} \qquad \therefore From\ Fig. \qquad r^2 = x^2+\left(\frac{d}{2}\right)^2$$

From the figure it is clear that the $y-$components of $\vec{E}_+$ and $\vec{E}_-$ will cancel the effect of each other, while the $x-$components of $\vec{E}_+$ and $\vec{E}_-$ added up to give of resultant electric field intensity $\vec{E}$ of electric dipole. Therefore,

$$|E| = |E_+|\cos\theta + |E_-|\cos\theta \qquad \therefore |E_+| = |E_-|$$

$$|E| = 2|E_+|\cos\theta \quad ---- \quad (2)$$

From Fig. $\cos\theta = \frac{(d/2)}{r} \Longrightarrow cos\theta = \frac{d/2}{\sqrt{x^2+\left(\frac{d}{2}\right)^2}}$

By putting the value of $|E_+|$ and $\cos\theta$ in equation (2), we get

$$|E| = 2|E_+|\left(\frac{d/2}{\sqrt{x^2+\left(\frac{d}{2}\right)^2}}\right) = |E_+|\left(\frac{d}{\sqrt{x^2+\left(\frac{d}{2}\right)^2}}\right)$$

$$|E| = \frac{1}{4\pi\varepsilon_0}\frac{q}{\left[x^2+\left(\frac{d}{2}\right)^2\right]}\left(\frac{d}{\sqrt{x^2+\left(\frac{d}{2}\right)^2}}\right) = \frac{1}{4\pi\varepsilon_0}\frac{qd}{\left[x^2+\left(\frac{d}{2}\right)^2\right]^{\frac{3}{2}}}$$

The quantity $p = qd$ is called dipole moment. Therefore,

$$|E| = \frac{1}{4\pi\varepsilon_0}\frac{p}{\left[x^2+\left(\frac{d}{2}\right)^2\right]^{\frac{3}{2}}} = \frac{1}{4\pi\varepsilon_0}\frac{p}{x^3\left[1+\frac{1}{x^2}\left(\frac{d}{2}\right)^2\right]^{\frac{3}{2}}}$$

$$|E| = \frac{1}{4\pi\varepsilon_0}\frac{p}{x^3}\left[1+\left(\frac{d}{2x}\right)^2\right]^{-\frac{3}{2}}$$

By using binomial expansion, we get

$$E = \frac{1}{4\pi\varepsilon_0}\frac{p}{x^3}\left[1+\left(-\frac{3}{2}\right)\left(\frac{d}{2x}\right)^2+\cdots\cdots\cdots\right]$$

If we neglect the 2nd and higher terms, then

$$E = \frac{1}{4\pi\varepsilon_0}\frac{p}{x^3}$$

This is the expression of electric field due to a dipole.

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Problem 40. Two point charges of magnitude $q_1 = 2.16\,\mu C$, $q_2 = 85.3\,nC$ are $11.7\,cm$ apart. Find magnitude of electric field that each produce on other. Find the magnitude of force on each charge?

Solution:

$q_1 = 2.16\,\mu C = 2.16 \times 10^{-6}\,C$

$q_2 = 85.3\,nC = 85.3 \times 10^{-9}\,C$

Distance $r = 11.7\,cm = 11.7 \times 10^{-2} m$

Electric field produce by $q_1 = E_{12}$ =?

$$E_{12} = k\frac{q_1}{r^2} = 9 \times 10^9 \times \frac{2.16 \times 10^{-6}}{(11.7 \times 10^{-2})^2} = 0.14 \times 10^7 \frac{N}{C}$$

Electric field produce by $q_2 = E_{21}$ =?

$$E_{21} = k\frac{q_2}{r^2} = 9 \times 10^9 \times \frac{85.3 \times 10^{-9}}{(11.7 \times 10^{-2})^2} = 5.62 \times 10^4 \frac{N}{C}$$

Force on charge $q_1 = F_1$ =?

$$F_1 = k\frac{q_1 q_2}{r^2} = 9 \times 10^9 \times \frac{2.16 \times 10^{-6} \times 85.3 \times 10^{-9}}{(11.7 \times 10^{-2})^2} = 12.139 \times 10^{-2}\,N$$

Force on charge $q_2 = F_2$ =?

$$F_2 = k\frac{q_1 q_2}{r^2} = 9 \times 10^9 \times \frac{2.16 \times 10^{-6} \times 85.3 \times 10^{-9}}{(11.7 \times 10^{-2})^2} = 12.139 \times 10^{-2}\,N$$

Problem 39. Two equal and opposite charges of magnitude $1.88 \times 10^{-7}\,C$ and held 15.2 cm apart. What is direction and magnitude of E at midway point between the charges? What is the force act on an electron placed here?

Solution:

$q_1 = 1.88 \times 10^{-7}\,C$

$q_2 = -1.88 \times 10^{-7}\,C$

Distance $r = 15.2\,cm = 15.2 \times 10^{-2} m$

Total electric field $E = |E_+| + |E_-|$ =?

Force on an electron placed at the same point F =?

$$|E_+| = k\frac{q_1}{r^2} = 9 \times 10^9 \times \frac{1.88 \times 10^{-7}}{(15.2 \times 10^{-2})^2} = 0.32 \times 10^6 \frac{N}{C}$$

$$|E_-| = k\frac{q_2}{r^2} = 9 \times 10^9 \times \frac{1.88 \times 10^{-7}}{(15.2 \times 10^{-2})^2} = 0.32 \times 10^6 \frac{N}{C}$$

Total electric field $E = |E_+| + |E_-| = 0.32 \times 10^6 + 0.32 \times 10^6 = 0.64 \times 10^6 \frac{N}{C}$

Now

Force on an electron placed at the same point $F = qE = -1.6 \times 10^{-19} \times 0.32 \times 10^6$

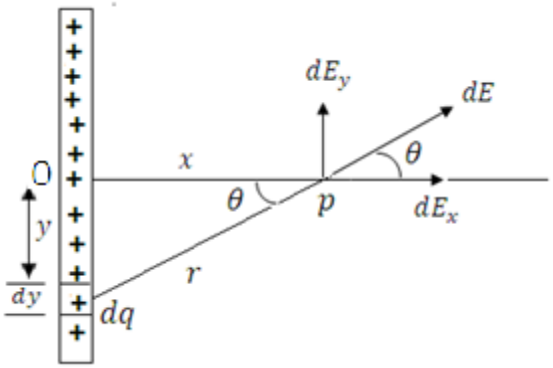
$$= -1.024 \times 10^{-13}\,N$$

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28.5 Electric Field Intensity due to an Infinite Line of Charges

Consider a portion of an infinite line of positive charge as shown in the figure below.



We want to calculate the electric field intensity at point 'P' at a perpendicular distance 'x' from the line of charge. As the charge is distributed uniformly over it, so it has constant linear charge density μ;

$$\mu = \frac{Charge}{Length} = \frac{q}{L}$$

For an infinitesimal length element 'dy' having charge 'dq'

$$\mu = \frac{dq}{dy} \Longrightarrow dq = \mu\, dy$$

The electric field intensity due to this length element at point 'P' is given by;

$$dE = \frac{1}{4\pi\varepsilon_0}\frac{dq}{r^2} = \frac{1}{4\pi\varepsilon_0}\frac{\mu\, dy}{r^2}$$

$$dE = \frac{1}{4\pi\varepsilon_0}\frac{\mu\, dy}{x^2 + y^2}$$

The rectangular components of 'dE' are

$$dE_x = dE\cos\theta \text{ and } dE_y = dE\sin\theta$$

If we consider identical charge elements symmetrically located on both sides of 'O', then the y −components of electrical field intensity will cancel out each other and its x −components are added up to give the total electric field intensity due to this continuous charge distribution. Therefore, the total electric field intensity will be

$$E = \int_{y=-\infty}^{y=\infty} dE_x = \int_{y=-\infty}^{y=\infty} dE\cos\theta = 2\int_{y=0}^{y=\infty} dE\cos\theta$$

Putting value of 'dE', we get

$$E = 2\int_{y=0}^{y=\infty} \frac{1}{4\pi\varepsilon_0}\frac{\mu\, dy}{x^2 + y^2}\cos\theta = \frac{1}{2\pi\varepsilon_0}\int_{y=0}^{y=\infty} \frac{\mu\, dy}{x^2 + y^2}\cos\theta$$

From Fig. $\frac{y}{x} = \tan\theta \Longrightarrow y = x\tan\theta$

$$dy = x\sec^2\theta\, d\theta$$

$$\therefore when\ y = 0,\ \theta = 0$$

$$\therefore when\ y = \infty,\quad \theta = \frac{\pi}{2}$$

$$\Longrightarrow E = \frac{\mu}{2\pi\varepsilon_0}\int_{\theta=0}^{\theta=\frac{\pi}{2}} \frac{x\sec^2\theta\, d\theta}{x^2 + x^2\tan^2\theta}\cos\theta = \frac{\mu}{2\pi\varepsilon_0}\int_{\theta=0}^{\theta=\frac{\pi}{2}} \frac{x\sec\theta\, d\theta}{x^2(1 + \tan^2\theta)}$$

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$$E = \frac{\mu}{2\pi\varepsilon_0}\int_{\theta=0}^{\theta=\frac{\pi}{2}} \frac{\sec\theta\, d\theta}{x\sec^2\theta} = \frac{\mu}{2\pi\varepsilon_0 x}\int_{\theta=0}^{\theta=\frac{\pi}{2}} \cos\theta\ d\theta$$

$$E = \frac{\mu}{2\pi\varepsilon_0 x}\,|sin\theta|_0^{\pi/2} = \frac{\mu}{2\pi\varepsilon_0 x}\left(sin\frac{\pi}{2} - \sin 0\right) = \frac{\mu}{2\pi\varepsilon_0 x}\,(1-0)$$

$$E = \frac{\mu}{2\pi\varepsilon_0 x}$$

This is the expression for electric field intensity at point 'P' due to an infinite line of charge.

Question. Explain the differences between linear, surface, and volume charge densities, and give examples of when each would be used.

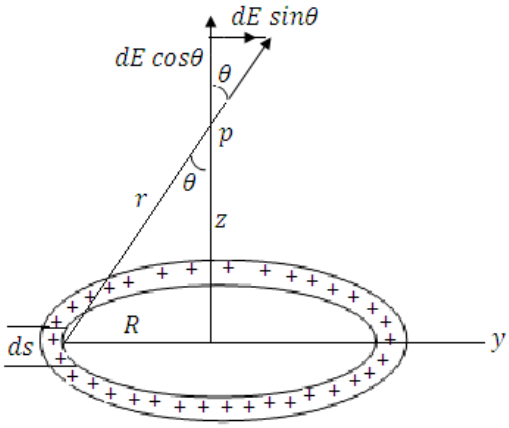
Ans. Linear charge density, μ, is charge per unit length. It is used when trying to determine the electric field created by a charged rod.

Surface charge density, σ, is charge per unit area. It is used when determining the electric field above a charged sheet or disk.

Volume charge density, ρ, is charge per unit volume. It is used when determining the electric field due to a uniformly charged sphere made of insulating material.

28.6 Electric Field Intensity due to a Ring of Charge

Consider a ring of positive charge of radius 'R' as shown in the figure below.



We want to find out the electric field intensity at point 'P' which is at the distance 'z' from the plane of ring.

As the charge is distributed uniformly over it, so it has constant linear charge density μ. For an infinitesimal length element 'ds' of ring,

$$\mu = \frac{dq}{ds} \Rightarrow dq = \mu\, ds$$

The electric field intensity due to the charge dq at point 'P' is given by;

$$dE = \frac{1}{4\pi\varepsilon_0}\frac{dq}{r^2} = \frac{1}{4\pi\varepsilon_0}\frac{\mu\, ds}{r^2}$$

$$dE = \frac{1}{4\pi\varepsilon_0}\ \frac{\mu\, ds}{z^2 + R^2}$$

The rectangular components of 'dE' are

$$dE_z = dE\cos\theta \text{ and } dE_y = dE\sin\theta$$

If we consider the identical charge elements located on the opposite end of tne diameter, then dE_y components will cancel out each other and dE_z components are added up to give the final value of electric field intensity at point P. Therefore, the total electric field intensity will be

$$E = E_z = \int dE_z = \int dE\ cos\theta$$

Putting value of 'dE', we get

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$$\Rightarrow E = \frac{1}{4\pi\varepsilon_0} \int \frac{\mu\, ds}{z^2 + R^2} cos\theta$$

From figure, $\cos\theta = \frac{z}{R}$

$$\Rightarrow E = \frac{1}{4\pi\varepsilon_0} \int \frac{\mu\, ds}{z^2 + R^2} \cdot \frac{z}{r}$$

From figure, $z^2 + R^2 = r^2$

$$\Rightarrow E = \frac{\mu z}{4\pi\varepsilon_0} \int \frac{ds}{r^3} = \frac{\mu z}{4\pi\varepsilon_0 r^3} \int ds = \frac{\mu z}{4\pi\varepsilon_0 r^3} (2\pi r) = \frac{(2\pi r\mu) z}{4\pi\varepsilon_0 (z^2 + R^2)^{3/2}}$$

$$\therefore \mu = \frac{total\ charge}{total\ length} = \frac{q}{2\pi r} \Rightarrow 2\pi r\, \mu = q$$

$$E = \frac{qz}{4\pi\varepsilon_0 (z^2 + R^2)^{3/2}}$$

This expression gives electric field intensity due to ring of charge.

Special Case:

When the point P is far away from ring, i.e., $z \gg R$ so that R^2 can be neglected.

$$E = \frac{qz}{4\pi\varepsilon_0 (z^2)^{3/2}} = \frac{qz}{4\pi\varepsilon_0 z^3}$$

$$\Rightarrow E = \frac{1}{4\pi\varepsilon_0} \cdot \frac{q}{z^2}$$

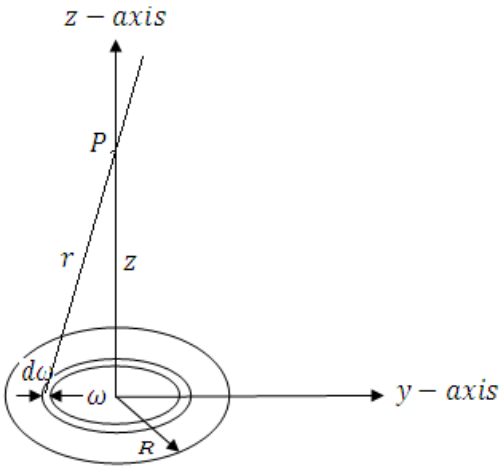
Which is the expression for the electric field intensity, when the field point is far away from the ring. Thus, the charged ring acts like a point charge when the field point is at the large distance.

Conceptual Q # 7. When is it valid to approximate a charge distribution by a point charge?

Ans. If a charge distribution is small compared to the distance of a field point from it, the charge distribution can be modeled as a single particle with charge equal to the net charge of the distribution. Further, if a charge distribution is spherically symmetric, it will create a field at exterior points just as if all of its charge were a point charge at its center.

28.8 Electric Field Intensity due to a Disk of Charge

Consider a circular disk of uniform surface charge density as shown in the figure below. We want to find out the electric field intensity at point 'P' which is at the distance 'z' from the plane of disk.



Consider a small element of the disk in the ring shape of radius 'ω' and width '$d\omega$'. If 'dq' is the charge on this element of ring having surface area dA, then

$$\sigma = \frac{dq}{dA} \Rightarrow dq = \sigma\, dA$$

$$\therefore dA = 2\pi\omega\, d\omega$$

$$\Rightarrow dq = \sigma\, (2\pi\omega\, d\omega)$$

We know the electric field intensity due to the ring of charge is given by;

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$$E = \frac{qz}{4\pi\varepsilon_0(z^2 + \omega^2)^{3/2}}$$

Consider the ring as the differential part of the disk. The above equation takes the form:

$$dE = \frac{z\,dq}{4\pi\varepsilon_0(z^2 + \omega^2)^{3/2}}$$

By putting the value of 'dq', we get

$$\Rightarrow dE = \frac{z\,\sigma\,(2\pi\omega\,d\omega)}{4\pi\varepsilon_0(z^2 + \omega^2)^{3/2}} = \frac{z\,\sigma}{4\varepsilon_0}\ \frac{(2\omega\,d\omega)}{(z^2 + \omega^2)^{3/2}}$$

$$\Rightarrow dE = \frac{z\,\sigma}{4\varepsilon_0}\ (z^2 + \omega^2)^{-3/2}\,(2\omega\,d\omega)$$

Now total electric field intensity at point P due to whole disk can be calculated integrating the above expression from '$\omega = 0$ to $\omega = R$'

$$E = \int_{\omega=0}^{\omega=R} dE$$

$$\Rightarrow E = \frac{z\,\sigma}{4\varepsilon_0}\int_{\omega=0}^{\omega=R}(z^2 + \omega^2)^{-3/2}\,(2\omega\,d\omega) = \frac{z\,\sigma}{4\varepsilon_0}\left|\frac{(z^2 + \omega^2)^{\frac{-3}{2}+1}}{\frac{-3}{2}+1}\right|_0^R$$

$$\Rightarrow E = \frac{z\,\sigma}{4\varepsilon_0}\left|\frac{(z^2 + \omega^2)^{\frac{-1}{2}}}{\frac{-1}{2}}\right|_0^R = -\frac{z\,\sigma}{2\varepsilon_0}\left|\frac{1}{\sqrt{z^2 + \omega^2}}\right|_0^R = -\frac{z\,\sigma}{2\varepsilon_0}\ \left(\frac{1}{\sqrt{z^2 + R^2}} - \frac{1}{\sqrt{z^2 + 0}}\right)$$

$$\Rightarrow E = -\frac{z\,\sigma}{2\varepsilon_0}\ \left(\frac{1}{\sqrt{z^2 + R^2}} - \frac{1}{z}\right) = \frac{z\,\sigma}{2\varepsilon_0}\left(\frac{1}{z} - \frac{1}{\sqrt{z^2 + R^2}}\right)$$

$$\Rightarrow E = \frac{\sigma}{2\varepsilon_0}\ \left(1 - \frac{z}{\sqrt{z^2 + R^2}}\right)$$

This is expression of electric field intensity due to disc of charge.

Special Case

If $R \gg z$; i.e., $z = 0$, Then $\frac{z}{\sqrt{z^2 + R^2}} = \frac{0}{\sqrt{0^2 + R^2}} = 0$

Hence $E = \frac{\sigma}{2\varepsilon_0}$

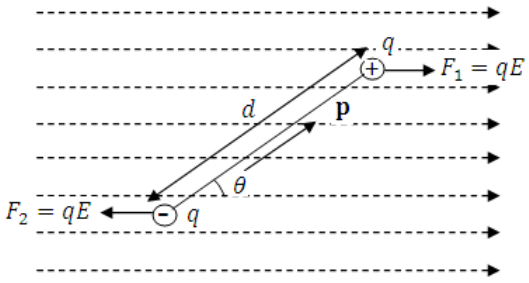
It means when $z = 0$, the disk of charge behaves like infinite sheet of charge.

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28.8 Torque on a Dipole in a Uniform Electric Field

Consider an electric dipole consists of $+q$ and $-q$ separated by a distance d is shown in the figure below.



When an electric dipole is placed in an external electric field, the force on the positive charge will be in one direction and the force on the negative charge in another direction. The force on $+q$ and $-q$ have the equal magnitude but opposite in direction. Therefore the net force on the dipole due to external field is zero. The magnitude of each force is

$$|F_1| = |F_2| = qE$$

These two forces make a couple and so the torque acts on the dipole. The magnitude of this torque is given as

Torque = (Force) (Moment Arm)

$$\tau = F\,(d\,sin\theta)$$

$$\tau = qEd\,sin\theta$$

$$\tau = pE\,sin\theta$$

$$\therefore Dipole\ Moment\ p = qd$$

In vector form,

$$\vec{\tau} = \vec{p} \times \vec{E}$$

The direction of torque of dipole in a uniform electric field is determined by right hand rule.

Problem 49. A dipole consists of two charges +2e and -2e separated by a distance of 0.78nm. It is in electric field strength of 3.4×10^6 N/C. Calculate the magnitude of torque when dipole moment is (a) parallel (b) at right angle and (c) opposite to electric field.

Solution:

$q_1 = 2e = 3.2 \times 10^{-19}$ C

$q_2 = -2e = -3.2 \times 10^{-19}$ C

$d = 0.78nm = 0.78 \times 10^{-9}$ m

$E = 3.4 \times 10^6$ N/C

Torque = ?

Case-1: When the dipole moment is parallel to the field

$$\tau = pE\sin\theta = qdE\sin\theta = 3.2 \times 10^{-19} \times 0.78 \times 10^{-9} \times 3.4 \times 10^6 \times \sin 0^0 = 0$$

Case-2: When the dipole moment is parallel to the field

$$\tau = pE\sin\theta = qdE\sin\theta = 3.2 \times 10^{-19} \times 0.78 \times 10^{-9} \times 3.4 \times 10^6 \times \sin 90^0 = 8.48 \times 10^{22}\ \text{Nm}$$

Case-3: When the dipole moment is parallel to the field

$$\tau = pE\sin\theta = qdE\sin\theta = 3.2 \times 10^{-19} \times 0.78 \times 10^{-9} \times 3.4 \times 10^6 \times \sin 180^0 = 0$$

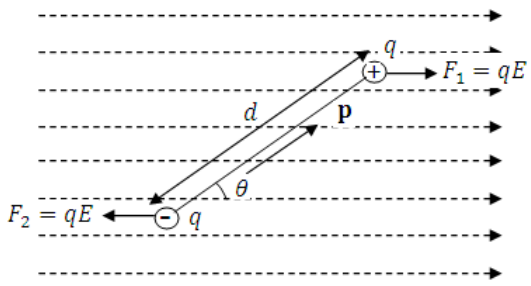
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28.10 Energy of Dipole in a Uniform Electric Field

Consider an electrical dipole is placed in a uniform electric field. We want to calculate the work done by the electric field in turning the dipole through an angle 'θ'. The work done by the electric field in turning the dipole from an initial angle $\theta_i$to the final angle θ_f is given by;



$$W = \int dw$$

$$W = -\int_{\theta_i}^{\theta_f} \tau \, d\theta$$

Where 'τ' is the torque exerted by the electric field. The negative sign is necessary because the torque tends to decrease θ. Also

$$\tau = pE \sin\theta$$

Therefore,

$$W = -\int_{\theta_i}^{\theta_f} pE \sin\theta \, d\theta$$

Assuming that the dipole is revolve from initial angle $\theta_i = \frac{\pi}{2}$ to the final angle $\theta_f = \theta$. Then

$$W = \int_{\frac{\pi}{2}}^{\theta} pE \sin\theta \, d\theta$$

$$W = pE \int_{\frac{\pi}{2}}^{\theta} \sin\theta \, d\theta$$

$$W = pE \, |\cos\theta \, |_{\pi/2}^{\theta}$$

$$W = pE \, [\cos\theta - \cos\frac{\pi}{2}]$$

$$W = pE \cos\theta$$

Since the work done by the agent that produces the external field is equal to the negative of the change in potential energy U of the system.

$$U = -W = -pE\cos\theta$$

Therefore

Potential Energy $U = -\mathbf{p}.\mathbf{E}$

This is the expression of potential energy of the dipole in a uniform electric field.

Problem 48. An electric dipole consisting of charges of magnitude $1.48\,nC$ separated by $6.23\mu m$ in an electric field of strength 1100 N/C. (a) What is the magnitude of electric dipole moment. (b) What is the difference in potential energy corresponding to dipole orientation parallel and anti parallel to the field?

Solution:

$$q = 1.48 \text{ nC} = 1.48 \times 10^{-9} \text{ C}$$

$$d = 6.23\mu\text{m} = 6.23 \times 10^{-6}\text{m}$$

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$$E = 1100\ N/C$$

As dipole moment $p = qd$ =?

$$p = 1.48 \times 10^{-9} \times 6.23 \times 10^{-6} = 9.22 \times 10^{-15}$$

Now, the potential energy of the dipole in an electric field can be field out by the expression:

$$U = -\mathbf{p}.\mathbf{E} = pE\cos\theta$$

Case 1.

When dipole is parallel to electric field, then $\theta = 0^0$

$$U_i = -pE\cos 0^0 = -9.22 \times 10^{-15} \times 1100 \times 1 = -10142 \times 10^{-15}$$

Case 2.

When dipole is parallel to electric field, then $\theta = 180^0$

$$U_f = -pE\cos 180^0 = -9.22 \times 10^{-15} \times 1100 \times (-1) = 10142 \times 10^{-15}$$

Difference between P.E

$$\Delta U = U_f - U_i = 10142 \times 10^{-15} - (-10142 \times 10^{-15}) = 20284 \times 10^{-15}$$

Problem 51. Find the work required to turn an electric dipole end to end in a uniform electric field E in terms of dipole moment p and initial angle θ_0 between p and E.

Solution:

Given that the dipole is placed in a uniform electric field E and initially, the dipole makes an angle θ_0 with it. The work will be done on dipole and it will attain inclination $\theta = 180^0$ with E after some time. Then

Work $W = -\Delta U = -(U_f - U_i)$ =?

$$U_i = pE\cos\theta_0$$

$$U_f = pE\cos(180 + \theta_0) = -[-pE\cos\theta_0] = pE\cos\theta_0$$

Now

$$W = -\Delta U = -(U_f - U_i)$$

$$= -(pE\cos\theta_0 - [-pE\cos\theta_0]) = -(pE\cos\theta_0 + pE\cos\theta_0) = -2pE\cos\theta_0$$

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GAUSS'S LAW

Coulomb's law can always be used to calculate the electric field intensity $\vec{E}$ for any discrete or continuous charge distribution of charges at rest. The sums or integrals might be complicated (and a computer might be needed to evaluate them numerically), but resulting electric field intensity $\vec{E}$ can always be found.

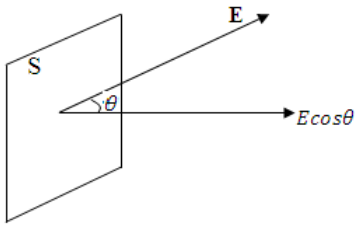
In this chapter, we discuss an alternative to Coulomb's law, called Gauss's law, that provides a more useful and instructive approach to calculating the electric field in the situations having certain symmetries.

The number of situations that can directly be analyzed using Gauss's law is small, but those cases can be done with extraordinary ease. Although Gauss's law and Coulomb's law gives identical results in the cases in which both can be used. Gauss's law is considered a more fundamental equation than Coulomb's law. It is fair to say that while Coulomb's law provides workhorse of electrostatics, Gauss's law provides the insight.

29.1 Electric Flux

The number of electric lines of force passing normally through a certain area is called the electric flux. It is measured by the product of area and the component of electric field intensity normal to the area. It is denoted by the symbol Φ_e.

Consider a surface 'S' placed in a uniform electric field of intensity '$\vec{E}$'. Let 'A' be the area of the surface. The component of $\vec{E}$ normal to the area A is $E\cos\theta$ as shown in the figure below.



The electric flux through the surface S is given by

$$\Phi_e = A\,(E\cos\theta)$$

$$\Phi_e = EA\cos\theta$$

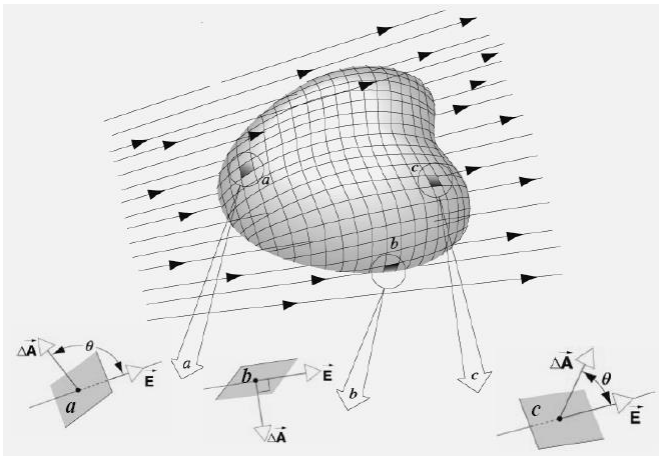
$$\Phi_e = \vec{E}.\vec{A}$$

Thus the electric flux is the scalar product of electric field intensity and the vector area. The SI unit of the electric flux is $\frac{Nm^2}{C}$.

29.2 Electric Flux through an Irregular Shaped Object

Consider an object of irregular shape placed in a non-uniform electric field. We want to find out the expression of electric flux through this irregular shaped object.

We divide the surface into n number of small patches having area $\Delta A_1, \Delta A_2, \Delta A_3, \ldots\ldots\ldots, \Delta A_n$. Let $\vec{E}_1, \vec{E}_2, \vec{E}_3, \ldots\ldots\ldots, \vec{E}_n$ are the electric field intensities which makes angle $\theta_1, \theta_2, \theta_3, \ldots\ldots\ldots, \theta_n$ with thenormal to the area elements $\Delta A_1, \Delta A_2, \Delta A_3, \ldots\ldots\ldots, \Delta A_n$, respectively. If $\Phi_1, \Phi_2, \Phi_3, \ldots\ldots\ldots, \Phi_n$ be the electric flux through $\Delta A_1, \Delta A_2, \Delta A_3, \ldots\ldots\ldots, \Delta A_n$, then the total electric flux Φ_e will be:



$$\Phi_e = \Phi_1 + \Phi_2 + \Phi_3, + \ \ldots\ldots\ldots + \Phi_n$$

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$$\Longrightarrow \Phi_e = \mathrm{E}_1(\Delta\mathrm{A}_1 \cos\theta_1) + \mathrm{E}_2(\Delta\mathrm{A}_2 \cos\theta_2) + \mathrm{E}_3(\Delta\mathrm{A}_3 \cos\theta_3) + \ \ldots\ldots\ldots + \mathrm{E}_n(\Delta\mathrm{A}_n \cos\theta_n)$$

$$\Longrightarrow \Phi_e = \mathrm{E}_1\Delta\mathrm{A}_1 \cos\theta_1 + \mathrm{E}_2\Delta\mathrm{A}_2 \cos\theta_2 + \mathrm{E}_3\Delta\mathrm{A}_3 \cos\theta_3 + \ \ldots\ldots\ldots + \mathrm{E}_n\Delta\mathrm{A}_n \cos\theta_n$$

$$\Longrightarrow \Phi_e = \vec{\mathrm{E}}_1.\Delta\vec{\mathrm{A}}_1 + \vec{\mathrm{E}}_2.\Delta\vec{\mathrm{A}}_2 + \vec{\mathrm{E}}_3.\Delta\vec{\mathrm{A}}_3 + \ \ldots\ldots\ldots + \vec{\mathrm{E}}_n.\Delta\vec{\mathrm{A}}_n$$

Where $\Delta\vec{\mathrm{A}}_1, \Delta\vec{\mathrm{A}}_2, \Delta\vec{\mathrm{A}}_3, \ldots\ldots\ldots, \Delta\vec{\mathrm{A}}_\mathrm{n}$ are the vector areas corresponding to area elements $\Delta\mathrm{A}_1, \Delta\mathrm{A}_2, \Delta\mathrm{A}_3, \ldots\ldots\ldots, \Delta\mathrm{A}_\mathrm{n}$respectively.

$$\Phi_e = \sum_{\mathrm{i}=1}^{\mathrm{n}} \vec{\mathrm{E}}_\mathrm{i}.\Delta\vec{\mathrm{A}}_\mathrm{i}$$

When $n \to \infty$, or $\Delta A \to 0$, then the sigma is replaced by the surface integral i.e.,

$$\Phi_e = \int_{\mathrm{S}} \vec{\mathrm{E}}.\mathrm{d}\vec{\mathrm{A}}$$

By convention, the outward flux is taken as positive and inward flux is taken as negative.

29.3 Gauss's Law

Statement

The total electric flux through any close surface is $\frac{1}{\varepsilon_0}$ times the total charge enclosed by the surface.

Explanation

The Gauss's law gives the relation between total flux and total charge enclosed by the surface. Consider a collection of positive and negative charges in a certain region of space. According to Gauss's law:

$$\Phi_e = \frac{q}{\varepsilon_0} \quad ----- \quad (\mathbf{1})$$

where q is the net charge enclosed by the surface. Also,

$$\Phi_e = \oint \vec{\mathrm{E}}.\mathrm{d}\vec{\mathrm{A}} \quad ----- \quad (\mathbf{2})$$

Comparing (1) and (2), we have:

$$\oint \vec{\mathrm{E}}.\mathrm{d}\vec{\mathrm{A}} = \frac{q}{\varepsilon_0}$$

Thus we can describe the Gauss's law as

The surface normal integral of electric field intensity is equal to

$\frac{1}{\varepsilon_0}$*times the total charge enclosed by the surface.*

Problem 5. A point charge of $1.84\ \mu C$ is at center of cubical Gaussian surface of $55cm$edge. Find flux through the surface.

Solution:

$$q = 1.84\mu C = 1.84 \times 10^{-6} C$$

Φ_e =?

Applying Gauss law:

$$\Phi_\mathrm{e} = \frac{q}{\varepsilon_0} = \frac{1.84 \times 10^{-6}}{8.85 \times 10^{-12}} = 2.07 \times 10^5 \frac{N - m^2}{C}$$

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29.4Differential Form of Gauss's Law

If the charge is distributed into a volume having uniform volume charge density 'ρ', then charge enclosed q by Gaussian surface is described by expression:

$$q = \int_v \rho \, dv$$

By Gauss's law:

$$\oint \vec{E}.d\vec{A} = \frac{q}{\varepsilon_0}$$

$$\Rightarrow \oint \vec{E}.d\vec{A} = \frac{1}{\varepsilon_0}\int_v \rho \, dv \quad ---- \quad (1)$$

By Gauss's Divergence theorem,

$$\oint \vec{E}.d\vec{A} = = \int_v div\, \mathbf{E}\, dv \quad ---- \quad (2)$$

Comparing (1) and (2), we have:

$$\int_v div\, \vec{E}\, dv = \frac{1}{\varepsilon_0}\int_v \rho \, dv$$

$$\Rightarrow \int_v div\, \vec{E}\, dv - \frac{1}{\varepsilon_0}\int_v \rho \, dv = \mathbf{0}$$

$$\Rightarrow \int_v \left(div\, \vec{E} - \frac{1}{\varepsilon_0}\rho\right) dv = 0$$

As $dv \neq \mathbf{0}$

$$div\, \vec{E} - \frac{1}{\varepsilon_0}\rho = 0$$

$$\Rightarrow div\, \vec{E} = \frac{1}{\varepsilon_0}\rho$$

This is differential form of Gauss's law.

29.4 Integral Form of Gauss's Law

By Gauss's law:

$$\oint \vec{E}.d\vec{A} = \frac{q}{\varepsilon_0} \qquad where\ 'q'\ is\ the\ total\ charge\ enclosed$$

If the charge is uniformly distributed into a volume having charge density 'ρ', then

$$\Phi_e = \oint \vec{E}.d\vec{A} = \frac{1}{\varepsilon_0}\int_v \rho \, dv \quad ---- \quad (1)$$

If the charge is uniformly distributed over a surface having a surface charge density 'σ', then

$$\Phi_e = \oint \vec{E}.d\vec{A} = \frac{1}{\varepsilon_0}\int_s \sigma \, dA \quad ---- \quad (2)$$

Equation (1) and (2) are the integral form of Gauss's law.

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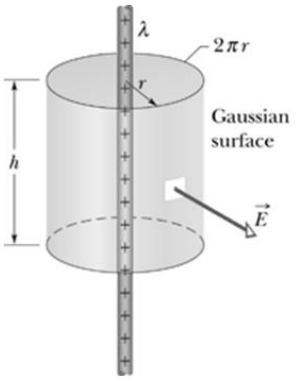
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29.5 Applications of Gauss's law

Gauss's law can be used to calculate the electric field intensity due to certain charge distributions if the charge distribution has the greater symmetry.

29.5.1 Electric Field due to Infinite Line of Charge

Consider a section of infinite line of charge having uniform linear charge density 'λ' as shown in the figure below.



We want to find out electric field intensity at any point 'P' which is at distance 'r' from the wire. For this we consider cylindrical Gaussian surface which passes through point 'P'. The electric flux passing through the cylinder is given as

$$\Phi_e = \oint_S \vec{E}.d\vec{A}$$

The surface 'S' of the cylinder consist of three parts i.e., $S_1, S_2 \text{and} S_3$, where

S_1 = Area of top cross section of cylindrical Gaussian surface

S_2 = Area of bottom cross section of cylindrical Gaussian surface

S_3 = Area of curved part of Gaussian surface

Thus

$$\Phi_e = \int_{S_1} \vec{E}.d\vec{A} + \int_{S_2} \vec{E}.d\vec{A} + \int_{S_3} \vec{E}.d\vec{A}$$

Now

$$\int_{S_1} \vec{E}.d\vec{A} = \int_{S_2} \vec{E}.d\vec{A} = 0 \qquad \therefore \vec{E} \perp d\vec{A} \; for \; S_1 \; and \; S_2$$

$$\text{And} \int_{S_3} \vec{E}.d\vec{A} = \int_{S_3} E\,dA\cos 0° = \int_{S_3} E\,dA \qquad \therefore \vec{E} \parallel d\vec{A} \; for \; S_3$$

Therefore

$$\Phi_e = \int_{S_3} E\,dA = E\int_{S_3} dA \quad ---- \quad (1) \qquad \therefore E \; is \; constant$$

For cylindrical symmetry

$$\int_{S_3} dA = 2\pi rh$$

So, equation (1) becomes:

$$\Phi_e = E(2\pi rh) = 2\pi rh\,E \quad ---- \quad (2)$$

By Gauss's law

$$\Phi_e = \frac{q}{\varepsilon_0} \quad ----- \quad (3)$$

As the line of charge has constant linear charge density λ, therefore:

$$\lambda = \frac{q}{h} \Longrightarrow q = \lambda h$$

So, the equation (3) becomes:

$$\Phi_e = \frac{\lambda\, h}{\varepsilon_0} \quad ----- \quad (4)$$

Comparing Eq. (2) and (4), we get:

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$$2\pi rh\,E = \frac{\lambda\, h}{\varepsilon_0}$$

$$E = \frac{\lambda}{2\pi r\varepsilon_0}$$

If '$\hat{r}$' gives the direction of electric field intensity, then

$$\vec{E} = \frac{\lambda}{2\pi r\varepsilon_0}\hat{r}$$

This expression gives the electric field intensity due to infinite line of charge.

Problem 20. An infinite line of charge produces a field of $4.52 \times 10^4 \frac{N}{C}$ at a distance of 1.96 m. calculate the linear charge density?

Solution: $E = 4.52 \times 10^4 \frac{N}{C}$

$r = 1.96\,m$

$\lambda = ?$

$$As\ E = \frac{\lambda}{2\pi\varepsilon_0 r}$$

$$\Longrightarrow \lambda = 2\pi\varepsilon_0 r.E$$

$$\Longrightarrow \lambda = 2 \times 3.14 \times 8.85 \times 10^{-12} \times 1.96 \times 4.52 \times 10^4 = 4.92 \times 10^{-6}\frac{C}{m}$$

Sample problem 5. A plastic rod whose length is 220 cm and whose radius is 3.6 mm carries a negative charge q of magnitude $3.8 \times 10^{-7} C$ spread uniformly over its surface. What is the electric field near the midpoint of the rod at a point on its surface?

Solution: $L = 220cm = 2.2\,m$

$r = 3.6 \times 10^{-3} m$

$q = -3.8 \times 10^{-7} C$

$E = ?$

As Electric field intensity due to infinite line of charge is:

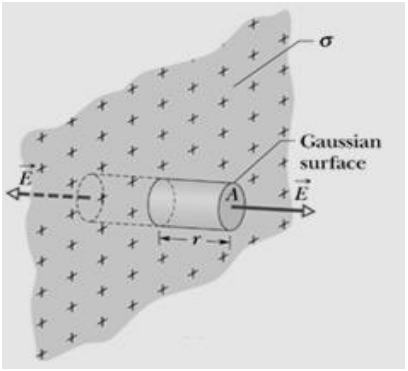
$$E = \frac{1}{2\pi\varepsilon_0}\frac{\lambda}{r} - - - - - (1)$$

$$\lambda = \frac{q}{L} = -\frac{3.8 \times 10^{-7}}{2.2} = -1.73 \times 10^{-7}\frac{C}{m}$$

$$Now E = \frac{1}{2\pi\varepsilon_0}\frac{\lambda}{r} = \frac{1}{2 \times 3.14 \times 8.85 \times 10^{-12}}\frac{(-1.73 \times 10^{-7})}{3.6 \times 10^{-3}}$$

$$= -8.6 \times 10^5 \frac{N}{C}$$

29.5.2 Electric Field at a Point Due to Infinite Sheet of Charge



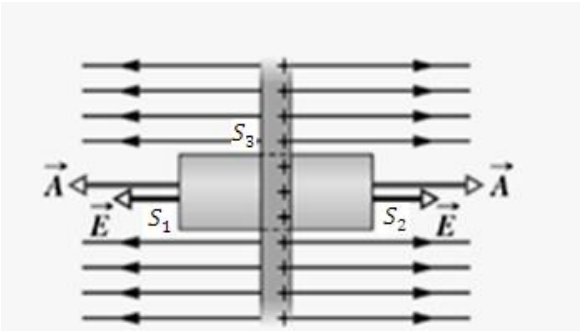
Consider an infinite sheet of charge having constant surface charge density 'σ'. The figure shows a small portion of such sheet.

We want to find electric field intensity at point 'P' which is at the distance 'r' from sheet. For this we consider a cylindrical Gaussian surface as shown in the figure below. The net electric flux passing through the cylinder is given as

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$$\Phi_e = \int_S \vec{E}.d\vec{A}$$



We divide the cylindrical Gaussian surface into three parts i.e., $S_1, S_2 \text{and} S_3$, where

S_1 = Left cross sectional area of cylindrical Gaussian surface

S_2 = Right cross sectional area of cylindrical Gaussian surface

S_3 = Area of curved of cylindrical Gaussian surface

Thus

$$\Phi_e = \int_{S_1} \vec{E}.d\vec{A}_1 + \int_{S_2} \vec{E}.d\vec{A}_2 + \int_{S_3} \vec{E}.d\vec{A}_3 \quad --- \quad (1)$$

$$Now \int_{S_3} \vec{E}.d\vec{A}_3 = 0 \qquad \therefore \vec{E} \perp d\vec{A} \; for \; S_3$$

Therefore, the equation (1) becomes:

$$\Phi_e = \int_{S_1} E.dA_1 + \int_{S_2} E.dA_2$$

In case of surfaces $S_1 \text{and} S_2$, $\vec{E} \parallel d\vec{A}$ are parallel to each other i.e., $\theta = 0^\circ$ and $|dA_1| = |dA_2| = dA$.

$$\Phi_e = \int_{S_1} E\,dA + \int_{S_2} E\,dA$$

$$\Phi_e = E\int_{S_1} dA + E\int_{S_2} dA \; \therefore E \; is \; constant$$

$$\Phi_e = E\,A + E\,A = 2\,E\,A \quad ----(2)$$

According to Gauss's law

$$\Phi_e = \frac{q}{\varepsilon_0} \qquad \therefore \sigma = \frac{q}{A} \Longrightarrow q = \sigma A$$

$$= \frac{\sigma A}{\varepsilon_0} \quad ---- \quad (3)$$

Comparing Eq. (2) and (3), we get

$$2\,E\,A = \frac{\sigma A}{\varepsilon_0}$$

$$\Longrightarrow E = \frac{\sigma}{2\varepsilon_0}$$

If '$\hat{r}$' gives the direction of electric field intensity, then

$$\vec{E} = \frac{\sigma}{2\varepsilon_0}\hat{r}$$

This is the expression of electric field intensity due to infinite sheet of charge.

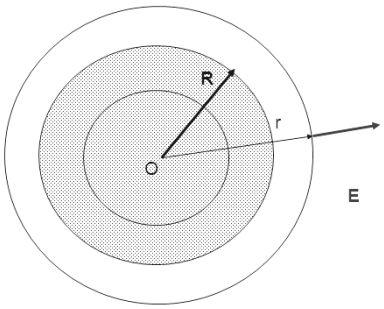
29.5.3 Electric Field due to Spherical Shell of Charge

Question: Show that the uniform spherical shell of charge behaves, for all external points, as if all its charge were concentrated at its center.

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Proof: Consider a thin spherical shell of radius 'R' which have the charge 'q' with constant surface charge density 'σ'. The surface charge density



$$\sigma = \frac{q}{A} \Longrightarrow \sigma = \frac{q}{4\pi r^2} \qquad \therefore A$$
$$= 4\pi r^2 \ (Surface\ area\ of\ sphere)$$
$$\Longrightarrow q = 4\pi r^2 \sigma$$

Consider a point 'P' outside the shell. We want to find out electric field intensity due to this charge distribution. For this we consider a spherical Gaussian surface of radius $r > R$ which passes through point 'P' as shown in the figure below.

According to Gauss's law,

$$\oint \vec{E}.d\vec{A} = \frac{q}{\varepsilon_0}$$

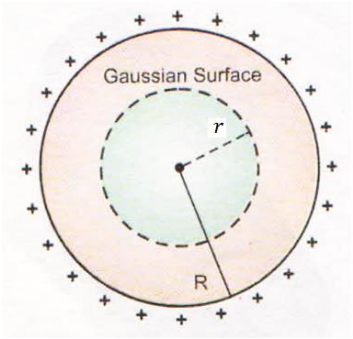
$$\oint E\, dA \cos 0° = \frac{q}{\varepsilon_0} \therefore \vec{E} \parallel d\vec{A}$$

$$E \oint dA = \frac{q}{\varepsilon_0} \therefore E\ is\ constant$$

$$E(4\pi r^2) = \frac{q}{\varepsilon_0}$$

$$E = \frac{1}{4\pi\varepsilon_0}.\frac{q}{r^2}$$

Thus the uniform spherical shell of charge behaves like a point charge for all the points outside the shell.



Question: Show that the uniform spherical shell of charge exerts no electrostatic force on a charged particle placed inside the shell.

Consider a point 'P' inside the shell. We want to fine out electric field intensity 'E' at point 'P' due to this symmetrical charge distribution. For this we consider a spherical Gaussian surface of radius $r < R$ which passes through point 'P' as shown in the figure below.

According to Gauss's law,

$$\oint \vec{E}.d\vec{A} = \frac{q}{\varepsilon_0}$$

Because the Gaussian surface enclose no charge, therefore 'q = 0',

$$\oint \vec{E}.d\vec{A} = 0$$

$$\oint E\, dA \cos 0° = 0 \qquad \therefore \vec{E} \parallel d\vec{A}$$

$$E \oint dA = 0 \qquad \therefore E\ is\ constant$$

As $dA \neq 0$, therefore

$$E = 0$$

So the electric field does not exist inside a uniform shell of charge. So the test charge placed inside the charged shell would experience no force.

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29.5.4 Electric Field due to Spherical Charge

Question.Find out the expression of electric field intensity outside solid sphere of charge.

Ans. Consider a spherical distribution of charge of radius 'R' with the uniform volume charge density 'ρ'. We want to find the electric field at point 'P' at a distance $r > R$ from the center of charged sphere. For we consider a spherical Gaussian surface which passes through point 'P' as shown in the figure below.

According to Gauss's law,

$$\oint \vec{E}.\,d\vec{A} = \frac{q}{\varepsilon_0}$$

$$\oint E\,dA\cos\theta = \frac{q}{\varepsilon_0}$$

As the electric field is radial, so the electric lines of force leave the Gaussian surface normally at all points. Therefore, the electric field intensity $\vec{E}$ and surface area element $d\vec{A}$are in same direction i.e., $\theta = 0°$.

$$\oint E\,dA = \frac{q}{\varepsilon_0}$$

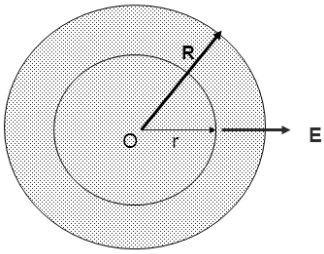
$$E\oint dA = \frac{q}{\varepsilon_0} \therefore E\ is\ constant$$

$$E(4\pi r^2) = \frac{q}{\varepsilon_0}$$

$$E = \frac{1}{4\pi\varepsilon_0}.\frac{q}{r^2}$$

Thus for all points outside the spherical charge distribution, the electric field has the same value as if the charge is concentrated at the center of sphere.

Q. Derive the expression of electric field intensity inside solid sphere of charge.



Ans. Consider a spherical distribution of charge of radius 'R' with the uniform volume charge density 'ρ'. The total charge in this uniform charge distribution is

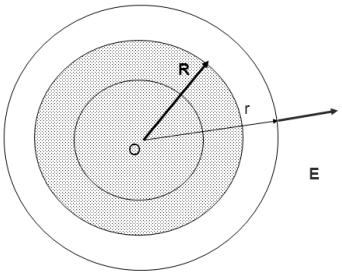
$$\rho = \frac{q}{V} \Longrightarrow q = \rho V$$

$$\Longrightarrow q = \rho\left(\frac{4}{3}\pi R^3\right) \quad ----\ (1)$$

We want to find the electric field at point 'P' at a distance $r < R$ from the center of charged sphere. For this we consider a spherical Gaussian surface of which passes through point Pas shown in the figure below.

Let the Gaussian surface encloses the charge $q' < q$ given by

$$q' = \rho\left(\frac{4}{3}\pi r^3\right) \quad ----\ (2)$$



Dividing eq. (1) and (2), we get

$$\frac{q'}{q} = \frac{\rho(\frac{4}{3}\pi r^3)}{\rho(\frac{4}{3}\pi R^3)}$$

$$\Longrightarrow \frac{q'}{q} = \frac{r^3}{R^3}$$

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$$\Rightarrow q' = \frac{r^3}{R^3}\, q \quad ----- \quad (3)$$

According to Gauss's law

$$\oint \vec{E}.\,d\vec{A} = \frac{q'}{\varepsilon_0}$$

$$\Rightarrow \oint E\, dA \cos 0° = \frac{q'}{\varepsilon_0} \qquad \therefore E \text{ is directed radially outward and } \vec{E} \parallel d\vec{A}$$

$$\Rightarrow E \oint dA = \frac{q'}{\varepsilon_0} \therefore E \text{ is constant}$$

$$\Rightarrow E(4\pi r^2) = \frac{q'}{\varepsilon_0}$$

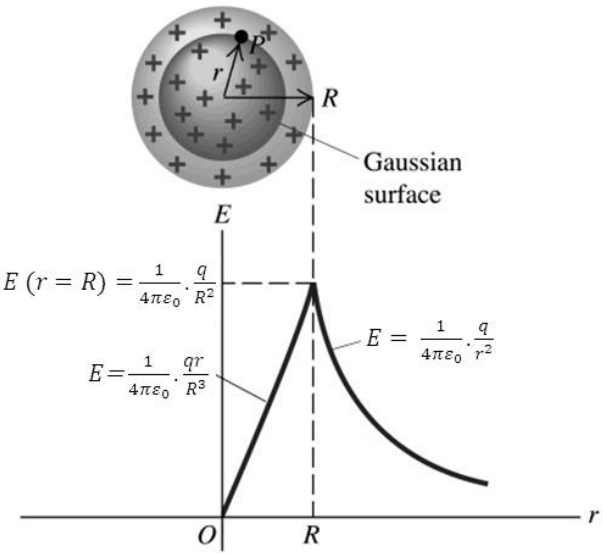
$$\Rightarrow E = \frac{1}{4\pi\varepsilon_0}.\frac{q'}{r^2}$$

Putting the value of q' from eq. (3), we get

$$E = \frac{1}{4\pi\varepsilon_0}.\frac{1}{r^2}.\frac{r^3}{R^3}\, q$$

$$\Rightarrow E = \frac{1}{4\pi\varepsilon_0}.\frac{r}{R^3}\, q$$

This is expression of electric field intensity inside solid sphere of charge.



Special case

To find out the expression of electric field intensity at the surface of solid sphere of charge, put $r = R$:

$$E\,(r = R) = \frac{1}{4\pi\varepsilon_0}\frac{q}{R^2}$$

Variation of Electric Field Intensity as a function of distance for Volume charge distribution

The graphical representation of the dependence of electric field strength on the radial distance 'r' from the center of this charge distribution is shown in the figure:

- Electric field intensity, inside solid sphere of charge, is directly proportional to distance as described by formula:

$$E = \frac{1}{4\pi\varepsilon_0}\frac{r}{R^3}\, q$$

So the graph between E and r is a straight line for the values of r from $0 \longrightarrow R$.

- The electric field intensity is maximum at the surface of sphere of charge:

$$E\,(r = R) = \frac{1}{4\pi\varepsilon_0}\frac{q}{R^2}$$

- The solid sphere of charge behaves as a point charge for all the points outside the solid sphere of charge i.e., electric field intensity is inversely proportional to the square of the distance from center of sphere of charge.

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Problem 21. The drum of a photocopying machine has the length of 42 cm and diameter of 12 cm having surface charge density equal to $2 \times 10^{-6} C/m^2$. What is the total charge on the drum? (b) The manufacturer wants to produce desktop version of machine. This requires reducing the length of drum to 28 cm and diameter of 8 cm. the electric field must remain unchanged. Calculate the charge of new drum.

Solution:

(a) $\sigma = 2 \times 10^{-6} C/m^2$

$h = 42\ cm = 0.42\ m$

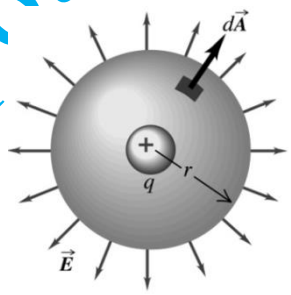
$d = 12\ cm = 0.12\ m$

$r = \frac{d}{2} = \frac{0.12}{2} = 0.06\ m$

q =?

$\sigma = \frac{q}{A}$

$\Rightarrow q = \sigma A = \sigma . 2\pi r h = 2 \times 10^{-6} \times 2 \times 3.14 \times 0.06 \times 0.42 = 322\ nC$



(b)

$\sigma = 2 \times 10^{-6}\ C/m^2$

$h' = 0.28\ m$

$d' = 0.08\ m$

$r' = 0.04m$

As $\sigma = \frac{q'}{A'}$

$\Rightarrow q' = \sigma A' = \sigma . 2\pi r' h' = 2 \times 10^{-6} \times 4 \times 3.14 \times 0.04 \times 0.028 = 141\ nC$

29.6 Deduction of Coulomb's Law from Gauss's Law

Coulomb's law can be deduced from Gauss's law under certain symmetry consideration. Consider positive point charge 'q'. In order to apply the Gauss's law, we assume a spherical Gaussian surface as shown in the figure below.

Considering the integral form of Gauss's law,

$$\oint \vec{E}.d\vec{A} = \frac{q}{\varepsilon_0}$$

Because the both vectors $\vec{E}$ and $d\vec{A}$ are directed radially outward, so

$$\oint E\ dA \cos 0° = \frac{q}{\varepsilon_0} \qquad \therefore E \text{ is directed radially outward and } \vec{E} \parallel d\vec{A}$$

As E is constant for all the points on the spherical Gaussian surface,

$$E \oint dA = \frac{q}{\varepsilon_0}$$

$$E(4\pi r^2) = \frac{q}{\varepsilon_0} \therefore \text{ For spherical symmetry } \oint dA = 4\pi r^2$$

$$E = \frac{1}{4\pi\varepsilon_0} . \frac{q}{r^2}$$

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This equation gives the magnitude of electric field intensity E at any point which is at the distance 'r' from an isolated point charge 'q'.

From the definition of electric field intensity, we know that

$$F = q_0 E$$

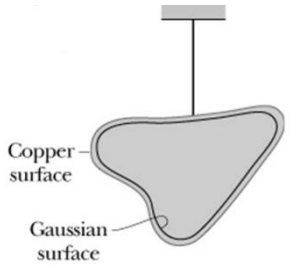
Where q_0 is the point charge placed at a point at which the value of electric field intensity has to be determined. Therefore

$$F = \frac{1}{4\pi\varepsilon_0}.\frac{qq_0}{r^2}$$

This is the mathematical form of Coulomb's law.

ہمیں اس کارِ خیر کو بہتر بنانے کے لئے آپ کی مدد کی ضرورت ہے۔اپنی قیمتی آراء اور نوٹس ای میل کریں یا فیس بک ایڈیس پر میسیج کریں۔

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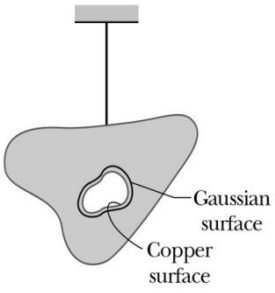


29.7 Prove that "An excess charge added to the isolated conductor moves entirely to its outer surface. None of the excess charge is found within the body of conductor".

Consider an isolated conductor (lump of copper) is hanging from a silk thread and carrying a net positive charge 'q' as shown in the figure below. The Gaussian surface lies inside the actual surface of the conductor.

Under the equilibrium conditions, the electric field inside the conductor must be zero. If it were not so, the field would exert the force on conduction electrons and the internal currents would be setup.

But there is no experimental evidence of such internal currents in isolated conductors. And if the extra charge is added to the surface, it redistribute itself on the surface in such a way that the electric field inside the conductor vanish.



If E is zero everywhere inside the conductor, it must be zero at all the points of Gaussian surface. This means that the flux through Gaussian surface must be zero. Gauss's law ($\Phi_e = \frac{q}{\varepsilon_0}$) then tells that the net charge inside the Gaussian surface must also be zero.

If the added charge is not inside the Gaussian surface, it can only be outside that surface. And the added charge must lie on the actual outer surface of the conductor.

29.8 Prove that the formation of cavity by cutting a natural material from the conductor does not change the distribution charge or pattern of electric field.

Consider an isolated conductor hanging from a silk thread carrying a net positive charge 'q'. Suppose a cavity is produced inside the conductor as shown in the figure below.

We draw a Gaussian surface around the cavity inside the conductor. Because E is zero inside the conductor, there is no flux though the Gaussian surface. So by Gauss's law, there is no net charge inside the Gaussian surface. So the total charge remains on the outer surface of conductor.

29.9 The External Electric Field

The electric field outside a charged isolated conductor can be find out by Gauss's law by considering the cylindrical Gaussian surface as shown in the figure below.

The flux through the interior end cap is zero, because E = 0 for all interior points of conductor. The flux through the cylindrical walls is also zero because the lines of E are parallel to the surface. But the flux through the outer cap will not be zero.

The total flux can then be calculated as:

$$\Phi_e = \int_{\substack{outer\\cap}} \vec{E}.d\vec{A} + \int_{\substack{inner\\cap}} \vec{E}.d\vec{A} + \int_{\substack{cylendrical\\walls}} \vec{E}.d\vec{A} \quad ----(1)$$

$$\int_{\substack{outer\\cap}} \vec{E}.d\vec{A} = \int_{\substack{outer\\cap}} E\,dA\cos 0° = \int_{\substack{outer\\cap}} E\,dA = E\int_{\substack{outer\\cap}} dA = EA \qquad \therefore \vec{E} \parallel d\vec{A} \text{ for outer surface}$$

$$\int_{\substack{inner\\cap}} \vec{E}.d\vec{A} = 0 \qquad \therefore \text{As E is zero inside the body of conductor}$$

$$\int_{\substack{cylendrical\\walls}} \vec{E}.d\vec{A} = 0 \qquad \therefore \text{As} \vec{E} \perp d\vec{A} \text{ for cylindrical walls}$$

Equation (1) will become:

$$\Phi_e = EA + 0 + 0 = EA \quad ----(2)$$

According to Gauss's law

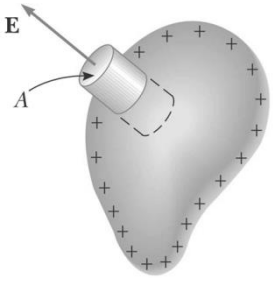
$$\Phi_e = \frac{q}{\varepsilon_0} = \frac{\sigma A}{\varepsilon_0} \quad ----(3) \therefore q = \sigma A$$

Comparing Eq. (2) and (3), we get

$$E = \frac{\sigma}{\varepsilon_0}$$

This showed that the electric field intensity at any point is doubled than the value of E for an infinite sheet of charge.

Sample problem 3. The electric field just above the surface of the charged drum of a photo copying machine has a magnitude E of $2.3 \times 10^5 N/C$. What is the surface charge density on the drum if it is a conductor?



Solution: $E = 2.3 \times 10^5 N/C$

For conductors,

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$$E = \frac{\sigma}{\varepsilon_0} \Longrightarrow \sigma = \varepsilon_0 E = 8.85 \times 10^{-12} \times 2.3 \times 10^5 = 2 \times 10^{-6} \frac{C}{m^2}$$

Sample problem 4. The magnitude of the average electric field normally present in the earth atmosphere just above the surface of the earth is $150\ N/C$ directed downward. What is the total net charge carried by the earth? Assume the earth to be conductor.

Solution: $E = -150 \frac{N}{C}$

q =?

We know

$$E = \frac{\sigma}{\varepsilon_0} \Longrightarrow \sigma = \varepsilon_0 E = 8.85 \times 10^{-12} \times (-150) = -1.33 \times 10^{-9} \frac{C}{m^2}$$

Now

$$\sigma = \frac{q}{A} = \frac{q}{4\pi r^2}$$

$$\Longrightarrow q = \sigma . 4\pi r^2 = -1.33 \times 10^{-9} \times 4 \times 3.14 \times (6.4 \times 10^6)^2 = -6.8 \times 10^5 C$$

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ELECTRIC POTENTIAL

The energy approach in the study of dynamics of the particles can yield not only the simplification but also new insights.

One advantage of energy method is that, although force is a vector, energy is a scalar. In problems involving vector forces and fields, calculations involving sums and integrals are often complicated.

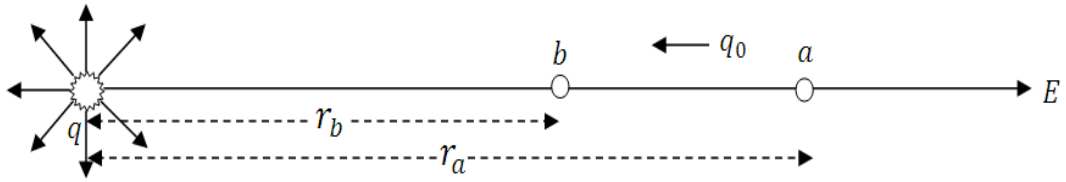
In this chapter, we introduce the energy method to the study of electrostatics.

30.1 Potential Difference

Potential difference 'ΔV' between two point is defined as "the amount of work done 'ΔW' per unit charge 'q_0' in moving it from one point to the other against the electric field and by keeping the system in equilibrium". Mathematically

$$\Delta V = \frac{\Delta W}{q_0}$$

Suppose a unit positive test charge 'q_0' is moved from one point 'a' to the point 'b' in the electric field '$\vec{\mathrm{E}}$' of a large positive charge 'q' as shown in figure below:



The work done in moving 'q_0' from point 'a' to the point 'b' against the electric field '**E**' is

$$W_{a\to b} = \int_a^b \vec{\mathrm{F}}.\overrightarrow{\mathrm{dr}}$$

The electrical force of magnitude '$\vec{\mathrm{F}} = -q_0\vec{\mathrm{E}}$' must have to supplied in order to move 'q_0' against the electric field. Therefore

$$W_{a\to b} = \int_a^b (-q_0\vec{\mathrm{E}}).\overrightarrow{\mathrm{dr}}$$

$$W_{a\to b} = \int_a^b (-q_0\vec{\mathrm{E}}).\overrightarrow{\mathrm{dr}}$$

$$W_{a\to b} = -q_0\int_a^b \vec{\mathrm{E}}.\overrightarrow{\mathrm{dr}}$$

$$\frac{W_{a\to b}}{q_0} = -\int_a^b \vec{\mathrm{E}}.\overrightarrow{\mathrm{dr}}$$

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$$\because \frac{W_{a\to b}}{q_0} = \Delta V = V_b - V_a$$

Therefore, the electrical potential difference between two points in an electrical field will be

$$V_b - V_a = -\int_a^b \vec{E}.\overrightarrow{dr}$$

30.2 Absolute Electrical Potential at a Point

Absolute electric potential at a point is defined as "the amount of work done per unit charge in moving it from infinity to a specific field point against the electric field and by keeping the system in equilibrium".

To find the absolute potential, the reference point is selected at which potential is zero. This point is situated at infinity i.e., out of the electric field. Thus, in equation (1)

$$V_a = V(\infty) = 0$$

$$\Rightarrow V_b - 0 = -\int_\infty^b \vec{E}.\overrightarrow{dr}$$

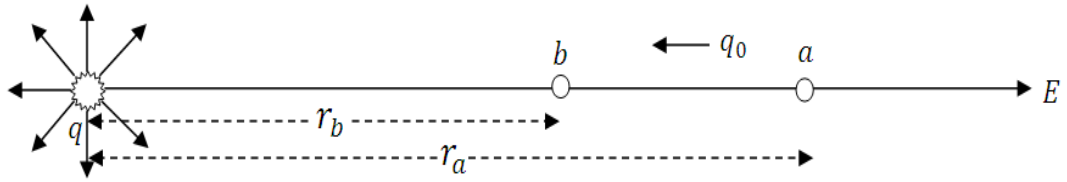
If the distance from the point 'b' to the charge 'q' is 'r', then in general

$$V(r) = -\int_\infty^r \vec{E}.\overrightarrow{dr}$$

30.3 Expression for the Electric Potential Difference due to a Point Charge

The potential difference between two points is the amount of work done per unit charge 'q_0' in moving it from one point to the other against the electric field 'E'. Mathematically, it is described as:

$$V_b - V_a = -\int_{r_a}^{r_b} \vec{E}.\overrightarrow{dr}$$



But the electric field intensity due to point charge: $\vec{E} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\hat{r}$

So, $V_b - V_a = -\int_{r_a}^{r_b} \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\hat{r}.\overrightarrow{dr}$

$$\Rightarrow V_b - V_a = -\frac{q}{4\pi\varepsilon_0}\int_{r_a}^{r_b} \frac{\hat{r}.\overrightarrow{dr}}{r^2}$$

$$\therefore \vec{E}\ is\ directed\ radially\ outward, therefore\ \hat{r}\ //\ \overrightarrow{dr}$$

$$\therefore\ \hat{r}\ .\ \overrightarrow{dr} = |\hat{r}|\ |\overrightarrow{dr}| \cos 0° = (1)(dr)(1) = dr$$

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$$\Longrightarrow V_b - V_a = -\frac{q}{4\pi\varepsilon_0}\int_{r_a}^{r_b}\frac{\mathrm{dr}}{r^2}$$

$$\Longrightarrow V_b - V_a = -\frac{q}{4\pi\varepsilon_0}\left|-\frac{1}{r}\right|_{r_a}^{r_b} = \frac{q}{4\pi\varepsilon_0}\left|\frac{1}{r}\right|_{r_a}^{r_b}$$

$$\Longrightarrow V_b - V_a = \frac{q}{4\pi\varepsilon_0}\left[\frac{1}{r_b} - \frac{1}{r_a}\right]$$

This is the expression for the potential difference between two points 'a' and 'b'.

30.4 Expression for the Absolute Electric Potential due to a Point Charge

The electric potential at any point is the amount of work done per unit charge in moving a unit positive charge (test charge) from infinity to that point, against the electric field.

If the point 'a' is at infinity then

$$V_a = V(\infty) = 0, \qquad r_a = \infty$$

Putting this value in equation in the expression of electric potential difference due to point charge, we get:

$$\Longrightarrow V_b - 0 = \frac{q}{4\pi\varepsilon_0}\left[\frac{1}{r_b} - \frac{1}{\infty}\right] \therefore \frac{1}{\infty} = 0$$

$$\Longrightarrow V_b = \frac{q}{4\pi\varepsilon_0}\left[\frac{1}{r_b}\right]$$

In general, the electric potential at point due to a point charge 'q' is

$$V = \frac{1}{4\pi\varepsilon_0}\frac{q}{r}$$

Sample Problem 1. Two protons in the nucleus of U^{238} are 6 fm apart. What potential energy associated with the electric force that acts between them?

Solution:

$$r = 6\ fm = 6\times10^{-15}m$$

ΔU =?

$$q_1 = q_2 = 1e = 1.6\times10^{-19}C$$

As $$\Delta V = k\frac{q}{r} = 9\times10^9\times\frac{1.6\times10^{-19}}{6\times10^{-15}} = 2.34\times10^5$$

Now

$$\Delta U = q_1.\Delta V = 1.6\times10^{-19}\times2.34\times10^5 = 3.744\times10^{-14}\ J$$

$$\Longrightarrow \Delta U = \frac{3.744\times10^{-14}}{1.6\times10^{-19}}eV = 2.34\times10^5\ eV$$

Sample problem 5. What must be the magnitude of an isolated positive point charge for the electric potential at 15 cm from the charge to be 120 V.

Solution:

$$r = 15\ cm = 15\times10^{-2}m$$

$$V = 120V$$

$$q = ?$$

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As $V = k\frac{q}{r}$

$$\Rightarrow q = \frac{V.r}{k} = \frac{120 \times 15 \times 10^{-2}}{9 \times 10^{9}} = 1.995 \times 10^{-9} C$$

Sample problem 5. What must be the magnitude of an isolated positive point charge for the electric potential at 15 cm from the charge to be 120 V.

Solution:

$r = 15\ cm = 15 \times 10^{-2} m$

$V = 120V$

$q = ?$

As $V = k\frac{q}{r} \Rightarrow q = \frac{V.r}{k} = \frac{120 \times 15 \times 10^{-2}}{9 \times 10^{9}} = 1.995 \times 10^{-9} C$

Sample problem 6. What is the electric potential at the surface of the gold nucleus. The radius of the gold nucleus is $7 \times 10^{-15} m$ and atomic number is 79.

Solution:

$r = 7 \times 10^{-15} m$

$Z = 79$

$q = 79e = 79 \times 1.6 \times 10^{-19} C$

$V = ?$

As $V = k\frac{q}{r} = 9 \times 10^{9} \times \frac{79 \times 1.6 \times 10^{-19}}{7 \times 10^{-15}} = 1.6 \times 10^{7}\ V$

Sample Problem 3. An alpha particle $q = 2e$ in a nuclear accelerator move from one terminal at a potential of $V_a = 6.5 \times 10^6 V$ to $V_b = 0$. What is the corresponding change in potential energy. (b) Assuming the terminals and their charges do not move and that no external force act on this system, what is the change in kinetic energy of the particle?

Solution:

$V_a = 6.5 \times 10^6 V$

$V_b = 0$

$\Delta U = ?$

$q = 2e$

(a) $\Delta U = q.\Delta V = q.(V_b - V_a) = 2e(0 - 6.5 \times 10^6) = -13 \times 10^6\ eV$

(b) $T.E = P.E + K.E$

$\Rightarrow 0 = P.E + K.E$

$\Rightarrow K.E = -P.E = -(-13 \times 10^6\ eV) = 13 \times 10^6\ eV$

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Problem 6. Two parallel flat conducting surfaces spacing $d = 1\ cm$ have a potential difference of 10.3 kV. An electron is projected from one plate directly towards second. What is the initial velocity of electron if it will come to rest just at the surface of the 2nd plate?

Solution: $q = 1.6 \times 10^{-19} C$

$d = 1\ cm = 0.01\ m$

$\Delta V = 10.3 \times 10^3\ V$

$velocity\ v = ?$

$\Delta U = q.\Delta V = 1.6 \times 10^{-19} \times 10.3 \times 10^3$

$Now \quad Total\ Loss\ of\ P.E = Gain\ in\ K.E$

$$\Rightarrow \Delta U = \frac{1}{2}mv^2$$

$$\Rightarrow v = \sqrt{\frac{2.\Delta U}{m}} = \sqrt{\frac{2 \times 1.6 \times 10^{-19} \times 10.3 \times 10^3}{9.1 \times 10^{-31}}} = _____ m/s$$

Problem 7. In a typical lighting flash, the potential difference between discharge points is about $1 \times 10^9\ V$ and the quantity of charge transfer is about 30 C. (a) How much energy is released? (b) If all energy released could be used to accelerate a 1200 kg automobile from rest, what would its final speed? (c) If it would be used to melt ice, how much ice it would melt at $0^0 C$?

Solution: (a) $\Delta V = 10^9\ V$

$q = 30C$

$\Delta U = q.\Delta V = 30 \times 10^9 J$

(b) Now

$Total\ Loss\ of\ P.E = Gain\ in\ K.E$

$$\Delta U = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{\frac{2.\Delta U}{m}} = \sqrt{\frac{2 \times 30 \times 10^9}{1200}} = _____ m/s$$

(c) $\Delta U = mH_f \Rightarrow m = \frac{\Delta U}{H_f} = \frac{30 \times 10^9}{336000} = _____ kg$

Problem 56. Find (a) charge and (b) charge density on the surface of conducting sphere of radius 15.2 cm whose potential is 215 V?

Solution: $q = ?$

$\sigma = ?$

$r = 0.15\ m$

$V = 215\ V$

(a) As $V = k.\frac{q}{r}$

$$\Rightarrow q = \frac{V.r}{k} = \frac{215 \times 0.15}{9 \times 10^9} C = 3.5 \times 10^{-9} C$$

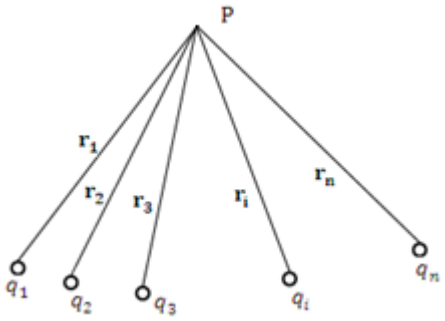
(b) $\sigma = \frac{q}{A} = \frac{q}{4\pi R^2} = \frac{3.5 \times 10^{-9}}{4 \times 3.14 \times (0.15)^2} = 12.3 \times\times 10^{-9} C$

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30.5 Electric Potential Due to a Collection of Point Charges

Let "$q_1, q_2, q_3, \ldots\ldots\ldots, q_n$" are the 'n' point charges which are at distances "$r_1, r_2, r_3, \ldots\ldots\ldots, r_n$" from a point '$P$', as shown in the figure:



Now if $V_1, V_2, V_3, \ldots\ldots\ldots, V_n$ be the electric potential at a field point 'P' due to the point charges $q_1, q_2, q_3, \ldots\ldots\ldots, q_n$ respectively. Then, the total electric potential at a field point 'P' due to assembly of 'n' point charges will be;

$$V = V_1 + V_2 + V_3 + \ldots\ldots\ldots + V_n \quad ----- \quad (1)$$

where

V_1 = Electric Potential at a Field Point 'P' due to Point Charge 'q_1' = $\frac{1}{4\pi\varepsilon_0}\frac{q_1}{r_1}$

V_2 = Electric Potential at a Field Point 'P' due to Point Charge 'q_2' = $\frac{1}{4\pi\varepsilon_0}\frac{q_2}{r_2}$

V_3 = Electric Potential at a Field Point 'P' due to Point Charge 'q_3' = $\frac{1}{4\pi\varepsilon_0}\frac{q_3}{r_3}$

$\vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots$

$\vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots$

V_n = Electric Potential at a Field Point 'P' due to Point Charge 'q_n' = $\frac{1}{4\pi\varepsilon_0}\frac{q_n}{r_n}$

Putting values in equation (1), we get

$$V = \frac{1}{4\pi\varepsilon_0}\frac{q_1}{r_1} + \frac{1}{4\pi\varepsilon_0}\frac{q_2}{r_2} + \frac{1}{4\pi\varepsilon_0}\frac{q_3}{r_3} + \ldots\ldots\ldots + \frac{1}{4\pi\varepsilon_0}\frac{q_n}{r_n}$$

$$= \frac{1}{4\pi\varepsilon_0}\left(\frac{q_1}{r_1} + \frac{q_2}{r_2} + \frac{q_3}{r_3} + \ldots\ldots\ldots + \frac{q_n}{r_n}\right)$$

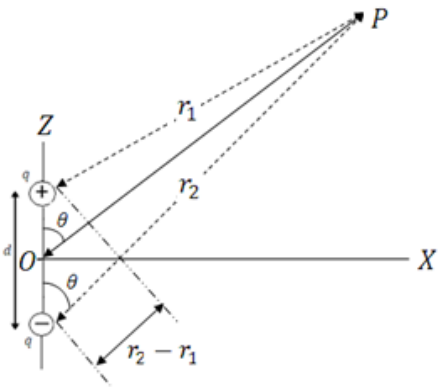
$$= \frac{1}{4\pi\varepsilon_0}\sum_{i=1}^{n}\frac{q_i}{r_i}$$

This is the net potential at any point 'P' due to collection of 'n' point charges.

30.6 Electric Potential due to a Dipole

Consider two point charges $+q$ and $-q$ of equal magnitude lying distance 'd' apart as shown in the figure.

The electric potential at point 'P' is the sum of potential due to '$+q$' and '$-q$'. If V_+ and V_- are the electric potential due to the charges $+q$ and $-q$. Therefore,



$$V = V_+ + V_-$$

$$\Rightarrow V = \frac{1}{4\pi\varepsilon_0}\frac{+q}{r_1} + \frac{1}{4\pi\varepsilon_0}\frac{-q}{r_2}$$

$$\Rightarrow V = \frac{q}{4\pi\varepsilon_0}\left[\frac{1}{r_1} - \frac{1}{r_2}\right]$$

$$\Rightarrow V = \frac{q}{4\pi\varepsilon_0}\left[\frac{r_2 - r_1}{r_1 r_2}\right]$$

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Normally for a dipole, $r \gg d$

Therefore, $r_1 r_2 \approx r^2$

And from figure, $r_2 - r_1 = d\cos\theta$

$$\Rightarrow V = \frac{q}{4\pi\varepsilon_0}\left[\frac{d\cos\theta}{r^2}\right]$$

$$\Rightarrow V = \frac{1}{4\pi\varepsilon_0}\frac{qd\cos\theta}{r^2}$$

$$\Rightarrow V = \frac{1}{4\pi\varepsilon_0}\frac{p\cos\theta}{r^2} \qquad \therefore p = qd = Dipole\ Moment$$

$$\Rightarrow V = \frac{1}{4\pi\varepsilon_0}\frac{p\, r\cos\theta}{r^3} \quad \therefore \text{Multiplying and dividing by } r$$

$$\Rightarrow V = \frac{1}{4\pi\varepsilon_0}\frac{\vec{p}.\vec{r}}{r^3}$$

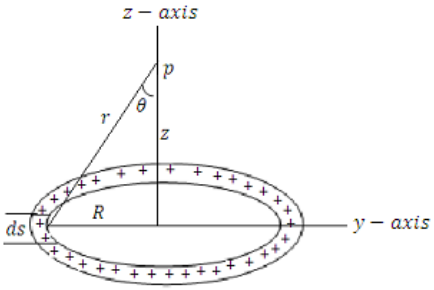
This is the expression of electric potential at any point 'P' due to a dipole.

30.7 Electric Potential of Continuous Charge Distribution

30.7.1 Electric Potential due to a Ring of Charge

Consider a ring of positive charge of radius 'R' as shown in the figure. We want to find out the electric potential at point 'P' which is at the distance 'z' from the plane of ring.

As the charge is distributed uniformly over it, so it has constant linear charge density λ. For an infinitesimal length element 'ds' of ring,



$$\lambda = \frac{dq}{ds}$$

$$dq = \lambda\, ds$$

The electric potential due to the charge dq at point 'P' is given by:

$$dV = \frac{1}{4\pi\varepsilon_0}\frac{dq}{r} = \frac{1}{4\pi\varepsilon_0}\frac{\lambda\, ds}{r}$$

$$\Rightarrow dV = \frac{1}{4\pi\varepsilon_0}\frac{\lambda\, ds}{\sqrt{z^2 + R^2}} \qquad \therefore From\ Figure, r = \sqrt{z^2 + R^2}$$

The total value of electric potential is

$$V = \int dV = \frac{1}{4\pi\varepsilon_0}\int \frac{\lambda\, ds}{\sqrt{z^2 + R^2}}$$

$$\Rightarrow V = \frac{1}{4\pi\varepsilon_0}\frac{1}{\sqrt{z^2 + R^2}}\int \lambda\, ds \qquad \therefore \int \lambda\, ds = q$$

Therefore,

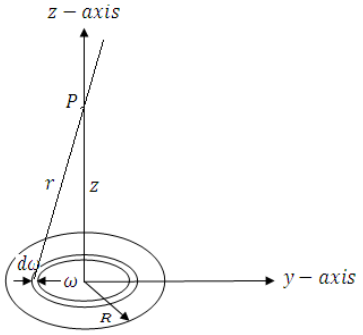
$$V = \frac{1}{4\pi\varepsilon_0}\frac{q}{\sqrt{z^2 + R^2}}$$

This is the expression of electric potential at any point 'P' due to a ring of charge.

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30.7.2 Electric Potential due to a Disk of Charge



Consider a circular disk of uniform surface charge density as shown in the figure below. We want to find out the electric potential at point 'P' which is at the distance 'z' from the plane of disk.

Consider a small element of the disk in the ring shape of radius 'ω' and width '$d\omega$'. If 'dq' is the charge on this element of ring, then

$$\sigma = \frac{dq}{dA} \qquad \therefore where\ dA\ is\ the\ is\ the\ area\ of\ length\ element$$

$$\Longrightarrow dq = \sigma\, dA$$

$$\Longrightarrow dq = \sigma\,(2\pi\omega\, d\omega) \qquad \therefore dA = 2\pi\omega\, d\omega$$

We know the electric potential due to the ring of charge which has the radius ω is given by;

$$dV = \frac{1}{4\pi\varepsilon_0}\frac{dq}{r}$$

$$\Longrightarrow dV = \frac{1}{4\pi\varepsilon_0}\frac{\sigma\,(2\pi\omega\, d\omega)}{z^2 + \omega^2}$$

The electrical potential at point 'P' due to whole disk is:

$$V = \frac{1}{4\pi\varepsilon_0}\int_{\omega=0}^{\omega=R}\frac{\sigma\,(2\pi\omega\, d\omega)}{z^2 + \omega^2}$$

$$\Longrightarrow V = \frac{\sigma}{4\varepsilon_0}\int_{\omega=0}^{\omega=R}(z^2 + \omega^2)^{-\frac{1}{2}}(2\omega)\, d\omega$$

$$\Longrightarrow V = \frac{\sigma}{4\varepsilon_0}\left|\frac{(z^2 + \omega^2)^{\frac{-1}{2}+1}}{\frac{1}{2}}\right|_0^R$$

$$\Longrightarrow V = \frac{\sigma}{2\varepsilon_0}\left[(z^2 + R^2)^{\frac{1}{2}} - (z^2)^{\frac{1}{2}}\right]$$

$$\Longrightarrow V = \frac{\sigma}{2\varepsilon_0}\left[\sqrt{z^2 + R^2} - z\right]$$

This is the expression of electric potential at point 'P' due to disk of charge.

Special Case:

If $z \gg R$, then $\sqrt{z^2 + R^2} = (z^2 + R^2)^{\frac{1}{2}} = z\left[1 + \frac{R^2}{z^2}\right]^{\frac{1}{2}} = z\left[1 + \frac{1}{2}\frac{R^2}{z^2} + \ldots\ldots\ldots\right]$

$$\Longrightarrow \sqrt{z^2 + R^2} = \left[z + \frac{R^2}{2z}\right] \qquad \therefore Neglecting\ Higher\ Terms$$

$$So, V = \frac{\sigma}{2\varepsilon_0}\left[z + \frac{R^2}{2z} - z\right] = \frac{\sigma\, R^2}{4\varepsilon_0\, z} = \frac{\sigma\,\pi R^2}{4\pi\varepsilon_0\, z} = \frac{\sigma\,(\pi R^2)}{4\pi\varepsilon_0\, z} \qquad \therefore \sigma\,(\pi R^2) = q\ (total\ charge\ on\ disk)$$

$$\Longrightarrow V = \frac{1}{4\pi\varepsilon_0}\frac{q}{z}$$

Thus, the disk of charge behave like a point charge for the case $z \gg R$.

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30.8 Equipotential Surfaces

If all the points on a surface have the same value of electric potential, then, it is known as equipotential surface. The examples of some equipotential surfaces are given below

- In case of uniform electric field, the equipotential surfaces are the planes as shown by the dashed line in fig. (a). The potential at points B_1, B_2 and B_3 is same. So, no work will be done in moving a test charge from B_1 to B_2 or B_3.
- The concentric spheres about a point charge '$+q$' are the equipotential surfaces as shown in the fig. (b) by dashed lines.
- The Equipotential surfaces of a dipole are shown by the dashed lines in fig. (c)

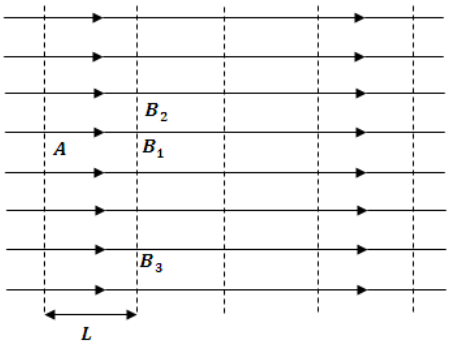


Fig. (a)

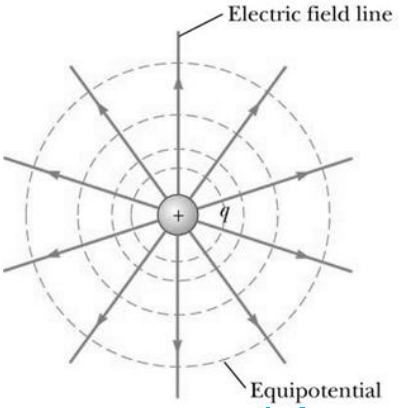


Fig. (b)

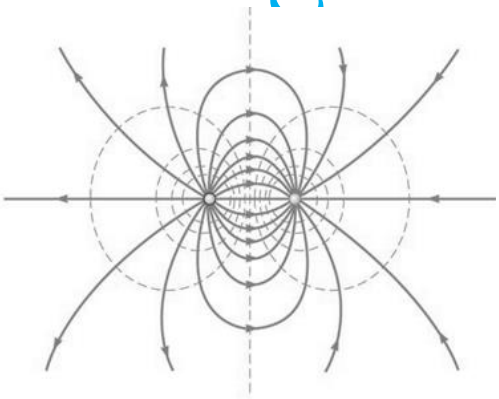


Fig. (c)

30.9 Calculating Electric Field from the Electric Potential

Consider a test charge 'q_0' is displaced in an electric field form point 'P' to point 'Q'. If the potential difference between the points 'P' and 'Q' is dV, then the work done 'dW' by the electric field is

$$dW = -q_0 dV \quad ----- \quad (1)$$

If the test charge is placed in an electric field, then the electrical force on the test charge is

$$\vec{F} = q_0 \vec{E}$$

So, the corresponding work done in diplacing it through displacemnet dx:

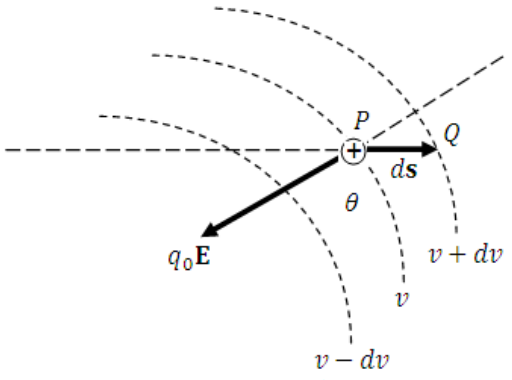
$$dW = \vec{F}.\overrightarrow{ds}$$

$$= q_0 \vec{E}.\overrightarrow{ds} \quad ---- \quad (2) \qquad \therefore \vec{F} = q_0 \vec{E}$$

Comparing eq. (1) and (2)

$$q_0 \vec{E}.\overrightarrow{ds} = -q_0 dV$$

$$q_0 E\, ds \cos\theta = -q_0 dV$$



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$$E\cos\theta = -\frac{dV}{ds}$$

Consider '$E\cos\theta = E_s$' which is the component of '**E**' along '$d\mathbf{s}$'

$$E_s = -\frac{dV}{ds}$$

There is only one direction in which the rate of change of electric potential w r t position is maximum, which is direction of electric field. Thus

$$E = -\left(\frac{dV}{ds}\right)_{max}$$

The maximum value of $\frac{dV}{ds}$ at a given point is called the 'potential gradient. Therefore,

$$E = -\vec{\nabla}V$$

$$\Rightarrow E = -\text{grad } V$$

Hence proved that the electric field is negative gradient of electric potential.

Question: Find the expression of electric field from electric potential due to a point charge.

Solution: the electric potential due to a point charge is

$$V = \frac{1}{4\pi\varepsilon_0}\frac{q}{r}$$

$$V = \frac{1}{4\pi\varepsilon_0}\frac{q}{\sqrt{x^2+y^2+z^2}} \quad ---- \quad (1) \qquad \therefore r = \sqrt{x^2+y^2+z^2}$$

Now $\text{grad}V = \left(\frac{\partial}{\partial x}\hat{\imath} + \frac{\partial}{\partial y}\hat{\jmath} + \frac{\partial}{\partial z}\hat{k}\right)V$

$$= \left(\frac{\partial V}{\partial x}\hat{\imath} + \frac{\partial V}{\partial y}\hat{\jmath} + \frac{\partial V}{\partial z}\hat{k}\right)$$

Now consider $\frac{\partial V}{\partial x} = \frac{\partial}{\partial x}\left[\frac{1}{4\pi\varepsilon_0}\frac{q}{\sqrt{x^2+y^2+z^2}}\right]$

$$= \frac{q}{4\pi\varepsilon_0}\frac{-\frac{1}{2}(2x)}{(x^2+y^2+z^2)^{3/2}} = -\frac{q}{4\pi\varepsilon_0}\frac{x}{(x^2+y^2+z^2)^{3/2}} = -\frac{q}{4\pi\varepsilon_0}\frac{x}{r^3}$$

Similarly

$$\frac{\partial V}{\partial y} = -\frac{q}{4\pi\varepsilon_0}\frac{y}{(x^2+y^2+z^2)^{3/2}} = -\frac{q}{4\pi\varepsilon_0}\frac{y}{r^3}$$

$$\frac{\partial V}{\partial z} = -\frac{q}{4\pi\varepsilon_0}\frac{z}{(x^2+y^2+z^2)^{3/2}} = -\frac{q}{4\pi\varepsilon_0}\frac{z}{r^3}$$

Putting these values in eq. (3)

$$\text{grad}V = -\frac{q}{4\pi\varepsilon_0}\left[\frac{x\hat{\imath}+y\hat{\jmath}+z\hat{k}}{r^3}\right]$$

$$\Rightarrow \text{grad}V = -\frac{q}{4\pi\varepsilon_0}\left[\frac{\vec{\mathrm{r}}}{r^3}\right] \qquad \therefore \vec{\mathrm{r}} = x\hat{\imath}+y\hat{\jmath}+z\hat{k}$$

$$\Rightarrow \text{grad}V = -\frac{q}{4\pi\varepsilon_0}\left[\frac{r\,\hat{r}}{r^3}\right] \qquad \therefore \vec{\mathrm{r}} = r\,\hat{r}$$

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$$\Longrightarrow \text{grad}V = -\frac{q}{4\pi\varepsilon_0}\left[\frac{\hat{r}}{r^2}\right]$$

$$\Longrightarrow -\text{grad}V = \frac{q}{4\pi\varepsilon_0}\left[\frac{\hat{r}}{r^2}\right]$$

$$\vec{\text{E}} = \frac{q}{4\pi\varepsilon_0}\left[\frac{\hat{r}}{r^2}\right] \qquad \therefore \vec{\text{E}} = -\text{grad}V$$

which is the expression of electric field intensity due to a point charge.

Problem. Using $V = \frac{\sigma}{2\varepsilon_0}\left(\sqrt{R^2+Z^2}-Z\right)$, find the values of electric field of uniformly charged disk.

Solution:

From statement

$$V = \frac{\sigma}{2\varepsilon_0}\left(\sqrt{R^2+Z^2}-Z\right)$$

From symmetry, E must lie along axis of the disk i.e., along z-axis.

$$E_z = -\frac{\partial V}{\partial Z} = -\frac{\partial}{\partial Z}\left[\frac{\sigma}{2\varepsilon_0}\left(\sqrt{R^2+Z^2}-Z\right)\right] = -\frac{\sigma}{2\varepsilon_0}\frac{\partial}{\partial Z}\left(\sqrt{R^2+Z^2}-Z\right)$$

$$\Longrightarrow E_z = -\frac{\sigma}{2\varepsilon_0}\left(\frac{1}{2}(R^2+Z^2)^{-1/2}.2Z-1\right)$$

$$\Longrightarrow E_z = -\frac{\sigma}{2\varepsilon_0}\left(\frac{Z}{\sqrt{R^2+Z^2}}-1\right) = \frac{\sigma}{2\varepsilon_0}\left(1-\frac{Z}{\sqrt{R^2+Z^2}}\right)$$

Problem 54. If the earth had a net charge equivalent to 1 $\frac{e}{m^2}$ of the surface area. (a) What would be the earth potential? (b) What would be the electric field earth just outside?

Solution:

$$\sigma = 1\frac{e}{m^2} = 1.6\times10^{-19}\frac{C}{m^2}$$

Radius of the earth $R = 6.4\times10^6 m$

$V = ?$

$E = ?$

As

$$\sigma = \frac{q}{A} \Longrightarrow q = \sigma A = \sigma.4\pi R^2 = 1.6\times10^{-19}\times4\times3.14\times(6.4\times10^6)^2$$

Now

$$V = k.\frac{q}{r} = 9\times10^9\times\frac{1.6\times10^{-19}\times4\times3.14\times(6.4\times10^6)^2}{6.4\times10^6} = 1201\times10^{-4}\,V$$

And $E = \frac{V}{d} = \frac{1201\times10^{-4}}{6.4\times10^6} = 1.80\times10^{-8}\frac{N}{C}$

ہمیں اس کار خیر کو بہتر بنانے کے لئے آپ کی مدد کی ضرورت ہے۔اپنی قیمتی آراءاور نوٹس ای میل کریں یا فیس بک ایڈیس پر میسیج کریں۔

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30.10 Electric Potential and Electric Field Inside and Outside an Isolated Conductor

Consider an isolated conductor in the form of a spherical shell having uniform surface charge density.

Electric Field Inside and Outside an Isolated Conductor

- If the excess of charge is placed on an isolated conductor, then it moves entirely on to the outer surface of the conductor. In equilibrium, none of the charge is inside the body of conductor or on any interior surface (even the conductor has internal cavities). Thus the electric field at any point inside sphere is zero.
- The electric field has maximum on the surface of the spherical shell.
- The electric field decreases as we move away from the sphere because

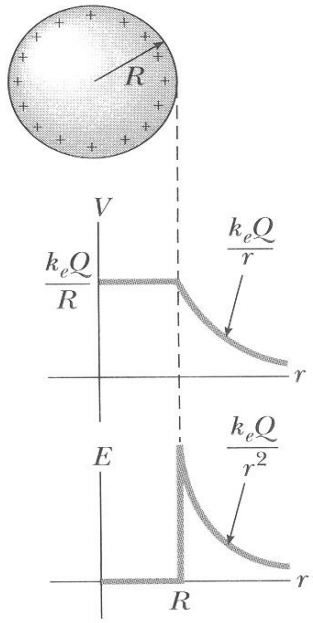
$$E \propto \frac{1}{r^2}$$

Electric Potential Inside and Outside an Isolated Conductor

- As the excess of charge placed on an isolated conductor distributes itself on the surface so that all points of conductor come to the same potential. So the electric potential remains constant at every point inside the sphere. Also, if a test charge is pushed inside the sphere through the hole, then it experiences no force and no work will be done.
- The value of electric potential is maximum at the surface of sphere.
- The electric potential decreases as we move away from the sphere, because

$$V \propto \frac{1}{r}$$

The variation of electric potential and electric field with respect to 'r' is shown in the figure:



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CAPACITORS AND DIELECTRICS

A capacitor is a device that stores energy in an electrostatic field. A flash bulb, for example, requires a short burst of electric energy that exceeds what a battery can generally provide. We can draw energy relatively slowly (over several seconds) from battery into the capacitor which releases the energy rapidly (within milliseconds) through the bulb. Much larger capacitors are used to provide intense laser pulses in attempts to induce thermonuclear fusion in tiny pellets of hydrogen. In this case the power level is about 10^{14}*, but it last for only about* $10^{-9}s$.

Capacitors can also be used to produce electric fields, such as the parallel plate device that deflects beam of charged particles.

In this chapter, we consider the electrostatic field and stored energy of capacitor.

Capacitors have other important functions in electronic circuits, especially for time varying voltages and currents. For transmitting and receiving radio and TV signals, capacitors are fundamental components of electromagnetic oscillators.

31.1 CAPACITOR

Capacitor is a device winch is used to store charge. A simple capacitor consists of two conductors which are separated a small distance. There may be vacuum or some dielectric medium between the conductors of a capacitor.

When the plates of a capacitor are connected with the terminals of the battery of emf V, then the charge q is stored in the capacitor. This charge stored is directly proportional to the potential difference applied between the plates.

$$q \propto V$$

$$q = CV$$

Here C is constant of proportionality, called the capacitance of a capacitor. The capacitance of a capacitor is its ability to store electrical charge. The SI unit of capacitance is farad which can be defined as "If one coulomb of charge given to the plates to produce a potential difference of one volt, then capacitance of the capacitor is one farad".

Problem 1. An electrometer is a device used to measure static charge. Unknown charge is placed on the plates of a capacitor and the potential difference is measured. What minimum charge can be measured by an electrometer with a capacitance of 50 pF and a voltage sensitivity of 0.15 V?

Solution:

$$C = 50\,pF = 50 \times 10^{-12} F$$

$$V = 0.15\,V$$

q =?

$$q = CV$$

$$= 50 \times 10^{-12} \times 0.15$$

$$= 7.5 \times 10^{-12}\,C$$

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Sample problem 1.A storage capacitor on a random access memory (VRAM) chip has a capacitance of 55 fF. If it is charged to 5.3 V, how many excess electrons are there on its negative plate?

Solution: $C = 55fF = 55 \times 10^{-15}F$,

$V = 5.3\,V$

$q = ?$

For case of a capacitor,

$$q = CV = 55 \times 10^{-15} \times 5.3 = 291 \times 10^{-15}C$$

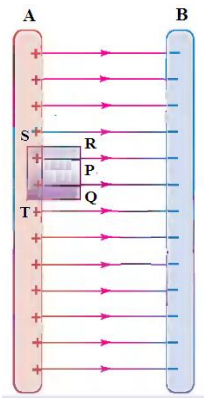
Number of electrons $n = ?$

As $q = ne$

$$\Rightarrow n = \frac{q}{e} = \frac{291 \times 10^{-15}C}{1.6 \times 10^{-19}C} = 181.8 \times 10^4 = 1.818 \times 10^6\ electrons$$

31.1.1 Electric Field between the Plates of Parallel Plate Capacitor

Let the two plates of a capacitor having uniform surface charge density σ. The plates are separated by a distance d. Suppose that the length of plate is very large as compared to the distance between the plates. So, E inside the plates of a capacitor is uniform. We want to find out the expression of electric field between the plates of capacitor. For this we consider a box shaped Gaussian surface as shown in the figure. Then by Gauss's law



$$\oint_{QRST} \vec{E}.d\vec{A} = \frac{q}{\varepsilon_0}$$

Where A area of the side of Gaussian surface through which the flux is passing, while dA is its infinitesimal element.

$$\Rightarrow \int_{SR} \vec{E}.d\vec{A} + \int_{RQ} \vec{E}.d\vec{A} + \int_{QT} \vec{E}.d\vec{A} + \int_{TS} \vec{E}.d\vec{A} = \frac{q}{\varepsilon_0} \quad ----\ (1)$$

As electric field is zero inside conducting plate A, therefore $\int_{TS} \vec{E}.d\vec{A} = 0$. Equation (1) becomes:

$$\Rightarrow \int_{SR} E\,dA\cos 90° + \int_{RQ} E\,dA\cos 0° + \int_{QT} E\,dA\cos 90° + 0 = \frac{q}{\varepsilon_0}$$

$$\Rightarrow 0 + \int_{RQ} E\,dA + 0 + 0 = \frac{q}{\varepsilon_0}$$

$$\Rightarrow EA = \frac{q}{\varepsilon_0}$$

$$\Rightarrow E = \frac{q}{A\varepsilon_0} = \frac{\sigma}{\varepsilon_0} \qquad \therefore \sigma = \frac{q}{A}$$

The equation is expression of electric field intensity inside the plates of capacitor.

31.1.2 Potential Difference between plates of Parallel Plate Capacitor

The potential difference between the plates of charged capacitor is

$$V_f - V_i = -\int_i^f \vec{E}.\overrightarrow{ds}$$

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where the integral is taken over the path which starts from the positive plate and terminate on negative plate. For the present case, E and ds are in same direction

$$V_f - V_i = -\int_i^f E\, ds\, cos0^\circ = -E\int_i^f ds$$

$$V_f - V_i = -E\, d \quad ----- \quad (1)$$

If the initial plate is positively charged and final plate is negatively charge, then

$$V_f - V_i = -V$$

Putting value in equation (1), we get:

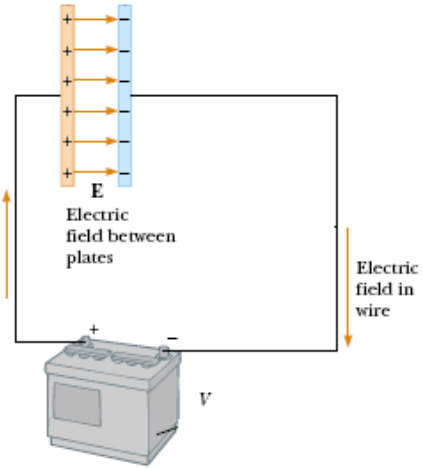
$$V = E\, d$$

31.1.3 Capacitance of Parallel Plate Capacitor

Consider a parallel plate capacitor. The size of the plate is very large and the distance between the plates is very small, so the electric field between the plates is uniform.

The electric field 'E' between the parallel plate capacitor is

$$E = \frac{q}{A\varepsilon_0} \quad ----- \quad (1)$$



Moreover, the potential difference between the parallel plate capacitor is related to the electric field is

$$V = E\, d$$

$$\Rightarrow E = \frac{V}{d} \quad ----- \quad (2)$$

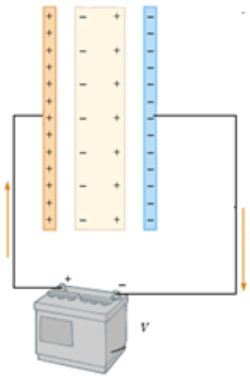
Putting this value in equation (1), we get

$$\frac{V}{d} = \frac{q}{A\varepsilon_0}$$

$$\Rightarrow \frac{q}{V} = \frac{A\,\varepsilon_0}{d} \qquad \therefore \frac{q}{V} = C\ (capacitance)$$

$$\Rightarrow C = \frac{A\,\varepsilon_0}{d}$$

This is the expression for capacitance of a parallel plate capacitor with the free space between the plates. When any dielectric medium having the dielectric constant 'κ_e' is placed between the plates of the parallel plate capacitor, then the capacitance of parallel plate capacitor will become



$$C = \frac{A\,\varepsilon_0\kappa_e}{d}$$

This equation describes that the capacitance of capacitor depends on the geometry of capacitor and medium between the plates.

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Sample problem 2. The plates of parallel plate capacitor are separated by a distance d=1mm. What must be the plate area if the capacitance to be 1 F.

Solution:

$d = 1mm = 1 \times 10^{-3}m$

$C = 1F$

$A = ?$

For a parallel plate capacitor,

$$C = \frac{A\varepsilon_0}{d}$$

$$\Rightarrow A = \frac{Cd}{\varepsilon_0} = \frac{1 \times 1 \times 10^{-3}}{8.85 \times 10^{-12}} = 1.1 \times 10^3 m^2$$

Problem 4. A parallel plate capacitor has circular plates of 8.22 cm radius and 1.31 mm separation. (a) Calculate the capacitance. (b) What charge will appear on the plates if a potential difference of 116 V is applied?

Solution:

$r = 8.22\ \text{cm} = 8.22 \times 10^{-2}\text{m}$

$\text{d} = 1.31\ \text{cm} = 1.31 \times 10^{-2}\text{m}$

(a) $C = ?$

Now the capacitance of a parallel plate capacitor

$$C = \frac{A\varepsilon_0}{d} = \frac{\pi r^2 \varepsilon_0}{d} = \frac{3.14 \times (8.22 \times 10^{-2})^2 \times 8.85 \times 10^{-12}}{1.31 \times 10^{-2}}$$

$$\Rightarrow C = 1433 \times 10^{-14}F = 14.33 \times 10^{-12}F = 14.33\ pF$$

(b) $q = ? \quad when\ V = 116\ V$

$$q = CV = 14.33 \times 10^{-12} \times 116$$

$$= 1662 \times 10^{-12} = 1.662 \times 10^{-9}C = 1.662\ nC$$

Problem 43. An air filled parallel plate capacitor has capacitance of 1.32 pF. The separation of the plate is doubled and wax inserted between them. The new capacitance is 2.57 pF. What is dielectric constant of wax.

Solution:

$\text{C}_0 = 1.32\ \text{pF} = 1.32 \times 10^{-12}\text{F}$

$\text{C} = 2.57\ \text{pF} = 2.57 \times 10^{-12}\text{F}$

(a) $\kappa_e = ?$

As $\text{C}_0 = \frac{\text{A}\varepsilon_0}{\text{d}}$

And $\text{C} = \frac{\text{A}\varepsilon_0\kappa_e}{2\text{d}} = \left(\frac{\text{A}\varepsilon_0}{\text{d}}\right)\frac{\kappa_e}{2} = \text{C}_0\frac{\kappa_e}{2}$

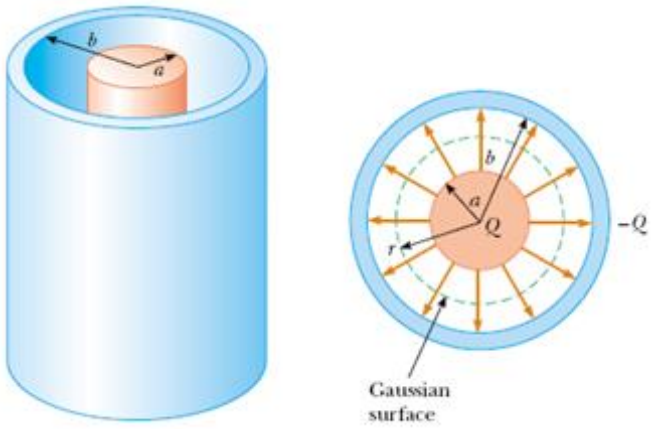
$$\Rightarrow \kappa_e = \frac{2\text{C}}{\text{C}_0} = 2 \times \frac{2.57 \times 10^{-12}}{1.32 \times 10^{-12}} = 3.89$$

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31.1.4 Capacitance of Cylindrical Capacitor

Consider a cylindrical capacitor of length L, formed by two coaxial cylinders of radii 'a' and 'b'. Suppose $L >> b$, such that there is no fringing field at the ends of cylinders. Let 'q' is the charge stored in the capacitor and 'V' is the potential difference between the plates. The inner cylinder is positively charged while the outer cylinder is negatively charged.



We want to find out the expression of capacitance for the cylindrical capacitor. For this we consider a cylindrical Gaussian surface of radius 'r' such that $a < r < b$.

If 'E' is the electric field intensity on any point of the cylindrical Gaussian surface, then by Gauss's law

$$\oint \vec{E}.d\vec{A} = \frac{q}{\varepsilon_0}$$

$$\Rightarrow \oint E\, dA \cos 0° = \frac{q}{\varepsilon_0} \qquad \therefore \vec{E} \parallel d\vec{A}$$

$$\Rightarrow E \oint dA = \frac{q}{\varepsilon_0} \qquad \therefore \text{E is constant at every point of Gaussian surface}$$

$$\Rightarrow E(2\pi rL) = \frac{q}{\varepsilon_0} \qquad \therefore 2\pi rL = \text{Area of curved part of Gaussian surface}$$

$$\Rightarrow E = \frac{q}{2\pi\varepsilon_0 rL}$$

If 'V' is the potential difference between the plates, then

$$V = \int_{+}^{-} E\, dr = \int_a^b \frac{q}{2\pi\varepsilon_0 rL}\, dr = \frac{q}{2\pi\varepsilon_0 L}\int_a^b \frac{dr}{r}$$

$$\Rightarrow V = \frac{q}{2\pi\varepsilon_0 L}|\ln r|_a^b$$

$$\Rightarrow V = \frac{q}{2\pi\varepsilon_0 L}[\ln b - \ln a]$$

$$\Rightarrow V = \frac{q}{2\pi\varepsilon_0 L}\ln\left(\frac{b}{a}\right)$$

$$\Rightarrow \frac{q}{V} = \frac{2\pi\varepsilon_0 L}{\ln\left(\frac{b}{a}\right)}$$

$$\Rightarrow C = \frac{2\pi\varepsilon_0 L}{\ln\left(\frac{b}{a}\right)} \qquad \therefore C = \frac{q}{V} = capacitance$$

This is the expression for the capacitance of a cylindrical capacitor.

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Problem 5. The plate and cathode of a vacuum tube diode are in the form of two concentric cylinders with the cathode as central cylinder. The cathode diameter is 1.62 mm and the plate diameter 18.3 mm with both elements having a length of 2.38 cm. Calculate the capacitance of the diode.

Solution:

$Diameter\ of\ cathode\ = a = 1.62\ \text{mm} = 1.62 \times 10^{-3}\text{m}$

$Diameter\ of\ anode\ = b = 18.3\ \text{mm} = 18.3\ \times 10^{-3}\text{m}$

$Length\ of\ cylindrical\ capacitor\ L = 2.38\ cm = 2.38 \times 10^{-2}\text{m}$

$Capacitance\ of\ cylkindrical\ capacitor \quad C = ?$

$$C = 2\pi\varepsilon_0 . \frac{L}{\ln\left(\frac{b}{a}\right)} = \frac{2 \times 3.14 \times 8.85 \times 10^{-12} \times 2.38 \times 10^{-2}}{\ln\left(\frac{2.1\times10^{-3}}{0.15\times10^{-3}}\right)} = 5.012 \times 10^{-13}\ F$$

Sample Problem 3. The space between conductors f a long coaxial cable, used to transmit TV signals, has an inner radius $a = 0.15$ mm and an outer radius $b = 2.1\ mm$. What is the capacitance per unit length of this cable?

Solution: $a = 0.15\ mm = 0.15 \times 10^{-3}\ m$

$b = 2.1\ mm = 2.1 \times 10^{-3}\ m$

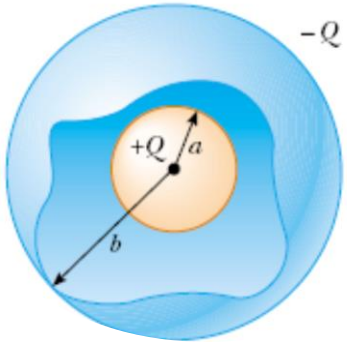
Capacitance per unit length $\frac{C}{L} = ?$

Now capacitance of cylindrical capacitor $C = 2\pi\varepsilon_0 . \frac{L}{\ln\left(\frac{b}{a}\right)}$

$$\Rightarrow \frac{C}{L} = \frac{2\pi\varepsilon_0}{\ln\left(\frac{b}{a}\right)} = \frac{2 \times 3.14 \times 8.85 \times 10^{-12}}{\ln\left(\frac{2.1\times10^{-3}}{0.15\times10^{-3}}\right)} = 2.105 \times 10^{-15}\ F/m$$

31.1.5 Capacitance of Spherical Capacitor

Consider a spherical capacitor which consist of two concentric spherical shells of radii 'a' and 'b'. Let 'q' is the charge stored in the capacitor and 'V' is the potential difference between the two spherical shells.



We want to find out the expression of capacitance for the spherical capacitor. For this we consider a spherical Gaussian surface of radius 'r' such that $a < r < b$. If 'E' is the electric field intensity on any point of the spherical Gaussian surface, then by Gauss's law

$$\oint \vec{E}.d\vec{A} = \frac{q}{\varepsilon_0}$$

$$\Rightarrow \oint E\ dA \cos 0° = \frac{q}{\varepsilon_0} \qquad \therefore \vec{E} \parallel d\vec{A}\ \text{(Radially Outward)}$$

$$\Rightarrow E \oint dA = \frac{q}{\varepsilon_0} \qquad \therefore \text{E is constant at every point of Gaussian surface}$$

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$$\Rightarrow E(4\pi r^2) = \frac{q}{\varepsilon_0} \qquad \therefore 4\pi r^2 = \text{surface area of Gaussian surface}$$

$$\Rightarrow E = \frac{q}{4\pi\varepsilon_0 r^2}$$

If 'V' is the potential difference between the plates, then

$$V = \int_{+}^{-} E\, dr = \int_a^b \frac{q}{4\pi\varepsilon_0 r^2}\, dr = \frac{q}{4\pi\varepsilon_0}\int_a^b \frac{dr}{r^2}$$

$$\Rightarrow V = \frac{q}{4\pi\varepsilon_0}\left|-\frac{1}{r}\right|_a^b$$

$$\Rightarrow V = \frac{q}{4\pi\varepsilon_0}\left[-\frac{1}{b} - (-\frac{1}{a})\right]$$

$$\Rightarrow V = \frac{q}{4\pi\varepsilon_0}\left[\frac{1}{a} - \frac{1}{b}\right]$$

$$\Rightarrow V = \frac{q}{4\pi\varepsilon_0}\left(\frac{b-a}{ab}\right)$$

$$\Rightarrow \frac{q}{V} = \frac{4\pi\varepsilon_0}{\left(\frac{b-a}{ab}\right)}$$

$$\Rightarrow C = 4\pi\varepsilon_0\left(\frac{ab}{b-a}\right) \qquad \therefore C = \frac{q}{V}\ (Capacitance)$$

This is the expression for the capacitance of a spherical capacitor.

Problem 7. The plates of a spherical capacitor have radii 38.0 mm and 40.0 mm. (a) Calculate capacitance. (b) What must be the plate area of a parallel-plate capacitor with the same plate separation and capacitance?

Solution:

$Radius\ of\ inner\ plate\ of\ spherical\ capacitor\ r_a = 38\ mm = 38\times10^{-3}m$

$Radius\ of\ outer\ plate\ of\ spherical\ capacitor\ r_b = 40mm = 40\times10^{-3}m$

(a) $Capacitance\ C = ?$

$$C = 4\pi\varepsilon_0\frac{r_a r_b}{r_b - r_a} = 2\times3.14\times8.85\times10^{-12}\times\left(\frac{38\times10^{-3}\times40\times10^{-3}}{40\times10^{-3}-38\times10^{-3}}\right) = 4.22\times10^{-11}\ F$$

(b) $d = 40\times10^{-3} - 38\times10^{-3} = 2\times10^{-3}\ m$

$A = ?$

For a parallel plate capacitor $C = \frac{A\varepsilon_0}{d} \Rightarrow A = \frac{C.d}{\varepsilon_0} = \frac{4.22\times10^{-11}\times2\times10^{-3}}{8.85\times10^{-12}} = 9.5\times10^{-3}\ m^2$

31.1.6 Capacitance of an Isolated Sphere

Consider a spherical capacitor which consists of a single isolated sphere of radius 'R'. The other (outer) sphere of this capacitor is the "missing plate" with an infinite radius = ∞.

The capacitance of spherical capacitor is

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$$C = 4\pi\varepsilon_0\left(\frac{ab}{b-a}\right) = 4\pi\varepsilon_0\left(\frac{a}{\frac{b}{b}-\frac{a}{b}}\right)$$

$$\Longrightarrow C = 4\pi\varepsilon_0\left(\frac{a}{1-\frac{a}{b}}\right)$$

Here 'a' and 'b' are the inner and outer radii of shells of spherical capacitor, respectively. Putting the values $b = \infty$and $a = R$, we get

$$\Longrightarrow C = 4\pi\varepsilon_0\left(\frac{R}{1-\frac{R}{\infty}}\right)$$

$$\Longrightarrow C = 4\pi\varepsilon_0 R$$

This is the expression for the capacitance of an isolate sphere.

Sample Problem 4. What is the capacitance of the Earth, viewed as an isolated conducting sphere of radius 6370 km?

Solution:

Capacitance C =?

Radius of Earth $R = 6370\ km = 6370 \times 10^3 m = 6.37 \times 10^6\ m$

Capacitance of an isolated capacitor

$$C = 4\pi\varepsilon_0 . R = 2 \times 3.14 \times 8.85 \times 10^{-12} \times 6.37 \times 10^6 = 7.1\ \mu F$$

31.2 Energy Stored in an Electric Field

Consider a capacitor with the capacitance 'C', which is connected to the battery of emf 'V'. If 'dq' charge is transferred from one plate to other, then the work done 'dW' will be

$$dW = Vdq$$

This work done is stored in the form of electric potential energy 'dU'

$$dU = Vdq$$

When the capacitor is fully charged then the total energy stored is

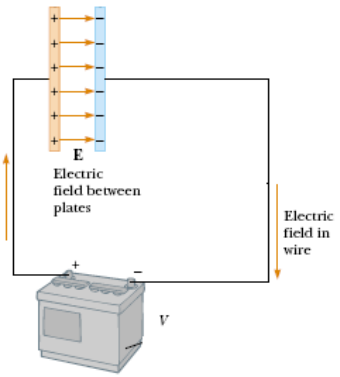
$$U = \int dU = \int_0^q Vdq$$

$$U = \int_0^q \frac{q}{C} dq = C\int_0^q q\, dq \qquad \therefore V = \frac{q}{C}$$

$$U = \frac{1}{C}\left|\frac{q^2}{2}\right|_0^q = \frac{1}{C}\left(\frac{q^2}{2} - 0\right)$$

$$U = \frac{1}{2}\left(\frac{q^2}{C}\right)$$

$$U = \frac{1}{2}\left(\frac{C^2V^2}{C}\right) \qquad \therefore q = CV$$



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$$U = \frac{1}{2}CV^2$$

The energy stored in the capacitor is the energy store in the electric field between its plates. So, the energy stored can be expressed in terms of electric field strength 'E'.

As $E = Vd$ and $C = \frac{A\,\varepsilon_0}{d}$, therefore

$$U = \frac{1}{2}\left(\frac{A\,\varepsilon_0}{d}\right)(E^2 d^2)$$

$$U = \frac{1}{2}\,\varepsilon_0 E^2 Ad$$

This is the expression of energy stored in the electric field between the plates of capacitor.

Energy Density

The energy density 'u' is described as the energy stored 'U' per unit volume 'V'. Mathematically

$$u = \frac{U}{V} = \frac{U}{Ad}$$

$$= \frac{\frac{1}{2}\,\varepsilon_0 E^2 Ad}{Ad}$$

$$u = \frac{1}{2}\,\varepsilon_0 E^2$$

Expression of Energy Stored in Capacitor and Energy Density with Dielectric Medium between Plates of Capacitor

If any dielectric medium having the dielectric constant 'κ_e' is placed between the plates of capacitor, then the expressions of energy stored in the electric field of capacitor 'U' and energy density 'u' will become,

$$U = \frac{1}{2}\,\varepsilon_0 \kappa_e E^2 Ad$$

$$u = \frac{1}{2}\,\varepsilon_0 \kappa_e E^2$$

Problem 29. How much energy is stored in $2.0\,m^3$ of air due to the "fair weather" electric field strength 150 V/m?

Solution

$$V = 2.0\ m^3$$

$$E = 150\,\frac{V}{m}$$

$U = ?$

$$U = \frac{1}{2}\varepsilon_0 E^2 V$$

$$= \frac{1}{2} \times 8.85 \times 10^{-12} \times (150)^2 \times 2$$

$$= 1.99 \times 10^{-7}\ \text{J}$$

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Problem 30. One capacitor provides 61 mF at 10 kV. Calculate the stored energy (a) in joules and (b) in kWh.

Solution

$C = 61\ mF = 61 \times 10^{-3} F$

$V = 10kV = 10^4\ V$

$U = ?$

(a) $U = \frac{1}{2} C\ V^2 = \frac{1}{2} \times 61 \times 10^{-3} \times (10^4)^2 = 30.5 \times 10^5\ J = 3.05\ MJ$

(b) $U\ in\ kWh = ?$

As $1\ kWh = 3.6 \times 10^6\ J \Longrightarrow 1\ J = \frac{1}{3.6 \times 10^6} kWh$

$$U = 3.05 \times 10^6\ J = \frac{3.05 \times 10^6}{3.6 \times 10^6} = 0.847\ kW\text{h}$$

Problem 31. A parallel plate air capacitor having area 42 cm^2 and spacing of 1.3 mm is charged to a potential difference of 625 V. (a) Find (a) capacitance, (b) the magnitude of charge on each plate, (c) stored energy, (d) electric field between plates, and (e) energy density between plates.

Solution:

$A = 42\ cm^2 = 42 \times 10^{-4} m^2$

$d = 1.3\ mm = 1.3 \times 10^{-3} m$

$V = 625\ V$

(a) $C = ?$

$$C = \frac{A\varepsilon_0}{d} = \frac{42 \times 10^{-4} \times 8.85 \times 10^{-12}}{1.3 \times 10^{-3}} = 285 \times 10^{-13} = 28.5\ pF$$

(b) $q = ?$

$q = CV = 28.5 \times 10^{-12} \times 625 = 1.78 \times 10^{-8} = 17.8\ nC$

(c) $U = ?$

$$U = \frac{1}{2} CV^2 = \frac{1}{2} \times 28.5 \times 10^{-12} \times (625)^2 = 5.5 \times 10^{-6}\ J$$

(d) $E = ?$

$$E = \frac{V}{d} = \frac{625}{1.3 \times 10^{-3}} = 4.8 \times 10^5 \frac{V}{m}$$

(e) *Energy Density* $\mu = ?$

$$\mu = \frac{1}{2} \varepsilon_0 E^2 = \frac{1}{2} \times 8.85 \times 10^{-12} \times (4.8 \times 10^5)^2 = 1.02\ \text{Jm}^{-3}$$

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31.3 Capacitance with Dielectrics

Consider a parallel plate capacitor which is connected with a battery of emf 'V'. Let 'A' is the area of each plate and 'd' is separation between the plates.

If 'q' charge is stored in the capacitor when there is vacuum or air as medium between the plates, then

$$q = CV$$

Where 'C' is the capacitance of the capacitor, which, for the case of parallel plate capacitor is expressed as

$$C = \frac{A\,\varepsilon_0}{d}$$

Micheal Faraday, in 1937, investigated that if the space between the plates of a parallel plate capacitor is filled with some dielectric medium, then the charge stored in the capacitor increased to 'q''. And, hence, the capacitance of the capacitor also increases to 'C''.

Therefore, the relation between the stored charge in the capacitor to the capacitance will become

$$q' = C'\,V$$

The factor by which the capacitance of capacitor increases as compared to the capacitance with air as the medium is called the dielectric constant 'κ_e'. The dielectric constant 'κ_e' is a dimensionless quantity. It is also called relative permittivity of the medium. The dielectric constant is described mathematically as:

$$\kappa_e = \frac{C'}{C} \qquad \therefore \kappa_e > 1$$

$$C' = \kappa_e C$$

$$C' = \kappa_e \left(\frac{A\,\varepsilon_0}{d}\right)$$

So, $C' > C$ by a factor of 'κ_e'.

31.3.1 Effects of Dielectric Medium

- When a dielectric medium, having dielectric constant 'κ_e', is placed between the point charges, then the electrical force between two point charges decreases by a factor of 'κ_e'. The expression of electrical force 'F_e' between two point charges 'q_1' and 'q_2', when the dielectric medium is place between them, is

$$F_e = \frac{1}{4\pi\varepsilon_0\kappa_e}\frac{q_1\,q_2}{r^2}$$

- The electrical field intensity 'E' due to a point charge between two point charges decreases, in the presence of a dielectric medium. If 'κ_e' is the dielectric constant of the corresponding dielectric medium, then the electric field intensity at any point due to point chage will be

$$E = \frac{1}{4\pi\varepsilon_0\kappa_e}\frac{q}{r^2}$$

- The electric field near the surface of charged conductor, which is immersed in a dielectric medium of dielectric constant 'κ_e' is

$$E = \frac{\sigma}{\varepsilon_0\kappa_e}$$

where 'σ' is the uniform charge density of conductor.

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31.4 Dielectrics: An Atomic View

Dielectrics are the insulating materials through which the electric current cannot pass easily, because these materials have very high value of electrical resistance. For example, paper, pyrex, polystyrence, transformer oil, pure water, silicon etc.

There are two types is dielectrics

(i) Polar Dielectrics

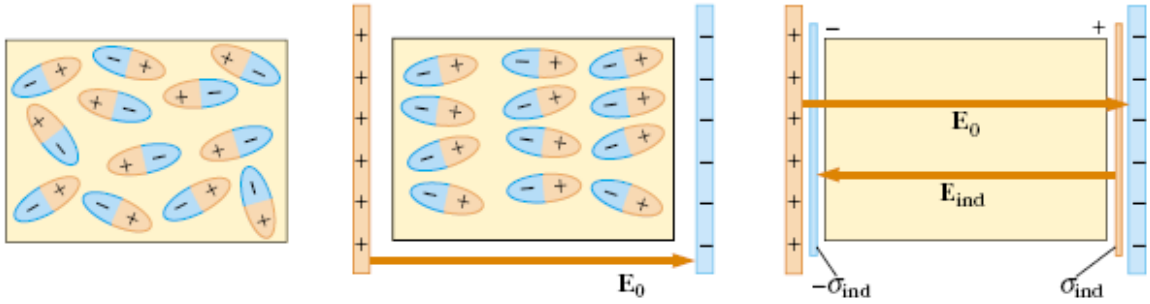
(ii) Non-Polar Dielectrics

31.4.1 Polar Dielectrics

The dielectric materials which have permanent electric dipoles moment are called polar dielectrics. These materials consist of molecules which are permanent dipoles.

In the absence of external electric field, the polar molecules are randomly oriented. As the result, these materials have no net dipole moment.

When the external electric field is applied, then all dipoles tend to align themselves with external electric field. But the thermal agitation tends to keep the dipoles randomly oriented. Hence the partial alignment of electric dipoles is produced in a polar dielectric medium for specific electric field strength. However, the alignment of dipoles can be increased by increasing the external electric field and decreasing the temperature.



31.4.2 Non-Polar Dielectrics

The dielectric materials which don't have permanent dipole moments are called non-polar dielectrics. In the absence of external electric field, the centers of positive and negative charges coincide. When the electric field of strength 'E_0' is applied, then it tends to separate to positive and negative charges on the atoms of molecules. As the result, the atoms and molecules of dielectric become dipoles, called induced dipoles and this process is called electric polarization.

31.4.3 Effect of Electric Polarization on Capacitance of Capacitor

- If the dielectric material is placed in a electric field having strength 'E_0', then another electric field 'E'' is produced due to the polarization of medium. The electric field produce due to the polarization of dielectric is always opposite to the direction of external electric field. So, the net electric field in the region is '$E = E_0 - E'$'. Hence, the net electric field is reduced due to polarization of dielectric medium.

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- The reduction of strength of electric field results in decrease in potential difference between plates of capacitor as electric field is directly proportional to potential difference (For Parallel Plate Capacitor $E = Vd$)
- As potential difference between conducting plates of capacitor is inversely proportional to capacitance of capacitor, as described by formula:

$$C = \frac{q}{V}$$

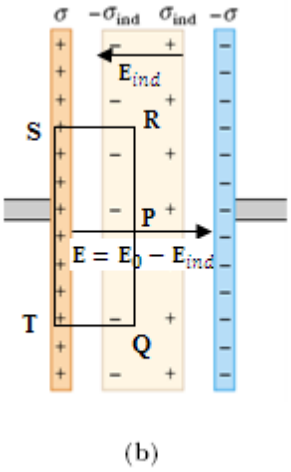
So decrease in potential difference will result in increase in capacitance of capacitor.

Hence, due to electric polarization of dielectric, the capacitance of capacitor increases.

31.5 Gauss's Law in Dielectrics

Consider a charged capacitor, having '$+q$' charge on one plate and '$-q$' on the other plate as shown in the figure below.

Let a dielectric medium of dielectric constant 'κ_e' is placed between the plates of charged capacitor. Due to the polarization of dielectric medium, an induced charge '$-q'$' appears near the positively charged plate and '$+q'$' appears near the negatively charged plate as shown in the figure below.



(b)

So, the net charge enclosed in the Gaussian surface decreases which remains '$q - q'$'. By applying the Gauss's law,

$$\oint \vec{E}.d\vec{A} = \frac{q - q'}{\varepsilon_0} \quad ----- \quad (1)$$

Where A area of the side of Gaussian surface through which the flux is passing, while dA is its infinitesimal element.

$$\Rightarrow \int_{SR} \vec{E}.d\vec{A} + \int_{RQ} \vec{E}.d\vec{A} + \int_{QT} \vec{E}.d\vec{A} + \int_{TS} \vec{E}.d\vec{A} = \frac{q - q'}{\varepsilon_0} \quad ----- \quad (2)$$

As electric field is zero inside conducting plate A, therefore $\int_{TS} \vec{E}.d\vec{A} = 0$. Equation (2) becomes:

$$\Rightarrow \int_{SR} E\,dA\cos 90° + \int_{RQ} E\,dA\cos 0° + \int_{QT} E\,dA\cos 90° + 0 = \frac{q - q'}{\varepsilon_0}$$

$$\Rightarrow 0 + \int_{RQ} E\,dA + 0 + 0 = \frac{q - q'}{\varepsilon_0}$$

$$\Rightarrow EA = \frac{q - q'}{\varepsilon_0}$$

$$\Rightarrow E = \frac{q - q'}{A\varepsilon_0}$$

$$\Rightarrow E = \frac{q}{A\varepsilon_0} - \frac{q'}{A\varepsilon_0} \quad ----(3)$$

Hence the electric field intensity decreased when the dielectric medium is placed between the plates of capacitor.

The electric field intensity between two oppositely charged plates, in the presence of a dielectric medium can also be find out by the expression,

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$$E = \frac{q}{\varepsilon_0 \kappa_e A} \quad ----(4)$$

Comparing eq. (3) and (4), we get

$$\Rightarrow \frac{q}{\varepsilon_0 \kappa_e A} = \frac{q}{A\varepsilon_0} - \frac{q'}{A\varepsilon_0}$$

$$\Rightarrow \frac{q}{\kappa_e} = q - q'$$

$$\Rightarrow q' = q - \frac{q}{\kappa_e}$$

This shows that the induced surface charge 'q'' is always less than the original free charge 'q'.

Putting values in equation (1), we have:

$$\oint \vec{E}.d\vec{A} = \frac{1}{\varepsilon_0}\left[q - \left(q - \frac{q}{\kappa_e}\right)\right]$$

$$\Rightarrow \oint \vec{E}.d\vec{A} = \frac{1}{\varepsilon_0}\left[q - q + \frac{q}{\kappa_e}\right]$$

$$\Rightarrow \oint \vec{E}.d\vec{A} = \frac{q}{\varepsilon_0 \kappa_e}$$

This is the Gauss's law in dielectrics.

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CURRENT AND RESISTANCE

In this chapter we shall study electric currents, i.e., of charges in motion. Example of electric currents abound, ranging from large currents that constitute lightning strokes to the tiny nerve current that regulate our muscular activity. We are familiar with currents resulting from charges flowing through solids conductors (household wiring, light bulbs), semiconductors (integrated circuits), gases (fluorescent lamps), liquids (automobile batteries), and even evacuated spaces (TV picture tubes)

32.1 ELECTRIC CURRENT

The time rate of flow of charge through a conductor is called current. If a charge 'dq' flows through any cross-section of a conductor in time 'dt', then the current 'I' is given by

$$I = \frac{dq}{dt}$$

The SI unit of current is Ampere, which can be defined as, "when one coulomb charge flows through a cross-section in one second, then the current flowing is one ampere".

Problem 1. A current of 4.82 A exist in a 12.4Ω resistor for 4.6 minutes. (a) Find out charge, (b) How many electrons pass through resistor in this time?

Solution. $I = 4.82\ A$

$R = 12.4\Omega$

$t = 4.6\ min = 4.6 \times 60\ s$

q =?

n =?

(a) As $I = \frac{q}{t}$

$\Rightarrow q = It = 4.82 \times 4.6 \times 60 = 1.33 \times 10^3\ C$

(b) As $q = ne \Rightarrow n = \frac{q}{e} = \frac{1.33 \times 10^3}{1.6 \times 10^{-19}} = 8.3 \times 10^{21}\ e$

Problem 2. The current in the electron beam of a typical video display terminal is $200\mu A$. How many electrons strike the screen each minute?

Solution:

$I = 200\ \mu A = 200 \times 10^{-6} A$

n =?

$t = 1\ min = 60\ s$

As $I = \frac{q}{t}$

$\Rightarrow q = It = 200 \times 10^{-6} \times 60 = 12000 \times 10^{-6}\ C$

Also $q = ne \Rightarrow n = \frac{q}{e} = \frac{12000 \times 10^{-6}}{1.6 \times 10^{-19}} = 7.5 \times 10^{16}\ e$

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32.2 CURRENT FLOW AND FLUID FLOW: A COMPARISON

When a steady current is flowing through idealized conducting wire, the electric current remains same for all cross-sections, even though the cross-sectional area may be different at different points. The condition of steady current flow is similar to the motion of incompressible fluid. The fluid that flows through any cross-section of the pipe is the same even if the cross-section varies. The fluid flows faster where the cross-section of the pipe is smaller and slower where it is larger, but the volume rate of flow remains constant.

32.3 DIRECTION OF CURRENT

In metals, the charge carriers are electrons. But in electrolytes, the current flow due to motion of negative and positive ions. A positive charge moving in one direction is equivalent in all external effects to a negative charge moving in the opposite the opposite direction. Hence for simplicity and algebraic consistency, we adopt the following convention:

The direction of current is the direction that positive charges would move, even if the actual charge carriers are negative. Thus, the direction of current is taken from the point of higher potential to the point of lower potential.

Even though we assign a direction, current is a scalar quantity, not a vector. The arrow that we draw to indicate the direction of current merely show the sense of charge flow through the wire and is not be taken as a vector. Current does not obey the law of vector addition. Changing the direction of wires does not change the way the currents are added.

32.4 CURRENT DENSITY

The current flowing per unit area is called the current density. It is a vector quantity and the SI unit of this quantity is Ampere per square meter ($\frac{A}{m^2}$). The electric current I can be described in terms of current density as

"The scalar product of current density '$\vec{J}$' and vector area '$\vec{A}$' is called the electric current".

Mathematically,

$$I = \vec{J}.\vec{A}$$

The electric current is macroscopic quantity, while the current density is its corresponding microscopic quantity. In general, if '$d\vec{A}$' is the small area element of conductor, then the current flowing through the whole area of conductor is.

$$I = \int \vec{J}.d\vec{A}$$

Sample Problem 1. One end of an aluminum wire whose diameter is 2.5 mm is welded to one end of copper wire whose diameter is 1.8mm. The composite wire carries a steady current of 1.3 A. What is current density in each wire?

Solution: Current flowing through composite wire $I = 1.3\ A$

For Aluminum

Diameter of wire $d = 2.5\ mm$

Radius $r = 1.25\ mm = 1.25 \times 10^{-3} m$

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Current Density in Al $J_{Al} = \frac{I}{A} = \frac{1.3}{3.14\times(1.25\times10^{-3})^2} = 2.64 \times 10^5 \frac{A}{m^2}$

For Copper

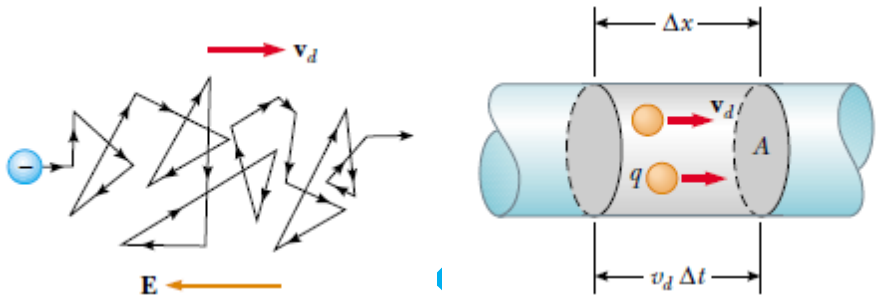
Diameter of wire $d = 1.8\ mm$

Radius $r = 0.5\ mm = 0.9 \times 10^{-3} m$

Current Density in Cu $J_{Cu} = \frac{I}{A} = \frac{1.3}{3.14\times(0.9\times10^{-3})^2} = 5.1 \times 10^5 \frac{A}{m^2}$

32.5 DETERMINATION OF CURRENT DENSITY OF CONDUCTOR

Let 'A' is the area of the cross-section of a conductor of length 'L' in which the current 'I' is flowing. The flow of current through a conductor is due to motion of electron in the direction opposite to electric field '$\vec{E}$'. The force on one electron due to electric field is '$-e\vec{E}$'. But this force does not produce any acceleration in the motion of electrons, because the conduction electrons keep on colliding with the lattice ions of conductor. Instead of this, the electrons acquire a constant drift speed 'v_d' in the direction of '$-\mathbf{E}$'.



Let

n = Number of free electrons per unit

AL = Volume of the conductor

nAL = Number of free electrons in the conductor

e = Charge on one electron

nALe = Δq = Total charge flowing in conductor

If the charge 'q' passes through conductor in time 'Δt', then

$$\Delta t = \frac{L}{v_d}$$

So, the current

$$I = \frac{\Delta q}{\Delta t} = \frac{nALe}{L/v_d} = nAev_d$$

The current density is

$$J = \frac{I}{A} = \frac{nAev_d}{A}$$

$$J = nev_d$$

The direction of '$\vec{J}$' is opposite to the direction of flow of electrons.

$$\vec{J} = -ne\,\vec{v}_d$$

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The mean drift velocity of electrons is very small i.e., of the order of 'cm/s'. While in random motion, the speed of electrons has a typical value of $10^6\ m/s$ in metals.

Sample Problem 2. What is drift speed of the conduction electron in copper wire of sample problem 1. The electron density in copper is $8.49 \times 10^{28}\ \frac{electrons}{m^3}$

Solution: Given $n = 8.49 \times 10^{28}\ \frac{electrons}{m^3}$

$$J_{Cu} = 5.1 \times 10^5\ \frac{A}{m^2}$$

The drift velocity can be find out by expression

$$v_d = \frac{J}{ne} = \frac{5.1 \times 10^5}{8.49 \times 10^{28} \times 1.6 \times 10^{-19}} = 0.37 \times 10^{-4}\ \frac{m}{s}$$

Sample problem 3. A strip of Si of width 3.2 cm and thickness $d = 250\ \mu m$ carries a current of 190 mA and $n = 8 \times 10^{21}\ m^{-3}$, (a) Find current density. (b) Find drift speed

Solution:(a) Current flowing through strip $I = 190\ mA = 190 \times 10^{-3} A$

Width $w = 3.2\ cm = 3.2 \times 10^{-2} m$

Thickness $d = 250\ \mu m = 250 \times 10^{-6} m$

Current Density in Si $J_{Si} = \frac{I}{A} = \frac{I}{w.d} = \frac{190 \times 10^{-3}}{3.2 \times 10^{-2} \times 250 \times 10^{-6}} = 2.4 \times 10^5\ \frac{A}{m^2}$

(b) Given $n = 8 \times 10^{21}\ m^{-3}$

The drift velocity can be find out by expression

$$v_d = \frac{J}{ne} = \frac{2.4 \times 10^5}{8 \times 10^{21} \times 1.6 \times 10^{-19}} = 190\ \frac{m}{s}$$

Problem 3. Suppose that we have 2.1×10^8 doubly charge positive ions per cubic centimeter, all moving north with a speed of $1.4 \times 10^5\ m/s$, (a) Find current density and direction. (b) Can you calculate total current? If not! What additional information is needed?

Solution:

$v_d = 1.4 \times 10^5\ m/s$

$n = 2.1 \times 10^8\ /cm^3 = 2.1 \times 10^{14}\ /m^3$

J =?

I =?

$$J = nev_d = 2.1 \times 10^{14} \times 1.6 \times 10^{-19} \times 1.4 \times 10^5 = 9.408\ \frac{A}{m^2}$$

$Now\ \ J = \frac{I}{A} \Longrightarrow I = JA$

So we can't find current because the value of area is not given.

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33.6 OHM'S LAW

It states that

"*The current flowing through a conductor is directly proportional to the applied potential difference if all physical states remain same*."

If 'V' is the potential difference between the ends of conductor and 'I' is the current flowing through it, then the Ohm's law is described mathematically as:

$$V \ \alpha \ I$$

$$V = R \ I$$

Where R is the constant of proportionality, called resistance of the conductor. It is described as the opposition offered by conductor to the flow of current. In system international, its unit is ohm. It is a macroscopic quantity. Its corresponding microscopic quantity is resistivity.

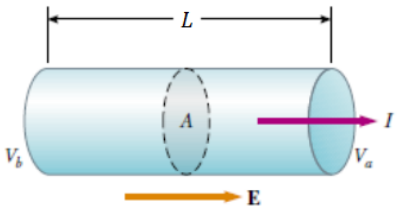
33.7 RESISTIVITY

The resistance of a meter cube of a substance is called resistivity or specific resistance. It is denoted by the symbol ρ. Its SI unit is ohm-meter $(\Omega - m)$.

Consider a conductor of length L and cross-sectional area A as shown in the figure. It is found experimentally that the resistance of conductor is directly proportional to length L of conductor and inversely proportional to cross-sectional are A:

$$R \propto L \quad ---- \ (1)$$

$$R \propto \frac{1}{A} \quad ---- \ (2)$$



Combining (1) and (2), we have:

$$R \propto \frac{L}{A}$$

$$R = \rho \frac{L}{A}$$

Where ρ is constant of proportionality, called resistivity or specific resistance of conductor.

$$\Rightarrow \rho = \frac{RA}{L}$$

This is the expression of resistivity for a conductor having length L and cross-sectional area A.

Problem 17. A steel trolley car-rail has a cross-sectional area of 56 cm^2. What is resistance of 11 km of rail? The resistance of steel is $3 \times 10^{-7} \Omega - m$.

Solution:

$$\mathrm{A} = 56\ \mathrm{cm}^2 = 56 \times 10^{-4}\mathrm{m}^2$$

$$\mathrm{L} = 11\ \mathrm{km} = 11 \times 10^{3}\mathrm{m}$$

$$\rho = 3 \times 10^{-7}\Omega\mathrm{m}$$

R =?

$$\mathrm{R} = \frac{\rho \mathrm{L}}{\mathrm{A}} = \frac{3 \times 10^{-7} \times 11 \times 10^{3}}{56 \times 10^{-4}} = 0.58\Omega$$

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Sample problem 4. A rectangular block of iron has dimension $1.2cm \times 1.2cm \times 15cm$. (a) What is resistance of block measured between two square ends? (b) What is resistance between two opposite rectangular face? The resistivity of iron at room temperature is $9.68 \times 10^{-8}\Omega m$.

Solution: (a) Area between two square faces $A = 1.2cm \times 1.2cm = 1.44\ cm^2 = 1.44 \times 10^{-4}m^2$

Length of conductor between two square ends $L = 15\ cm = 15 \times 10^{2}m$

Resistance $R = \rho\frac{L}{A} = 9.68 \times 10^{-8} \times \frac{15\times10^{2}}{1.44\times10^{-4}} = 1 \times 10^{-4}\Omega$

(b)Area between two rectangular faces $A' = 15cm \times 1.2cm = 18\ cm^2 = 18 \times 10^{-4}m^2$

Length of conductor between two square ends $L' = 1.2\ cm = 1.2 \times 10^{2}m$

Resistance $R' = \rho\frac{L}{A} = 9.68 \times 10^{-8} \times \frac{1.2\times10^{2}}{18\times10^{-4}} = 6.5 \times 10^{-7}\Omega$

33.8 MICROSCOPIC FORM OF OHM'S LAW

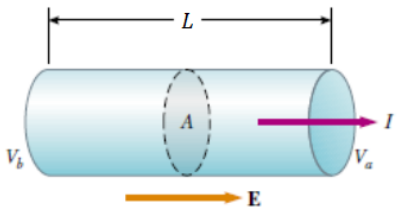
Consider a conductor of length L and cross-sectional area A as shown in the figure. The expression of resistivity ρ of such conductor is described by formula:

$$\rho = \frac{RA}{L}$$

By Ohm's law, we have $R = \frac{V}{I}$. Therefore,

$$\rho = \frac{VA}{IL}$$

$$\Rightarrow \rho = \frac{V}{\left(\frac{I}{A}\right)L}$$



where $J = \frac{I}{A}$ is the current density.

$$\rho = \frac{V}{JL}$$

$\therefore$ *For present case, the potential difference applied and the electric field developed across of ends conductors is* $V = EL$

$$\Rightarrow \rho = \frac{EL}{JL}$$

$$\Rightarrow \rho = \frac{E}{J}$$

$$E = \rho J$$

As resistivity 'ρ' is reciprocal of conductivity 'σ'. So we can write:

$$\rho = \frac{1}{\sigma}$$

Therefore:

$$E = \left(\frac{1}{\sigma}\right)J$$

$$J = \sigma E$$

In vector form:

$$\vec{J} = \sigma\vec{E}$$

This is known as microscopic form of ohm's law.

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Problem 33. When 115 V is applied across a 9.66 m long wire, the current density is $1.4\frac{A}{cm^2}$. Calculate the conductivity of the wire material.

Solution:

$$V = 115\ V$$

$$L = 9.66\ m$$

$$J = 1.4\frac{A}{cm^2} = 1.4 \times 10^4 \frac{A}{m^2}$$

σ =?

$$J = \sigma E \Longrightarrow \sigma = \frac{J}{E} = \frac{J}{(V/L)} = \frac{JL}{V} = \frac{1.4 \times 10^{-4} \times 9.66}{115} = 1.19 \times 10^3\ \frac{C}{m^2}$$

TEMPERATURE VARIATION OF RESISTIVITY

The resistivity of the conductor increases with increase in temperature. The temperature dependence of resistivity 'ρ' is shown in the figure.

Let 'ρ_0' and 'ρ' be the values of resistivity at temperature 0°C and T°C, respectively. The change of resistivity $(\rho - \rho_0)$ is directly proportional to the resistivity 'ρ_0' and change of temperature $(T - T_0)$. That is

$$(\rho - \rho_0) \propto \rho_0 \quad ---- \ (1)$$

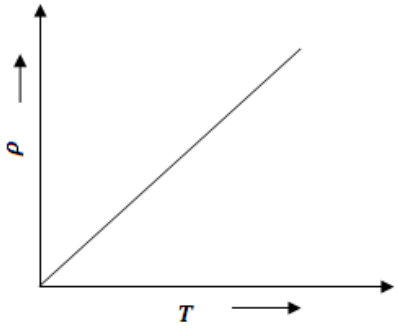
$$(\rho - \rho_0) \propto (T - T_0) \quad ---- \ (2)$$

Combining (1) and (2), we have:

$$(\rho - \rho_0) \propto \rho_0 (T - T_0)$$

$$\Longrightarrow (\rho - \rho_0) = \bar{\alpha}\, \rho_0 (T - T_0)$$

Where $\bar{\alpha}$ is the mean temperature coefficient of resistivity.



$$\bar{\alpha} = \frac{(\rho - \rho_0)}{\rho_0 (T - T_0)}$$

Thus we can describe resistivity as the fractional change in resistivity per unit rise in temperature. The temperature coefficient of resistivity depends upon the nature of material and it is measure in the units $°C^{-1}$ or K^{-1}. The general formula for temperature coefficient of resistivity is

$$\alpha = \frac{1}{\rho}\frac{d\rho}{dT}$$

where $\frac{d\rho}{dT}$ is the rate of change of resistivity with respect to temperature.

ہمیں اس کار خیر کو بہتر بنانے کے لئے آپ کی مدد کی ضرورت ہے۔اپنی قیمتی آراءاور نوٹس ای میل کریں یا فیس بک ایڈیس پر میسیج کریں۔

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ANALOGY BETWEEN THE CURRENT AND HEAT FLOW

Derivation of Expression for Current Flow

Consider a thin electrically conducting slab of thickness 'Δx' and area 'A'. When potential difference 'ΔV' is applied across the ends of a conductor, the current 'I' will flow through it.

By Ohm's law

$$I = \frac{\Delta V}{R}$$

As $R = \frac{\rho \Delta x}{A}$

$$\Longrightarrow I = \frac{\Delta V}{\left(\frac{\rho \Delta x}{A}\right)} = \frac{A}{\rho}\frac{\Delta V}{\Delta x}$$

But $I = \frac{dq}{dt}$. So,

$$\frac{dq}{dt} = \frac{A}{\rho}\frac{dV}{dx} \Longrightarrow \frac{dq}{dt} = A\sigma\frac{dV}{dx} \qquad \therefore \frac{1}{\rho} = \sigma$$

As the current flows in the direction of decreasing potential, so we place a minus sign to encounter this phenomenon. Therefore,

$$\frac{dq}{dt} = -A\sigma\frac{dV}{dx} \quad ----(1)$$

The equation (1) describe current flow through conductor.

Equation for Heat Flow

If "dQ" is the heat flows through the area 'A' in the small interval of time 'dt', then the rate of flow of heat $\frac{dQ}{dt}$ is described as:

$$\frac{dQ}{dt} = -kA\frac{dT}{dx} \quad ---- \ (2)$$

Comparison of Current Flow and Heat Flow

The rate of flow charge is given by eq. (1) and the rate of flow of heat is described in eq. (2). Hence there is a close analogy. The current flows due to the difference in the potential and heat flows due to the difference in temperature. Here

$\frac{dV}{dx}$ = Potential Gradient

$\frac{dT}{dx}$ = Temperature Gradient

Similarly, the electrical conductivity in eq. (1) has the same effect as that of thermal conductivity in eq. (2).

DERIVATION OF EXPRESSION OF RESISTIVITY OF CONDUCTOR

In metals, the valance electrons are not attached to the individual atoms but are free to move about within the lattice called conduction electrons. According to free electron model, the conduction electrons are assumed to move freely throughout the conducting material, somewhat like the molecules of gas in a container. These conduction electron moves randomly like the molecules of gas. The electrons make collisions with atoms and molecules during their random motion. In case of copper, the average speed of electrons in random motion is 1.6×10^6 ms^{-1}.

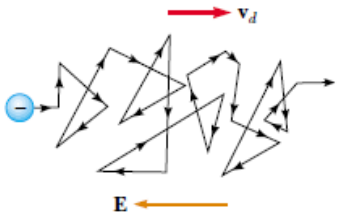
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If the electric field is applied, then the motion of electrons slightly shifted in the direction opposite of that of $\vec{E}$. Then the force '$\vec{F}$' acting on the free electron is find out using relation

$$F = eE$$

where 'e' is the charge on electron. By using Newton's second law of motion, eq. (1) will become



$$ma = eE$$

$$\Rightarrow a = \frac{eE}{m}$$

During the collision of free electrons and the atom (or ion core), the tendency of the electron to drift is destroyed. Therefore the average drift speed of the electron will be

$$v_d = a\tau = \frac{eE\tau}{m}$$

where a is the average acceleration of electrons and τ is the mean free time.

Thus the current density J is given by

$$J = nev_d$$

Where n is the number of electrons per unit volume

$$J = ne\left(\frac{eE\tau}{m}\right)$$

$$\Rightarrow \frac{E}{J} = \frac{m}{ne^2\tau} \quad ---- \quad (1)$$

According to microscopic form of Ohm's law, the resistivity 'ρ' of a conductor is expressed as

$$\rho = \frac{E}{J}$$

Thus equation (1) becomes:

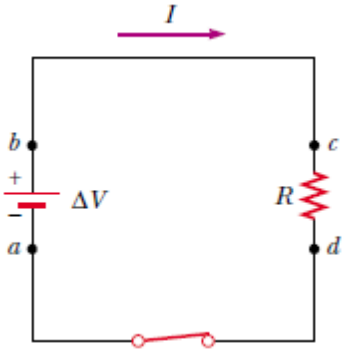
$$\rho = \frac{m}{ne^2\tau}$$

This equation gives the value of electrical resistivity.

OHMIC DEVICES

A material of a circuit element that obeys the ohm law is called the ohmic. Therefore a conducting device obeys ohm's law, if the resistance between the two points is independent of magnitude and polarity of potential difference.

ENERGY TRANSFER IN AN ELECTRIC CIRCUIT AND POWER DISSIPATION



Let a battery is connected between the terminals 'a' and 'b' of an electric circuit as shown in the figure.

Let 'V' is the potential difference applied by the battery between the points 'a' and 'b'. As the result the current 'I' flow through the circuit. During this process, energy is transfer from battery to the electrical circuit. Let a small amount of charge 'dq' during the small interval of time 'dt'.

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Using the meaning of potential difference, the work done ΔW in moving ΔQ up through the potential difference V is:

$$\Delta W = V \times \Delta Q$$

This work done will be appear the energy supplied by the battery. The rate at which the battery is supplying electrical energy is called the electrical power of the battery.

$$Electrical\ Power = \frac{Energy\ Supplied}{Time\ Taken} = V\ \frac{\Delta Q}{\Delta t}$$

Since $I = \frac{\Delta Q}{\Delta t}$

$$Electrical\ Power = VI$$

By the principal of conservation of energy, the electrical power of the battery is dissipated in the resistor R. Therefore,

$$Power\ Dissipated\ (P) = VI$$

From Ohm's law, substituting $V = IR$ and $I = \frac{V}{R}$

$$Power\ Dissipated\ (P) = VI = IR * I = I^2R$$

$$Power\ Dissipated\ (P) = VI = V * \frac{V}{R} = \frac{V^2}{R}$$

JOULE HEATING

The electrical energy consumed in a resistor appears in the form of heat, which is also called 'Joule Heating'. The heat energy produced in t interval of time is given by

Heat Energy = (Power)(Time)

$$= (VI)(t)$$

$$= VIt = I^2R\,t = \frac{V^2}{R}t$$

Sample problem 6. You are given length of heating wire made of nickel-chromium-iron Alloy called Nichrome. It has a resistance of 72Ω. Under what circumstances, the wire will dissipates more power. (a) It is to be connected across a 120 V line. (b) The wire is cut in half pieces and two halves are connected in parallel across the line?

Solution: (a) Power dissipation in the wire:

$$R = 72\Omega$$

$$P = \frac{V^2}{R} = \frac{(120)^2}{72} = 200\ \text{Watt}$$

(b) The power dissipation when the wire is cut in half pieces and two halves are connected in parallel across the line:

$$R' = 36\Omega$$

$$P' = \frac{V^2}{R'} = \frac{(120)^2}{36} = 400\ \text{Watt}$$

The power dissipation in second case in more than 1st case

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Problem 45. A student has a portable radio $9V, 7.5W$ was left on from 9:00 pm to 3:am. How much charge pass through it.

Solution: $V = 9V$

$P = 7.5W$

$t = 6h = 6 \times 3600\ s = 21600\ s$

q =?

As

$$P = VI \Longrightarrow I = \frac{P}{V} \Longrightarrow \frac{q}{t} = \frac{P}{V} \Longrightarrow q = \frac{Pt}{V}$$

$$\Longrightarrow q = \frac{Pt}{V} = \frac{7.5 \times 21600}{9} = 18000\ C$$

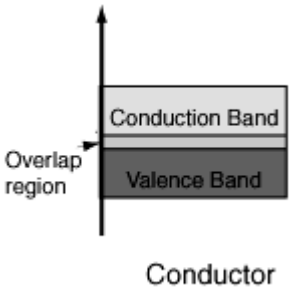
ENERGY BAND THEORY AND ELECTRICAL BEHAVIOR OF CONDUCTORS, INSULATORS AND SEMICONDUCTORS

Energy band theory

The electrons of an atom have the discrete values of energy which are also called quantized energy levels. The concept of discrete energy level is related to an isolated atoms i.e. the atom that does not interact with other atoms. But if the atom is not alone and is under the influence of its neighboring atoms, then each energy level splits into sub-levels. This group of sub-level is called the energy band. Within the energy band, there are permitted energy states, which are so close together that they are virtually continuous. But there exist an energy gap between these bands, which contains no states that an individual electron may occupy. It is also called forbidden energy gap. The electron may jump from one energy band to another by acquiring energy equal to the energy of forbidden energy gap.

Conductors

All the metals are good conductors of electricity and their resistivity is of the order of $10^{-8}\ \Omega - m$. In case of conductors, there is no forbidden energy gap between the valance and the conduction band. The valance band and conduction band are partially filled at room temperature. So the electrons can easily jump from valance band to the conduction band. Due to this reason, the current can easily pass through conductors.

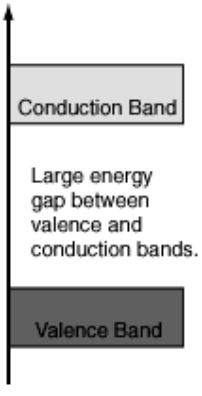


Conductor

The temperature coefficient resistivity is positive. It means that the resistance of conductors increases by increasing the temperature.

Insulators

The insulators have the very large value of resistivity which is of the order of $10^{10}\ \Omega - m$. In case of insulators, the valance band is completely filled and the conduction band is empty. The energy gap between the valance and conduction band is very large. Thus, no electron can jump from valence band to conduction band. As there are no free electrons in insulator, hence no current can pass through insulators.

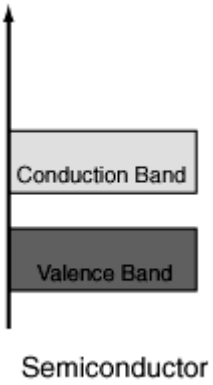


Insulator

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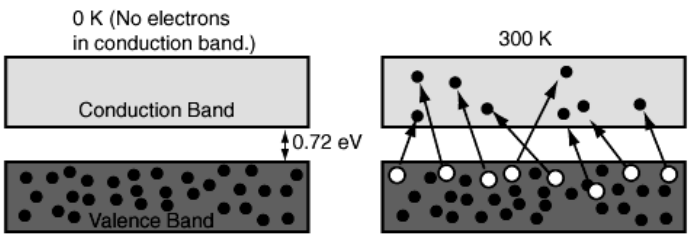
Semiconductors: The materials which have intermediate values of resistivity (of the order of $10^{2}\ \Omega - m$) called semiconductor materials. The energy gap between the valance and conduction band is very small. The two most important are germanium and silicon.



The semiconducting materials have negative temperature coefficient of resistivity. At low temperatures, the valence band is completely filled and conduction band is completely empty. Thus the semiconducting materials behave like insulator at low temperatures.

At comparatively higher temperature, the electrons in valance band acquire sufficient energy to jump in conduction band. As the temperature increases, the probability of the electrons to jump from valance to conduction band increases. Therefore, the conductivity of semiconductors increases with increase in temperature.

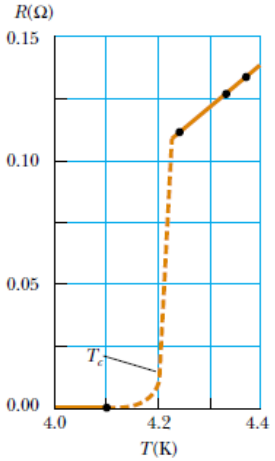
The probability of the electrons to jump from valance band to conduction band depends upon the energy distribution factor $e^{\Delta E/kT}$. Here ΔE is the energy gap, k is the boltzman constant and T is the absolute temperature.



SUPERCONDUCTORS

There are some materials whose resistivity becomes zero below a certain critical temperature T_C called critical temperature. below the critical temperature, such materials are called super conductors. Once the resistance of material drops to zero, no energy is dissipated in the material during the flow of current through material. The current established in such materials will continue to exist indefinitely without the source of emf.

In 1911, a Dutch Physicist K. Onnes observed that below 4.2 K, mercury lost its resistivity and became perfect conductor. The resistance versus temperature graph of for a sample of mercury (Hg) is shown in the figure below



The resistance versus temperature graph shows that the electrical resistance of mercury (Hg) drops to zero below the critical temperature $T_C = 4.2\ K$.

In 1986, a series of ceramic materials were discovered which have comparatively high critical temperature of $90\ K$. The research is in progress to discover such materials which show superconducting behavior at room temperature. the interest in the field of superconductivity is due to its following application

- The energy can be stored and transported without any resistive loss.
- Superconducting electromagnets can large magnetic field in surrounding space.
- Superconducting components in electronic circuits would generate no joule heating and will permit further miniaturization of the circuits

Important Note:

The best known conductors at room temperature don't show any show any superconductivity at all.

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DC CIRCUITS

In this chapter, the study the behavior of specific electric circuits that include resistive elements which may be individual resistors or may be the internal resistance of circuit elements such as batteries or wires

We will confine out study in this chapter to direct current (DC) circuits in which the direction of the current doesn't change with time. In DC circuit, that contain only batteries and resistors, the magnitude of the current doesn't vary with time. While in DC circuits containing capacitors, the magnitude of current may be time dependent.

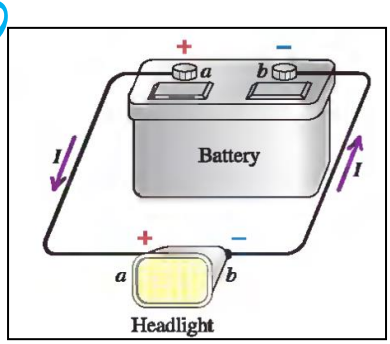
33.1 Electromotive Force

The amount of energy supplied per unit charge in order to move it in a circuit is called electromotive force. The electromotive force is not actually a force i.e., we don't measure it in newton N. The name originates from the early history of the subject. The unit of emf is joule/coulomb, which is volt:

1 volt = 1 joule/coulomb

33.2 Source of Electromotive Force

The devices that provide emf in electrical circuits, to sustain steady current flow in the circuit, are called sources of emf. These sources provide energy to charge carries to make the steady flow of current possible. Source of current converts some non-electrical energy (chemical, mechanical, heat or solar energy) to electrical energy.

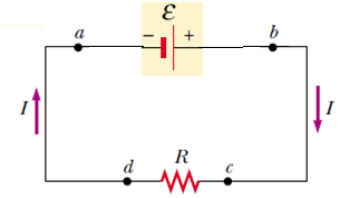


Consider a source of electromotive force $\mathcal{E}$ is connected to a resistor R as shown in the figure. The source of emf maintains its one terminal at a high potential and its other terminal at low potential, as indicated by the + and – signs. Therefore, the emf of the battery would cause positive charge carriers to move in the external circuit as shown by the arrows marked I.

In its interior, the source of emf acts to move positive charges from the point of low potential to the point of high potential. The charges then move through the external circuit, dissipating energy in the process, and return to the negative terminal, from which the emf raises them to the positive terminal again, and the cycle continues again and again to make the steady flow of current possible.

33.3 Determination of Current in a Single Loop Using Energy Conservation Principle

Consider a single loop circuit consists of one source of emf $\mathcal{E}$ and one resistor R as shown in the figure. In time dt a charge $dq(= Idt)$ moves through the source of emf. The energy supplied by the source of emf to the charge carries can be find out by expression:



$$Electromotive\ Force = \frac{Energy\ Supplied}{Charge}$$

$$\Longrightarrow Energy\ Supplied = (Electromotive\ Force)(Charge)$$

$$\Longrightarrow Energy\ Supplied = (\mathcal{E})(dq) = \mathcal{E}\, I\, dt$$

The energy supplied by the battery is dissipated in the resistor R. The total energy dissipated is determined as:

$$Power\ Dissipated = \frac{Energy\ Dissipated}{Time}$$

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$$\Longrightarrow Energy\ Dissipated = (Power\ Dissipated)(Time)$$

$$\Longrightarrow Energy\ Dissipated = (I^2R)(dt) = I^2R\ dt$$

From the conservation of energy principle, the work done by the source must equal to the internal energy dissipated in the resistor, i.e.,

$$\mathcal{E}\ I\ dt = I^2R\ dt$$

Solving for I, we obtain

$$I = \frac{\mathcal{E}}{R}$$

This is the expression of electric current moving through a single loop circuit.

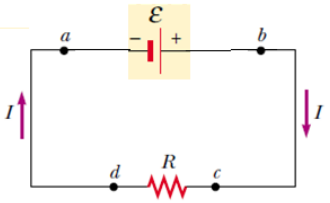
33.4 Kirchhoff's Voltage Rule

Kirchhoff's Voltage rule is a particular way of stating the law of conservation of energy for a charge carrier travelling in a closed circuit. The Kirchhoff's second rule is described as:

The algebraic sum of the changes in potential encountered in a complete traversal of any close circuit is zero.

33.5 Determination of Current in a Single Loop Using Kirchhoff Rule

Consider a single loop circuit consists of one source of emf $\mathcal{E}$ and one resistor R as shown in the figure. Let a steady current I flow through the loop. Applying Kirchhoff's voltage rule, we have:



$$\mathcal{E} - IR = 0$$

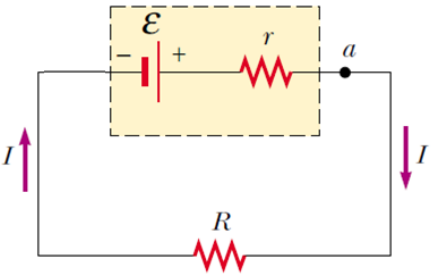
$$\Longrightarrow I = \frac{\mathcal{E}}{R}$$

This expression gives the current flow through a circuit containing single loop.

33.6 Determination of Current in a Single Loop Circuit by Considering the Internal Resistance of a Source of EMF

All the sources of emf have an intrinsic internal resistance, which can't be removed because it is the inherent part of device.

Consider a single loop circuit consists of one source of emf $\mathcal{E}$ having internal resistance r and one resistor R as shown in the figure. Let a steady current I flow through the loop. Applying Kirchhoff's voltage rule, we have:



$$\mathcal{E} - Ir - IR = 0$$

Thus the expression of the current from this single loop of current becomes:

$$\Longrightarrow I = \frac{\mathcal{E}}{R+r}$$

Note that the internal resistance r reduces the current that the emf can supply to the external circuit.

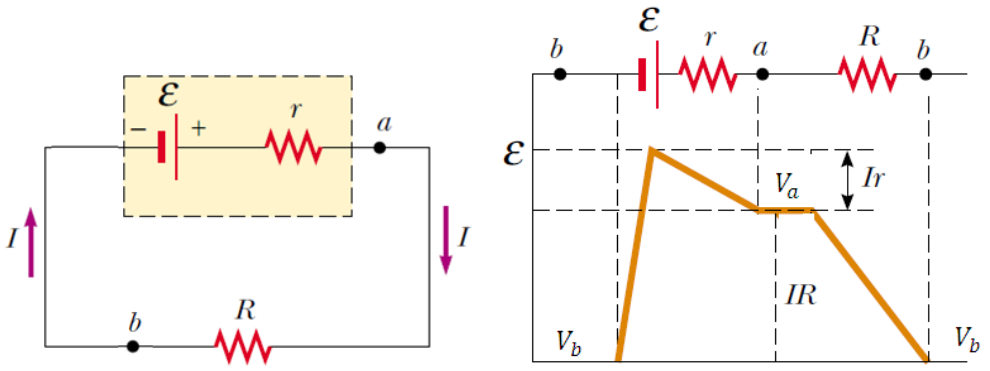
ہمیں اس کار خیر کو بہتر بنانے کے لئے آپ کی مدد کی ضرورت ہے۔اپنی قیمتی آراءاور نوٹس ای میل کریں یا فیس بک ایڈیس پر میسیج کریں۔

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33.7 DETERMINATION OF POTENTIAL DIFFERENCES BETWEEN THE POINTS OF CIRCUIT

Consider a circuit which consists of a resistor R and a source of emf $\mathcal{E}$ with internal resistance r as shown in the figure below:



We want to find out the expression of potential difference V_{ab} between the two points a and b of an electrical circuit. If V_a and V_b are the potential at points a and b, respectively, than we have

$$V_b + IR = V_a$$
$$\Longrightarrow V_{ab} = V_a - V_b = IR$$
$$\Longrightarrow V_{ab} = IR$$

As for the single loop, given above, we have

$$I = \frac{\mathcal{E}}{R+r}$$

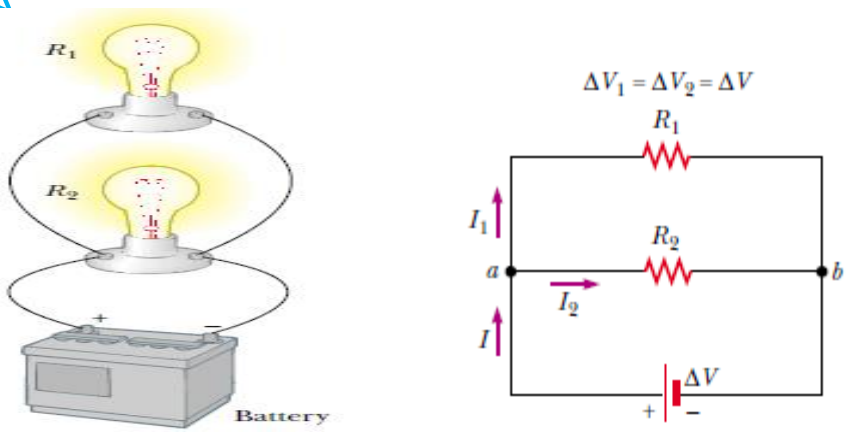
Thus eq. (1) will become:

$$V_{ab} = \left(\frac{\mathcal{E}}{R+r}\right) R$$
$$\Longrightarrow V_{ab} = \mathcal{E}\frac{R}{R+r}$$

This is the expression of electric potential of electric potential difference between the two points of an electrical circuit.

33.8 EQUIVALENT RESISTANCE CONNECTED IN PARALLEL

In parallel arrangement a number of resistors are connected side by side with their ends joined together at a common point as shown in the figure below:



According to the properties of a parallel circuit, the total current is shared among the branches, so

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$$I = I_1 + I_2 \quad \text{---------- (1)}$$

In case of parallel combination of resistors, the potential difference across each resistor is V, therefore the current through each resistor is:

$$I_1 = \frac{V}{R_1} \text{ and } I_2 = \frac{V}{R_2}$$

Thus equation (1) will become:

$$I = \frac{V}{R_1} + \frac{V}{R_2} \quad \text{---------- (2)}$$

If we replace the parallel combination by a single equivalent resistance R_{eq}, the same current I must flow through circuit, i.e.,

$$I = \frac{V}{R_{eq}} \quad \text{---------- (3)}$$

Comparing eq. (1) and (3), we get

$$\frac{V}{R_{eq}} = \frac{V}{R_1} + \frac{V}{R_2}$$

$$\frac{V}{R_{eq}} = V\left(\frac{1}{R_1} + \frac{1}{R_2}\right)$$

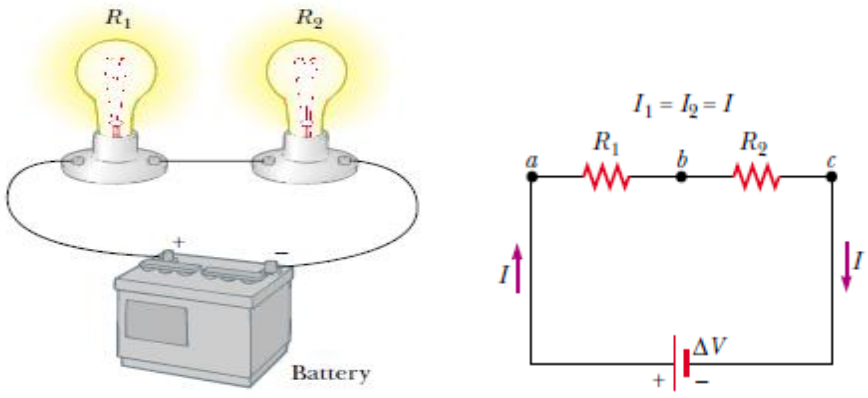
or $$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$$

The general expression for the equivalent resistance of a parallel combination of any number of resistor is:

$$\frac{1}{R_{eq}} = \sum_n \frac{1}{R_n}$$

33.9 RESISTORS CONNECTED IN SERIES

If the resistors are connected end to end such that the same current passes through all of them, they are said to be connected in series as shown in the figure below:



Suppose a battery of potential difference V is connected with the series combination of resistors such that the current I passes through each resistor. The potential difference across each resistor is:

$$V_1 = IR_1 \text{ and } V_2 = IR_2$$

The sum of potential drop across each resistor must be equal to the potential difference supplied by the battery, i.e.,

$$V = V_1 + V_2 \quad \text{---------- (1)}$$

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If we replace the combination of resistances with the equivalent resistance, the same current would be established and so

$$V = IR_{eq}$$

Thus, the equation (1) will become

$$IR_{eq} = IR_1 + IR_2$$

$$R_{eq} = R_1 + R_2$$

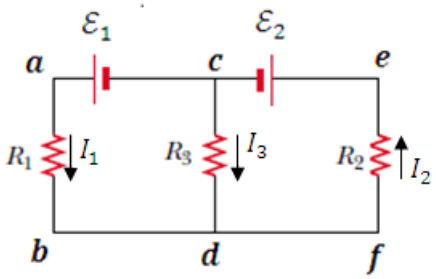
Expending this result to a seires combination of any number of resistors, we obtain:

$$R_{eq} = \sum_n R_n$$

33.10 MULTILOOP CIRCUIT

The circuits having more than one loop are called multiloop circuit.

Consider a complex network consisting of three resistors circuit consist of R_1, R_2 and R_3 and two batteries of emf $\mathcal{E}_1$ and $\mathcal{E}_2$. We want to find out the unknown currents I_1, I_2 and I_3 move through the circuit, as shown in the figure below:



At junction d, the total current entering the junction is $I_1 + I_3$, and the current moving away from junction d is I_2. Thus according to the Kirchoff's 1st rule

$$I_1 + I_3 = I_2$$

$$I_1 = I_2 - I_3 \quad \text{---------- (1)}$$

Applying the Kirchoff's 2nd rule on the loop abcd, we get

$$\mathcal{E}_1 - I_1R_1 + I_3R_3 = 0 \quad \text{---------- (2)}$$

$$\Longrightarrow I_3 = \frac{I_1R_1 - \mathcal{E}_1}{R_3} \quad \text{---------- (3)}$$

Using the kirchoff rule for the loop cdef:

$$-\mathcal{E}_2 - I_3R_3 - I_2R_2 = 0 \quad \text{---------- (4)}$$

Putting value of I_3 in eqution (3), we get

$$-\mathcal{E}_2 - \left(\frac{I_1R_1 - \mathcal{E}_1}{R_3}\right)R_3 - I_2R_2 = 0$$

$$-\mathcal{E}_2 - I_1R_1 + \mathcal{E}_1 - I_2R_2 = 0$$

$$\Longrightarrow I_2 = \frac{\mathcal{E}_1 - \mathcal{E}_2 - I_1R_1}{R_2} \quad \text{---------- (5)}$$

Putting the values of I_2 and I_3 from equation (4) and (5) in equation (1), we get

$$I_1 = I_2 - I_3$$

$$I_1 = \left(\frac{\mathcal{E}_1 - \mathcal{E}_2 - I_1R_1}{R_2}\right) - \left(\frac{I_1R_1 - \mathcal{E}_1}{R_3}\right)$$

$$I_1 = \left(\frac{\mathcal{E}_1 - \mathcal{E}_2 - I_1R_1}{R_2}\right) - \left(\frac{I_1R_1 - \mathcal{E}_1}{R_3}\right)$$

$$I_1 = \frac{R_3(\mathcal{E}_1 - \mathcal{E}_2 - I_1R_1) - R_2(I_1R_1 - \mathcal{E}_1)}{R_2R_3}$$

$$I_1 = \frac{\mathcal{E}_1(R_2 + R_3) - \mathcal{E}_2R_3}{R_1R_2 + R_2R_3 + R_3R_1} \quad \text{---------- (6)}$$

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Now adding equation (2) and (4), we obtain:

$$\mathcal{E}_1 - I_1R_1 - I_2R_2 - \mathcal{E}_2 = 0$$

$$I_1R_1 = \mathcal{E}_1 - I_2R_2 - \mathcal{E}_2$$

$$I_1 = \frac{\mathcal{E}_1 - I_2R_2 - \mathcal{E}_2}{R_1} \quad \text{---------- (7)}$$

From equation (4)

$$I_3 = \frac{-\mathcal{E}_2 - I_2R_2}{R_3} \quad \text{---------- (8)}$$

Putting value of I_1 and I_3 from equation (7) and (8) in (1), we get

$$I_2 = I_1 + I_3$$

$$I_2 = \left(\frac{\mathcal{E}_1 - I_2R_2 - \mathcal{E}_2}{R_1}\right) + \left(\frac{-\mathcal{E}_2 - I_2R_2}{R_3}\right)$$

Solving this equation, we get

$$I_2 = \frac{\mathcal{E}_1 R_3 - \mathcal{E}_2 (R_1 + R_3)}{R_1R_2 + R_2R_3 + R_3R_1} \quad \text{---------- (9)}$$

Putting value of I_1 and I_2 from equation (6) and (9) in (1), we get

$$I_3 = I_2 - I_1$$

$$I_3 = \left(\frac{\mathcal{E}_1 R_3 - \mathcal{E}_2 (R_1 + R_3)}{R_1R_2 + R_2R_3 + R_3R_1}\right) - \left(\frac{\mathcal{E}_1 (R_2 + R_3) - \mathcal{E}_2 R_3}{R_1R_2 + R_2R_3 + R_3R_1}\right)$$

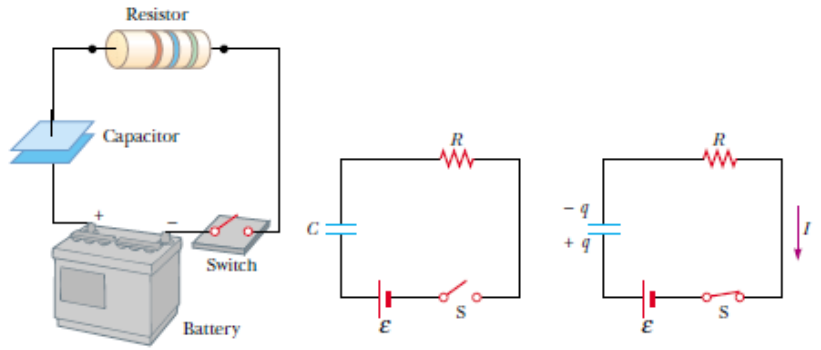
Solving the above equation, we get

$$I_3 = \frac{-\mathcal{E}_1 R_2 - \mathcal{E}_2 R_1}{R_1R_2 + R_2R_3 + R_3R_1} \quad \text{---------- (10)}$$

Thus equation (6), (9) and (10) are the required expression for unknown currents I_1, I_2 and I_3 in a circuit.

33.11 GROWTH OF CHARGE IN AN RC CIRCUIT

A circuit containing a series combination of a resistor and a capacitor is called an RC circuit. Consider an RC circuit in series with the battery of emf $\mathcal{E}$ as shown in the figure below:



When the switch S is closed, the growth of charge starts in capacitor. by using the Kirchoff 2nd rule, we get:

$$\mathcal{E} - V_R - V_C = 0$$

$$\Longrightarrow \mathcal{E} = V_R + V_C \quad \text{---------- (1)}$$

Since R and C are in series, therefore

$$V_R = IR \text{ and } V_C = \frac{q}{C}$$

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Putting values in equation (9), we get

$$\mathcal{E} = IR + \frac{q}{C} \qquad \text{---------- (2)}$$

where q and I denote the current and charge at any time t.

Now as $I = \frac{dq}{dt}$, therefore, equation (1) becomes

$$\mathcal{E} = \frac{dq}{dt}R + \frac{q}{C}$$

$$\Rightarrow \mathcal{E}C = \frac{dq}{dt}RC + q$$

$$\Rightarrow \frac{dq}{\mathcal{E}C - q} = \frac{1}{RC}dt$$

$$\Rightarrow -\frac{dq}{\mathcal{E}C - q} = -\frac{1}{RC}dt$$

Integrating, we get

$$\Rightarrow \int \frac{-dq}{\mathcal{E}C - q} = -\frac{1}{RC}\int dt$$

$$\Rightarrow \ln(\mathcal{E}C - q) = -\frac{1}{RC}t + A \qquad \text{---------- (3)}$$

Where A is the constant of integration. To find A, we makes use of initial conditions:

At $t = 0; q = 0$, we have

$$\ln(\mathcal{E}C) = A$$

The equation (3) will become:

$$\ln(\mathcal{E}C - q) = -\frac{1}{RC}t + \ln(\mathcal{E}C)$$

$$\Rightarrow \ln(\mathcal{E}C - q) - \ln(\mathcal{E}C) = -\frac{1}{RC}t$$

$$\Rightarrow \ln\left(\frac{\mathcal{E}C - q}{\mathcal{E}C}\right) = -\frac{1}{RC}t$$

$$\Rightarrow \frac{\mathcal{E}C - q}{\mathcal{E}C} = e^{-\frac{1}{RC}t}$$

$$\Rightarrow \mathcal{E}C - q = \mathcal{E}Ce^{-\frac{1}{RC}t}$$

Or

$$q = \mathcal{E}C - \mathcal{E}Ce^{-\frac{1}{RC}t}$$

$$q = \mathcal{E}C - \mathcal{E}Ce^{-\frac{1}{RC}t}$$

$$q = \mathcal{E}C\left(1 - e^{-\frac{1}{RC}t}\right)$$

This equation shows that at $t = \infty; q = \mathcal{E}C = q_0$, where q_0 is the maximum value of charge on the capacitor. therefore

$$q = q_0\left(1 - e^{-\frac{1}{RC}t}\right) \qquad \text{---------- (4)}$$

This equation gives the growth of charge in an RC circuit. The equation shows that the charge q goes on increasing and ultimately attains the maximum value q_0 after a long time.

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Capacitive time constant

In the equation (4), the factor RC has the dimensions of time and is called capacitive time constant. It is denoted by τ_C.

The equation (4) is written as:

$$q = q_0\left(1 - e^{-\frac{t}{\tau_C}}\right)$$

Special Cases:

- At $t = 0; q = 0$
- At $t = \infty; q = q_0 = \mathcal{E}C$
- At $t = \tau_C; q = q_0\left(1 - e^{-\frac{\tau_C}{\tau_C}}\right)$

$$q = q_0(1 - e^{-1}) = q_0\left(1 - \frac{1}{e}\right) = q_0(1 - 0.37)$$

$$q = 0.63\, q_0$$

So the capacitance time constant is the time after which the charge on the capacitor grows to 63% of its maximum value.

Sample Problem 7: A resistor $R = 6.2\ M\Omega$ and a capacitor $C = 2.4\ \mu F$ are connected in series and a 12 V battery of negligible internal resistance is connected across their combination. (a) What is capacitive time constant of this circuit. (b) At what time after the battery is connected does the potential difference across the capacitor equal to 5.6 V.

Solution:

$$R = 6.2\ M\Omega = 6.2 \times 10^6 \Omega$$

$$C = 2.4\ \mu F = 2.4 \times 10^{-6} F$$

$$\varepsilon = 12\ V$$

(a) Capacitive time constant τ =?

$$\tau = RC = 6.2 \times 10^6 \times 2.4 \times 10^{-6} = 15\ s$$

(b) t =? when potential across capacitor is 5.6 V

As $q = C\varepsilon\left(1 - e^{-\frac{t}{\tau}}\right)$

$$\Rightarrow \frac{q}{C} = \varepsilon\left(1 - e^{-t/\tau}\right) \Rightarrow V = \varepsilon\left(1 - e^{-t/\tau}\right)$$

$$\Rightarrow \frac{V}{\varepsilon} = \left(1 - e^{-\frac{t}{\tau}}\right) \Rightarrow e^{-\frac{t}{\tau}} = 1 - \frac{V}{\varepsilon}$$

$$\Rightarrow -\frac{t}{\tau} = \ln\left(1 - \frac{V}{\varepsilon}\right)$$

$$\Rightarrow t = -\tau \ln\left(1 - \frac{V}{\varepsilon}\right)$$

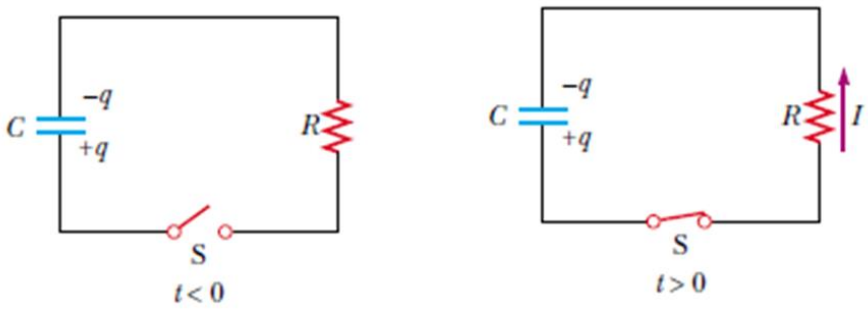
$$\Rightarrow t = -15 \ln\left(1 - \frac{5.6}{12}\right) = 9.4\ \text{s}$$

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33.12 DECAY OF CHARGE IN AN RC CIRCUIT

Consider the circuit which consists of a capacitor carrying an initial charge q, a resistor R, and a switch S as shown in figure below:



When the switch is open, a potential difference $V_C = \frac{q}{C}$ exists across the capacitor and $V_R = 0$ because $I = 0$. If the switch is closed at $t = 0$, the capacitor begins to discharge through the resistor. Applying the loop rule:

$$-V_R - V_C = 0$$

$$\because V_R = IR \text{ and } V_C = \frac{q}{C}$$

$$-IR - \frac{q}{C} = 0$$

$$IR = -\frac{q}{C} \Rightarrow I = -\frac{q}{RC}$$

$$\because I = \frac{dq}{dt}$$

$$\frac{dq}{dt} = -\frac{q}{RC}$$

$$\Rightarrow \frac{dq}{q} = -\frac{1}{RC}dt$$

Integrating both sides, we obtain

$$\int \frac{dq}{q} = -\frac{1}{RC}\int dt$$

$$\ln q = -\frac{1}{RC}t + A \qquad \text{---------- (1)}$$

By applying the initial conditions, i.e., at $t = 0$; $q = q_0$, we get:

$$A = \ln q_0$$

The equation (1) implies:

$$\ln q = -\frac{1}{RC}t + \ln q_0$$

$$\Rightarrow \ln q - \ln q_0 = -\frac{t}{RC}$$

$$\Rightarrow \ln \frac{q}{q_0} = -\frac{t}{RC}$$

$$\Rightarrow \frac{q}{q_0} = e^{-\frac{t}{RC}}$$

$$\Rightarrow q = q_0 e^{-\frac{t}{RC}} \qquad \text{---------- (2)}$$

This is the expression for the decay of charge of capacitor in an RC circuit.

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Capacitive Time Constant

The factor $RC = \tau_C$ is called the capacitive time constant. The equation (2) will become: $q = q_0 e^{-\frac{t}{\tau_C}}$

Special Cases:

- At $t = 0; q = q_0$
- At $t = \infty; q = 0$
- At $t = \tau_C; q = q_0 e^{-\frac{\tau_C}{\tau_C}} = q_0 e^{-1} = \frac{q_0}{e} = 0.37\, q_0$

So after time τ_C, the charge of capacitor reduces to 37% of theoratical value.

Sample Problem 8. A capacitor C discharge through a resistor R. (a) After how many time constant does its charge fall to one half of its initial value? (b) After how many time constants does the stored energy drop to half its initial value.

Solution:

(a) As we know $q = q_0 e^{-t/\tau}$

As we want to find out the time in which $= \frac{q_0}{2}$. Therefore,

$$\frac{q_0}{2} = q_0 e^{-\frac{t}{\tau}} \Longrightarrow \frac{1}{2} = e^{-\frac{t}{\tau}}$$

$$\Longrightarrow -\frac{t}{\tau} = \ln\left(\frac{1}{2}\right) \Longrightarrow \frac{t}{\tau} = \ln(2)$$

$$\Longrightarrow t = \ln(2)\tau = 0.693\,\tau$$

(b) As $U = \frac{q^2}{C}$

Also $q = q_0 e^{-\frac{t}{\tau}} \Longrightarrow q^2 = (q_0)^2 \left(e^{-\frac{t}{\tau}}\right)^2$

$$\Longrightarrow q^2 = q_0^2 e^{-2t/\tau} \Longrightarrow \frac{q^2}{C} = \frac{q_0^2}{C} e^{-2t/\tau}$$

$$\Longrightarrow U = U_0 e^{-2t/\tau}$$

We want to find out the time in which $= \frac{U_0}{2}$. Therefore,

$$\frac{U_0}{2} = U_0 e^{-\frac{2t}{\tau}}$$

$$\Longrightarrow \frac{1}{2} = e^{-\frac{2t}{\tau}} \Longrightarrow -\frac{2t}{\tau} = \ln\left(\frac{1}{2}\right)$$

$$\Longrightarrow t = -\frac{\tau}{2}\ln\left(\frac{1}{2}\right)$$

$$\Longrightarrow t = \frac{\tau}{2}\ln(2)$$

$$\Longrightarrow t = \frac{\tau}{2}(0.693) = 0.35\,\tau$$

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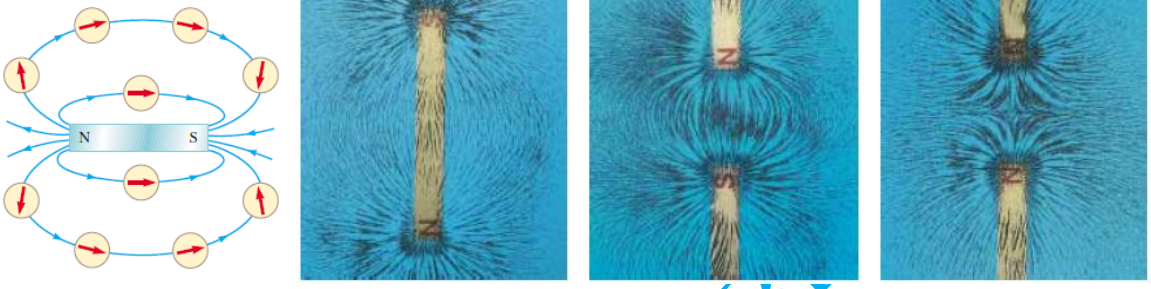
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MAGNETIC FIELD EFFECTS

34.1 Magnetic Field

Magnetic field is the region or space around any charge within which its influence can be felt by other magnetic substances.

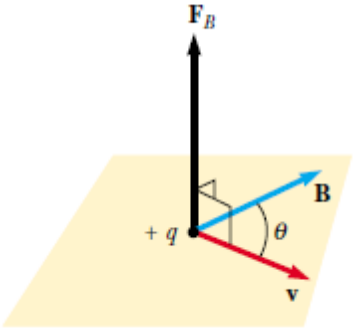
The magnetic field around any magnet is considered as closely spaced magnetic field lines. The magnetic field lines of a bar magnet can be traced with the aid of a compass as shown in the figure below:



In addition to a bar magnet, a moving charge or a current creates a magnetic field in the surrounding space (in addition to its electric field). The magnetic field exerts a force F on any other moving charge or current that is present in the field.

34.2 Magnetic Force on a Charged Particle

Consider a charged object having charge q is projected a uniform magnetic field of flux density **B** with velocity **v**. The magnetic force **F** acting on the object can be expressed as:



$$\mathbf{F} = q(\mathbf{v} \times \mathbf{B})$$

$$\mathbf{F} = qvB \sin\theta\, \hat{n} \qquad \text{---------- (1)}$$

Here $\hat{n}$ is the unit vector that is perpendicular to the plane of **v** and **B**, and is used to describe the direction of magnetic force **F** on the charged object.

It is clear from equation (1) that the maximum magnetic force will act on the charged object when it will be projected perpendicular to the magnetic field. The maximum magnetic force on the charged object will be

$$F = qvB$$

$$\Longrightarrow B = \frac{F}{qv}$$

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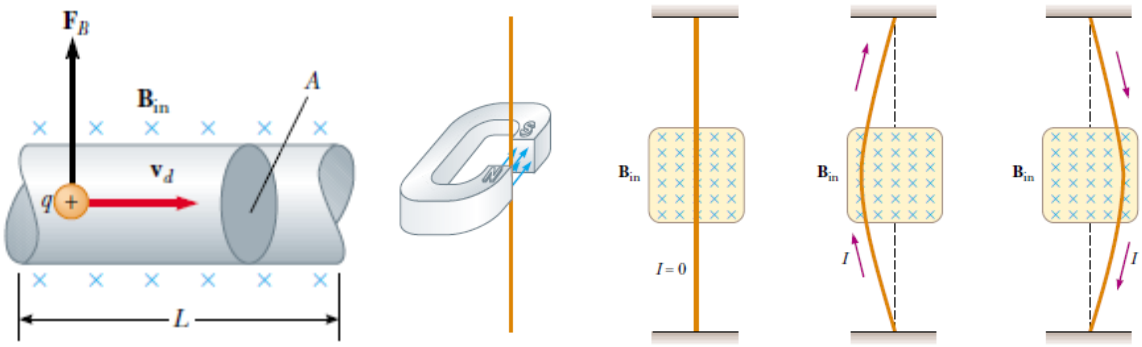
B is also called the magnetic field induction which can be defined as:

"*The force on a unit positive charge moving perpendicular to the magnetic field with uniform velocity*". The SI unit of magnetic field induction **B** is tesla, while in cgs system of units **B** is measured in gauss.

$$1\ tesla\ =\ 10^4\ gauss$$

34.3 Magnetic Force on a Current Carrying Conductor

Consider a current carrying wire of length L and cross-sectional area A, is placed in a uniform magnetic field of flux density **B** as shown in the figure below:



If n is the number of free charges per unit volume of a conductor (each having charge e), then the total charge flowing through the wire is $q = nAle$.

Suppose the charges are moving with drift velocity $\mathbf{v_d}$ and cover length L in t seconds, then

$$t = \frac{L}{v_d}$$

If I is the current flowing through the conductor, then

$$I = \frac{q}{t} = \frac{nAle}{L/v_d} = nAev_d$$

$$\Longrightarrow v_d = \frac{I}{nAe}$$

Now

Force on one charge $F' = evB_\perp$

Force on nAL charges $F = nALF' = nALevB_\perp$

Putting value of v_d, we get

$$F = nALe\left(\frac{I}{nAe}\right)B_\perp = ILB_\perp$$

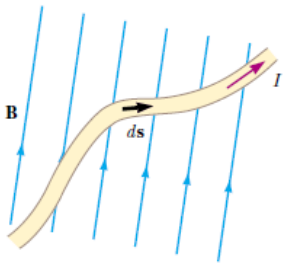
In vector form:

$$\mathbf{F} = I(\mathbf{L} \times \mathbf{B})$$

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The direction of F will be perpendicular to the plane of **L** and **B**.



If the wire is not straight or field is not uniform, then we divide the wire into small elements of length '**ds**'. Then the force on each segment is written as:

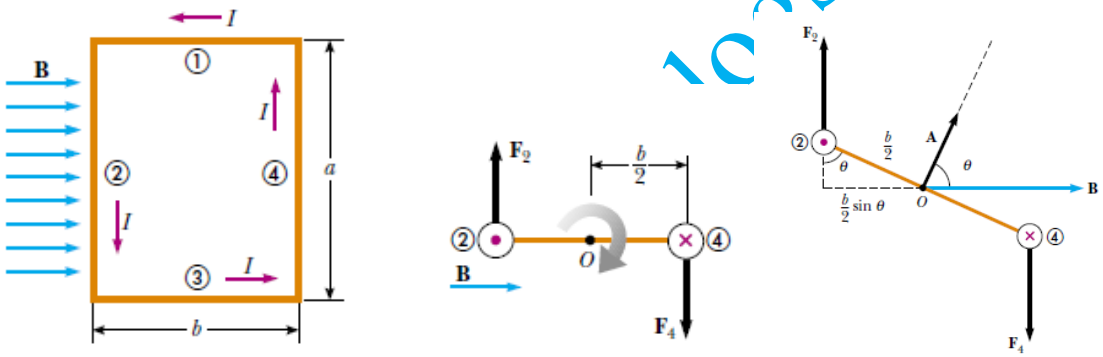
$$d\mathbf{F} = I(d\mathbf{s} \times \mathbf{B})$$

Then the force on whole wire is obtained by integrating over the whole length L.

$$\mathbf{F} = \int d\mathbf{F} = \int I(d\mathbf{s} \times \mathbf{B})$$

34.4 Torque on a Current Loop in Magnetic Field

Consider a loop of wire carrying current suspended in a uniform magnetic field of flux density **B** as shown in the figure below:



We want to find out the expression of torque produced in current carrying coil due to the effect of magnetic field.

The side 1 and 3 of coil are oriented parallel, while the sides 2 and 4 are held perpendicular to the magnetic field. No magnetic forces act on sides 1 and 3 because these wires are parallel to the field; hence $\mathbf{L} \times \mathbf{B} = 0$, for these sides. However, magnetic forces do act on sides 2 and 4 because these sides are oriented perpendicular to the field. The magnitude of these forces is

$$F_2 = F_4 = F = ILB \sin\theta = IaB \sin 90^\circ = IaB \qquad \because L = a$$

The direction of F_2, the magnetic force exerted on wire 2, is out of the page and that of F_4, the magnetic force exerted on wire 4, is into the page. These two anti parallel forces are separated by a small distance form a couple. The couple tends to rotate the loop about point O. The magnitude of torque of this couple will be:

$$Torque = (Force)\,(Perpendicular\ Distance\ between\ line\ of\ action\ of\ Force)$$

$$\tau = (F)(b \sin\theta) = (IaB)(b \sin\theta) = IabB \sin\theta$$

where θ is the angle between the magnetic field **B** and vector area **A** of loop.

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$\tau = IAB \sin\theta$ $\because A = ab = area\ of\ loop$

This is the expression of torque produced in a loop with single turn. For a loop with N turns, the torque will be:

$\tau = NIAB \sin\theta$

This result shows that the torque has its maximum value IAB when the field is perpendicular to the normal to the plane of the loop ($\theta = 90°$), and is zero when the field is parallel to the normal to the plane of the loop ($\theta = 0°$).

In vector form:

$\boldsymbol{\tau} = NI(\mathbf{A} \times \mathbf{B})$

This is the expression of the torque on a current carrying loop in a magnetic field.

34.5 The Magnetic Dipole

The electric dipole placed in an electric field will experience a torque, which is expressed as:

$\boldsymbol{\tau} = \mathbf{p} \times \mathbf{E}$

where $p = qd$ is the electric dipole moment. The magnitude of the torque produce in an electric dipole in an electric field is expressed as:

$\tau = pE \sin\theta$

where θ is the angle between **p** and **E**.

Similarly, the expression of torque acting on a magnetic dipole placed in a magnetic field of flux density **B** is described as:

$\boldsymbol{\tau} = NI(\mathbf{A} \times \mathbf{B})$

The magnitude of the torque is

$\tau = NIAB \sin\theta$ ---------- (1)

By analogy with the electrical case, we define a vector **μ**, the magnetic dipole moment, to have the magnitude:

$\mu = NIA$ ---------- (2)

Thus, the equation (1) becomes

$\tau = \mu B \sin\theta$

In vector form,

$\boldsymbol{\tau} = \boldsymbol{\mu} \times \mathbf{B}$ ---------- (3)

Equation (3) gives the torque on a current carrying loop in a magnetic field of flux density **B**.

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The magnetic dipole consist of two opposite magnetic poles (north and south), which are separated by a small distance. And the magnetic dipole moment is a vector associated with a magnet or a current loop, whose cross product with magnetic field strength is equal to the torque exerted on the system by the field.

The work done on a magnetic dipole to change its orientation in the magnetic field is stored as potential energy of magnetic dipole which is given as:

$$P.E = \text{Work Done}$$

$$P.E = \int \tau \, d\theta$$

$$= \int \mu B \sin\theta \, d\theta$$

$$= \mu B \int \sin\theta \, d\theta$$

$$= \mu B(-\cos\theta)$$

$$= -\mu B \cos\theta$$

$$\Longrightarrow P.E = -\boldsymbol{\mu}.\mathbf{B} \quad \text{---------- (4)}$$

This is the expression for P.E. of magnetic dipole.

The equation (4) tells us that the unit of magnetic dipole moment is obtained by dividing by energy unit by the unit of magnetic induction. Therefore,

Unit of Dipole Moment $\mu = \frac{joule}{tesla} = \frac{J}{T} = JT^{-1}$

The other unit of dipole moment is described by using the equation (2):

$$\mu = NIA$$

Unit of Dipole Moment $\mu = (ampere) \times (meter)^2 = A - m^2$

Question. Prove that $\frac{\boldsymbol{joule}}{\boldsymbol{tesla}} = (\boldsymbol{ampere}) \times (\boldsymbol{meter})^2$

$$\text{L. H. S.} = \frac{joule}{tesla} = \frac{(newton) \times (meter)}{tesla}$$

$$\because work\ W = (force\ F) \times (displacement\ d)$$

$$= \frac{(ampere \times meter \times tesla) \times (meter)}{tesla}$$

$$\because force\ F = (current\ I) \times (length\ L) \times (magnetic\ induction\ B)$$

$$= \frac{ampere \times (meter)^2 \times tesla}{tesla} = ampere \times (meter)^2 = \text{R. H. S.}$$

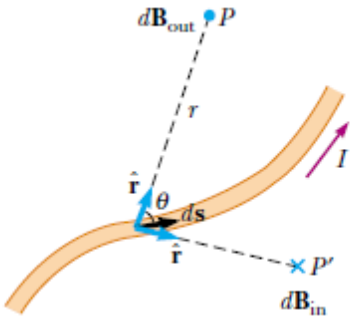
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AMPERE'S LAW

35.1 The Biot–Savart Law

Shortly after Oersted's discovery in 1819 that a compass needle is deflected by a current-carrying conductor, Biot (1774–1862) and Savart (1791–1841) performed quantitative experiments on the force exerted by an electric current on a nearby magnet. The experimental observations of Biot and Savart about the magnetic field produced by a current carrying conductor at some point in space are as follows:



- The vector $d\mathbf{B}$ is perpendicular both to $d\mathbf{s}$ (which points in the direction of the current) and to the unit vector $\hat{\mathbf{r}}$ directed from $d\mathbf{s}$ toward P.
- The magnitude of $d\mathbf{B}$ is inversely proportional to r^2, where r is the distance from $d\mathbf{s}$ to P.
- The magnitude of $d\mathbf{B}$ is proportional to the current and to the magnitude $|d\mathbf{s}|$ of the length element $d\mathbf{s}$.
- The magnitude of $d\mathbf{B}$ is proportional to $\sin\theta$, where θ is the angle between the vectors $d\mathbf{s}$ and $\hat{\mathbf{r}}$.

These observations are summarized in the mathematical expression known today as the Biot–Savart law:

$$|d\mathbf{B}| \ \alpha \ \frac{Id\mathbf{s}\sin\theta}{r^2}$$

$$\Longrightarrow |d\mathbf{B}| = \frac{\mu_0}{4\pi}\frac{I\,ds\sin\theta}{r^2}$$

Where $\frac{\mu_0}{4\pi}$ is the constant of proportionality and $\mu_0 = 4\pi \times 10^{-7}\ \frac{Wb}{A-m}$ is called the permeability of free space.

In vector form:

$$d\mathbf{B} = \frac{\mu_0}{4\pi}\frac{I\,ds\sin\theta}{r^2}\,\hat{\mathbf{n}} \qquad \text{---------- (1)}$$

Where $\hat{\mathbf{n}}$ is the unit vector showing the direction of field and is determined by right band rule.

If $\hat{\mathbf{i}}$ is a unit vector along current element and $\hat{\mathbf{r}}$ is the unit vector along position vector $\mathbf{r}$ of point P, then

$$\hat{\mathbf{i}}\times\hat{\mathbf{r}} = |\hat{\mathbf{i}}||\hat{\mathbf{r}}|\sin\theta\,\hat{\mathbf{n}} = (1)(1)\sin\theta\,\hat{\mathbf{n}} = \sin\theta\,\hat{\mathbf{n}}$$

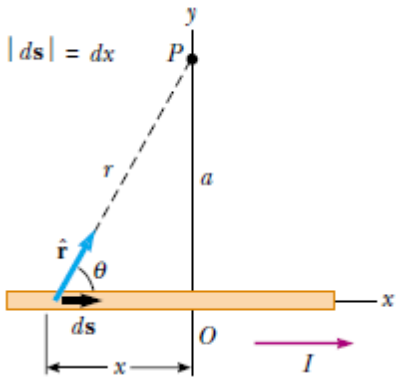
The equation (1) will become:

$$d\mathbf{B} = \frac{\mu_0}{4\pi}\frac{I\,ds\sin\theta}{r^2}\,\hat{\mathbf{n}} = \frac{\mu_0}{4\pi}\frac{I\,ds\,(\hat{\mathbf{i}}\times\hat{\mathbf{r}})}{r^2} = \frac{\mu_0}{4\pi}\frac{I\,ds\,\hat{\mathbf{i}}\times\hat{\mathbf{r}}}{r^2}$$

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As $\hat{\mathbf{i}}$ is a unit vector along current element, therefore, $d\mathbf{s} = ds\,\hat{\mathbf{i}}$



$$d\mathbf{B} = \frac{\mu_0}{4\pi}\frac{I\,d\mathbf{s}\times\hat{\mathbf{r}}}{r^2}$$

$$\Rightarrow d\mathbf{B} = \frac{\mu_0}{4\pi}\frac{I\,d\mathbf{s}\times\left(\frac{\mathbf{r}}{r}\right)}{r^2} \qquad \because\ \hat{\mathbf{r}} = \frac{\mathbf{r}}{r}$$

$$\Rightarrow d\mathbf{B} = \frac{\mu_0}{4\pi}\frac{I\,d\mathbf{s}\times\mathbf{r}}{r^3} \qquad \text{---------- (2)}$$

The magnetic induction **B** at point *P* due to whole wire is obtained by integrating eq. (2):

$$\mathbf{B} = \int d\mathbf{B} = \frac{\mu_0}{4\pi}\int\frac{I\,d\mathbf{s}\times\mathbf{r}}{r^3}$$

This is known as Biot–Savart law.

35.2 Application of Biot–Savart law

35.2.1 Magnetic Field due to Current in a Straight Conductor

Consider a long straight conductor carries current I as shown in the figure below:

We want to find out magnetic field strength at point P due to this current carrying conductor.

The perpendicular distance of point P from the wire is 'a'.

For this we consider a small length element ds, which is at the distance x from point O (taken as the origin). The magnetic field strength due to length element ds by Biot–Savart law will be:

$$dB = \frac{\mu_0}{4\pi}\frac{I\,ds\sin\theta}{r^2} \qquad \text{---------- (1)}$$

$$\because\ r^2 = x^2 + a^2 \ \ and \ \ ds = dx$$

Thus equation (1) will become

$$dB = \frac{\mu_0}{4\pi}\frac{I\,dx\sin\theta}{x^2+a^2} \qquad \text{---------- (2)}$$

From figure:

$$\tan\theta = \frac{a}{-x} \Rightarrow x = -\frac{a}{\tan\theta}$$

$\because$ negative sign is necessary because *d*s is located at a negative value of x

$$\Rightarrow x = -a\cot\theta$$

And $dx = a\csc^2\theta\ d\theta$

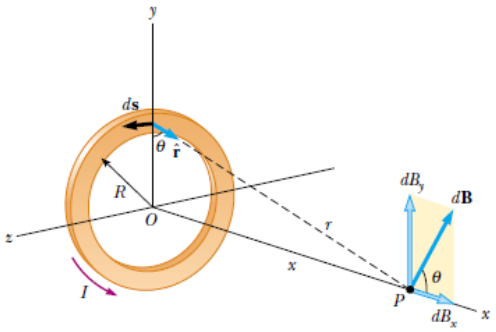
The equation (2) will become:

$$dB = \frac{\mu_0}{4\pi}\frac{I\left(a\csc^2\theta\,d\theta\right)\sin\theta}{a^2\cot^2\theta + a^2} = \frac{\mu_0}{4\pi}\frac{I\left(a\csc^2\theta\,d\theta\right)\sin\theta}{a^2\left(1+\cot^2\theta\right)} = \frac{\mu_0}{4\pi}\frac{I\left(a\csc^2\theta\,d\theta\right)\sin\theta}{a^2\csc^2\theta}$$

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$$dB = \frac{\mu_0}{4\pi} \frac{I \sin\theta d\theta}{a} \quad \text{----------} \quad (3)$$



To find out the field due to the whole wire can be obtained by integrating equation (3):

$$B = \int_{-\infty}^{+\infty} dB = \frac{\mu_0}{4\pi} \int_{-\infty}^{+\infty} \frac{I \sin\theta d\theta}{a} =$$

$$\frac{\mu_0 I}{4\pi a} \int_{-\infty}^{+\infty} \sin\theta \, d\theta$$

As $x = -a\cot\theta$

$when\ x \to -\infty$: $\theta \to 0$

$when\ x \to \infty$: $\theta \to \pi$

$$B = \frac{\mu_0 I}{4\pi a} \int_0^{\pi} \sin\theta \, d\theta = \frac{\mu_0 I}{4\pi a} |-\cos\theta|_0^{\pi}$$

$$B = -\frac{\mu_0 I}{4\pi a} (\cos\pi - \cos 0) = -\frac{\mu_0 I}{4\pi a} (-2)$$

$$B = \frac{\mu_0 I}{2\pi a}$$

This is the expression of magnetic field induction due to a current carrying conductor.

In vector form:

$$\mathbf{B} = \frac{\mu_0 I}{2\pi a} \hat{\mathbf{n}}$$

Where $\hat{n}$ is the unit vector which represents the direction of tangent to the circle at point P.

35.2.2 Magnetic Field due to a Circular Current Loop

Consider a circular current loop of radius R carrying current I as shown in the figure below:

We want to find out the value of magnetic field strength at point P on at the distance x from the center of the loop. For this we consider a small element of length $d\mathbf{s}$ such that the angle between $\mathbf{r}$ and $d\mathbf{s}$ is 90°. Therefore,

$$dB = \frac{\mu_0}{4\pi} \frac{I\,ds \sin\theta}{r^2}$$

By Biot–Savart law, we find that magnetic induction $d\mathbf{B}$ is perpendicular to $\mathbf{r}$ and $d\mathbf{s}$. So by putting $\theta = 90^\circ$, we get:

$$dB = \frac{\mu_0}{4\pi} \frac{I\,ds}{r^2} \quad \text{----------} \quad (1)$$

We resolve the $d\mathbf{B}$ into rectangular components dB_x and dB_y. From the symmetry, we find that dB_y has no contribution to the field at point P, because dB_y components of element will cancel out each other. Only the x-components are added up to give the magnetic field strength at point P. Therefore,

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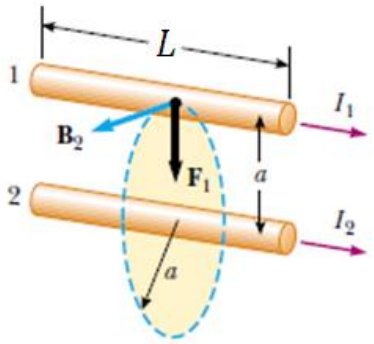
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$$B = \oint dB_x = \oint dB\cos\theta = \oint \frac{\mu_0}{4\pi}\frac{I\,ds}{r^2}\cos\theta = \frac{\mu_0 I}{4\pi}\oint \frac{ds}{r^2}\cos\theta$$

From figure $r^2 = x^2 + R^2$

And

$\cos\theta = \frac{R}{\sqrt{x^2+R^2}}$ $\frac{R}{r} =$



$$B = \frac{\mu_0 I}{4\pi}\oint \frac{ds}{x^2+R^2}\quad \frac{R}{\sqrt{x^2+R^2}} = \frac{\mu_0 IR}{4\pi\,(x^2+R^2)^{3/2}}\oint ds$$

$$B = \frac{\mu_0 IR}{4\pi\,(x^2+R^2)^{3/2}}\oint ds \qquad \because$$

$$\oint ds \quad = 2\pi R$$

$$B = \frac{\mu_0 IR}{4\pi\,(x^2+R^2)^{3/2}}(2\pi R)$$

$$B = \frac{\mu_0 IR^2}{2\,(x^2+R^2)^{3/2}} \qquad \text{---------- (2)}$$

This is the expression of magnetic induction at point P due to circular loop of current.

Special Cases.

Case 1. *Magnetic field induction at center of circular current carrying loop*

At the center of the loop $x = 0$. Putting value in equation (2), we get

$$B = \frac{\mu_0 IR^2}{2\,(R^2)^{3/2}} = \frac{\mu_0 IR^2}{2\,R^3}$$

$$B = \frac{\mu_0 I}{2\,R}$$

Case 2. *Magnetic field induction at very large distance from loop*

At very large distance from the center of the current loop i.e., $x \gg R$, the magnetic field induction will be:

$$B = \frac{\mu_0 IR^2}{2\,(x^2+R^2)^{3/2}} \qquad \because x \gg R; \quad x^2 + R^2 \approx x^2$$

$$B = \frac{\mu_0 IR^2}{2\,(x^2)^{3/2}} = \frac{\mu_0 IR^2}{2\,x^3}$$

$$B = \frac{\mu_0 I\pi R^2}{2\pi\,x^3} = \frac{\mu_0 IA}{2\pi\,x^3} \qquad \because \textit{Multipying and Dividing by } \pi$$

This is the expression of magnetic induction for a current carrying coil of one loop. For a current carrying loop having N turns:

$$B = \frac{\mu_0 NIA}{2\pi\,x^3} = \frac{\mu_0 \mu}{2\pi\,x^3} \qquad \because \text{Dipole Moment } \mu = NIA$$

35.2.3 Force between Long Parallel Current Carrying Conductor

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Consider two long, straight, parallel wires separated by a small distance 'a' carrying currents I_1 and I_2, respectively. The current passing through each wire produces a magnetic field around it and each wire is placed in the magnetic field produced by the other.

The wire 2, which carries a current I_2 creates a magnetic field B_2 at the location of wire 1, which can be find out by using the expression:

$$B_2 = \frac{\mu_0 I_2}{2\pi a} \quad \text{---------- (1)}$$

The direction of B_2 is perpendicular to wire 1, as shown in Figure. The magnetic force on wire 1, due to magnetic force of wire 2, will be:

$$\mathbf{F_1} = I_1\mathbf{L} \times \mathbf{B_2}$$

The magnitude of magnetic force on wire 1 is given by:

$$\Longrightarrow F_1 = I_1 L B_2 \sin 90^\circ \qquad \because \mathbf{L} \text{ and } \mathbf{B_2} \text{ are perpendicular to each other}$$

$$\Longrightarrow F_1 = I_1 L B_2$$

$$\Longrightarrow F_1 = I_1 L \left(\frac{\mu_0 I_2}{2\pi a}\right) = \frac{\mu_0\, I_1\, I_2}{2\,\pi\, a} L$$

where L is the length of conductor. The direction of force F_1 is towards wire 2.

Similarly, the force on wire 2, due to the magnetic force of wire 1, carrying current I_2 is expressed as:

$$F_2 = \frac{\mu_0\, I_1\, I_2}{2\,\pi\, a} L$$

The force F_2 is directed towards wire 1. The forces are equal and opposite, so, will attract each other.

If the direction of element in one wire is opposite to that in the other, the wires will repel each other.

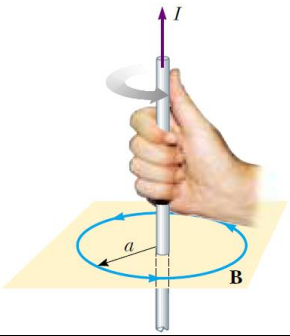
35.3 Ampere's Circuital Law

The ampere law is stated as,

"The line integral of magnetic induction **B** *due to current I around any close loop is μ_0 times the current enclosed"*

35.3.1 Integral Form of Ampere's Law

Consider a straight current carrying wire. The magnetic field will produced around the conductor as the current flow through it.



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The magnetic field strength due to this current carrying wire at any point at the distance a is expressed as:

$$\mathbf{B} = \frac{\mu_0 I}{2\pi R}\hat{\mathbf{k}} \qquad \text{---------- (1)}$$

The symmetry shows that the magnetic induction is same everywhere on the circumference of circle. The equation (1) can be expressed in more general form as:

$$\oint \mathbf{B}.d\mathbf{l} = \oint \frac{\mu_0 I}{2\pi a}\hat{\mathbf{k}}.d\mathbf{l}$$

$$= \frac{\mu_0 I}{2\pi a}\oint \hat{\mathbf{k}}.d\mathbf{l}$$

$$\because \hat{\mathbf{k}} \text{ is partallel to } d\mathbf{l}$$

$$= \frac{\mu_0 I}{2\pi a}\oint dl \qquad \because dl = ad\theta$$

$$\Rightarrow \oint \mathbf{B}.d\mathbf{l} = \frac{\mu_0 I}{2\pi a}\oint ad\theta = \frac{\mu_0 Ia}{2\pi a}\oint d\theta = \frac{\mu_0 I}{2\pi}(2\pi)$$

$$\Rightarrow \oint \mathbf{B}.d\mathbf{l} = \mu_0 I$$

This is called Ampere's circuital law.

As $I = \int \mathbf{J}.d\mathbf{s}$, therefore

$$\oint \mathbf{B}.d\mathbf{l} = \mu_0 \int \mathbf{J}.d\mathbf{s}$$

This is integral form of Ampere's law.

35.3.2 Differential Form of Ampere's Law

The Ampere's law is expressed in integral form as:

$$\oint \mathbf{B}.d\mathbf{l} = \mu_0 \int_s \mathbf{J}.d\mathbf{s} \qquad \text{---------- (2)}$$

Using Stoke's theorem on L.H.S., we get

$$\oint \mathbf{B}.d\mathbf{l} = \int_s curl\,\mathbf{B}.d\mathbf{s}$$

Thus equation (2) will become:

$$\int_s curl\,\mathbf{B}.d\mathbf{s} = \mu_0 \int_s \mathbf{J}.d\mathbf{s}$$

$$\int_s curl\,\mathbf{B}.d\mathbf{s} - \mu_0 \int_s \mathbf{J}.d\mathbf{s} = 0$$

$$\int_s (curl\,\mathbf{B} - \mu_0\mathbf{J}).d\mathbf{s} = 0$$

$$\Rightarrow curl\,\mathbf{B} - \mu_0\mathbf{J} = 0$$

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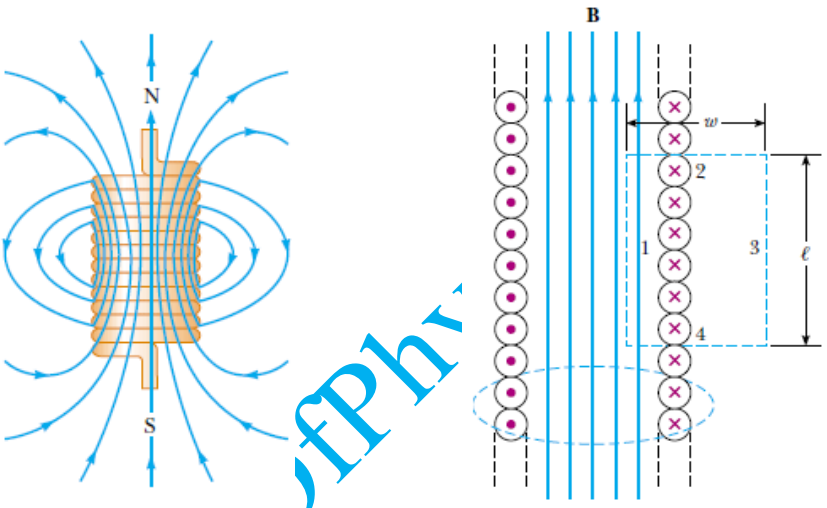
$$\Longrightarrow curl\ \mathbf{B} = \mu_0 \mathbf{J}$$

This is differential form of Ampere's law.

35.4 Applications of Ampere's Law

35.4.1 Magnetic Field due to a Solenoid

A solenoid is a cylindrical frame tightly wounded by an insulated wire. The magnetic field produced by the current carrying solenoid is like the field of a bar magnet. The magnetic field strength outside the solenoid is negligible as compared to the field inside it.



We want to find out magnetic field intensity **B** at point *P* inside a current carrying solenoid. For this we consider an Amperian loop. The symmetry shows that the close path can be divided into four elements1, 2, 3 and 4. By Ampere's law:

$$\oint \mathbf{B}.d\mathbf{l} = \mu_0 \times (Current\ Enclosed)$$

The integral $\oint \mathbf{B}.d\mathbf{l}$ can be written as the sum of four integrals:

$$\int_1 \mathbf{B}.d\mathbf{l} + \int_2 \mathbf{B}.d\mathbf{l} + \int_3 \mathbf{B}.d\mathbf{l} + \int_4 \mathbf{B}.d\mathbf{l} = \mu_0 \times (Current\ Enclosed)$$

Here $\int_2 \mathbf{B}.d\mathbf{l} = \int_4 \mathbf{B}.d\mathbf{l} = 0$ because the angle between **B** and $d\mathbf{l}$ is 90°.

Also $\int_3 \mathbf{B}.d\mathbf{l} = 0$ as the field outside the solenoid is negligible.

Therefore,

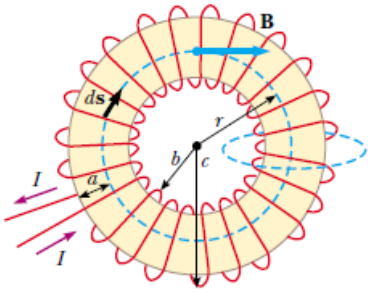
$$\int_1 \mathbf{B}.d\mathbf{l} = \mu_0 \times (Current\ Enclosed)$$

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$$\Rightarrow \int_1 B\, dl\, \cos 0^\circ = \mu_0 \times (Current\ Enclosed)$$

$$\Rightarrow \int_1 B\, dl = \mu_0 \times (Current\ Enclosed)$$



Since the magnetic field strength **B** for the loop element 1 is constant inside the solenoid. So,

$$B \int_1 dl = \mu_0 \times (Current\ Enclosed)$$

$$B\, l = \mu_0 \times (Current\ Enclosed) \quad \text{---------- (1)}$$

where l is the length of element 1 of Amperian loop.

To find out the current enclosed by the loop, we consider that there are n turns per unit length of a solenoid. Then

$$Number\ of\ turns\ in\ length\ l\ of\ solinoid = n\, l$$

If I is the current flowing through solenoid, then

$$Current\ Enclosed\ by\ the\ Amperian\ Loop = n\, l\, I$$

Hence the equation (1) will become:

$$B\, l = \mu_0 \times (n\, l\, I)$$

$$B = \mu_0 n\, I$$

This is the expression of magnetic field strength due to a current carrying solenoid. This relation shows that the magnetic field inside a solenoid depend only on current I and number of turns per unit length n.

35.4.2 Magnetic field due to a Toroid

A toroid is used to create an almost uniform magnetic field in some enclosed area. The device consists of a conducting wire wrapped around a ring (a *torus*) made of a nonconducting material as shown in the figure.

We want to find out magnetic field strength at any point inside a toroid. For this we consider circular Amperian loop of radius r. By symmetry, we see that the magnitude of the field is constant on this circle and tangent to it, therefore:

$$\mathbf{B}.d\mathbf{s} = B\, ds$$

Where $d\mathbf{s}$ is the small element of amperian loop.

By applying Ampere's law:

$$\oint \mathbf{B}.d\mathbf{s} = \mu_0 \times (Current\ Enclosed)$$

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$$\Longrightarrow \oint B\, ds \;=\; \mu_0 \times (Current\ Enclosed)$$

$$\Longrightarrow B \oint ds \;=\; \mu_0 \times (Current\ Enclosed)$$

$$\because \oint ds \;= 2\pi r$$

$$\Longrightarrow B \times 2\pi r \;=\; \mu_0 \times (Current\ Enclosed) \quad \text{---------- (1)}$$

If N are the number of turns of the toroid and I is the current in the toroid, then

$$Current\ Enclosed\ by\ the\ Amperian\ Loop = N\ I$$

The equation (1) will become:

$$B \times 2\pi r \;=\; \mu_0 \times (N\ I\ )$$

$$B \;=\; \frac{\mu_0 N\ I}{2\pi r}$$

This is the expression of magnetic field strength inside a current carrying toroid.

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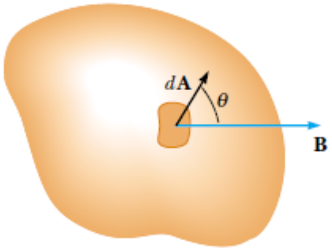
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FARADAY'S LAW OF ELECTROMAGNETIC INDUCTION

36.1 Magnetic Flux

The number of magnetic lines of force passing normally through certain area is called magnetic flux. It is denoted by Φ_B. It is a scalar quantity and its SI unit is weber (Wb). It is measured by the product of magnetic field strength and the component of vector area parallel to magnetic field.

If d**A** is the vector area element of the surface placed in uniform magnetic field of magnetic field strength **B** as shown in the figure below.



The magnetic flux $d\Phi_B$ through d**A** is given by:

$$d\Phi_B = \mathbf{B}.\mathrm{d}\mathbf{A}$$

$$\Phi_B = \int \mathbf{B}.\mathrm{d}\mathbf{A}$$

$$\Phi_B = \int B\, dA \cos\theta$$

where θ is the angle between magnetic field strength and vector area element.

36.2 Faraday's Law of Electromagnetic Induction

Statement:

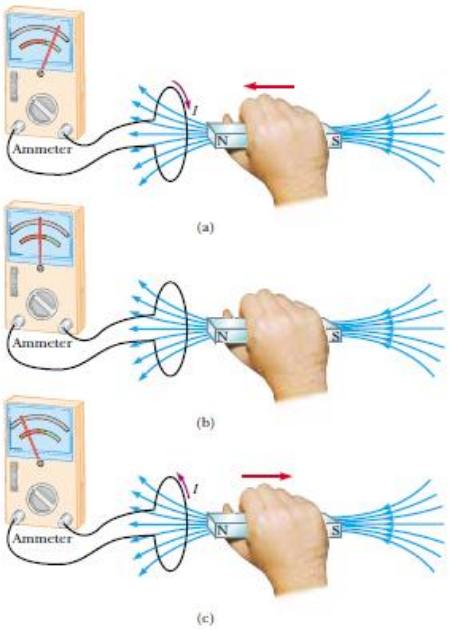
The induced emf in a circuit is equal to the negative of rate at which the magnetic flux through the circuit is changing with time.

Or

The magnitude of induced emf in a circuit is directly proportional to the rate of change of magnetic flux.

Explanation

When a magnet is moved toward the loop, the ammeter needle deflects in one direction, as shown in the figure (a). When the magnet is brought to rest and held stationary relative to the loop figure (b), no deflection is observed. When the magnet is moved away from the loop, the needle deflects in the opposite direction, as shown in figure (c). Finally, if the magnet is held stationary and the loop is moved either toward or away from it, the needle deflects.



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From these observations, we conclude that the loop detects that the magnet is moving relative to it and we relate this detection to a change in magnetic field. Thus, it seems that a relationship exists between current and changing magnetic field.

These results are quite remarkable in view of the fact that a current is set up even though no batteries are present in the circuit. We call such a current the induced current which is produced by an induced emf. This phenomenon is called electromagnetic induction.

36.2.1 Integral form of Faraday's Law of Electromagnetic Induction

If ε is the induce emf due to change in flux $d\Phi_B$ in time dt, then we can describe the Faraday's law of electromagnetic induction as:

$$\varepsilon \ \alpha \ -\frac{d\Phi_B}{dt}$$

$$\varepsilon = -\text{constant}\ \frac{d\Phi_B}{dt}$$

$$\varepsilon = -\frac{Nd\Phi_B}{dt}$$

Where N is constant and called number of turns in a coil.

Now as $\Phi_B = \int_s \mathbf{B}.\mathrm{d}\mathbf{A}$, therefore

$$\varepsilon = -N\frac{d}{dt}\int_s \mathbf{B}.\mathrm{d}\mathbf{A}$$

For a single loop coil, $N = 1$, we have:

$$\varepsilon = -\frac{d}{dt}\int_s \mathbf{B}.\mathrm{d}\mathbf{A} \quad \text{---------- (1)}$$

Also,

$$\varepsilon = \int \mathbf{E}.d\mathbf{r} \quad \text{---------- (2)}$$

Combining equation (1) and (2), we get:

$$\int \mathbf{E}.d\mathbf{r} = -\frac{d}{dt}\int_s \mathbf{B}.\mathrm{d}\mathbf{A} \quad \text{---------- (3)}$$

This is the integral form of Faraday's law of electromagnetic induction.

The negative sign is due to the fact that the direction of induced current is such that it opposes the cause producing it.

36.2.2 Differential form of Faraday's Law

Applying the Stoke's theorem on L.H.S of equation (3),

$$\int_s curl\ \mathbf{E}.d\mathbf{A} = -\frac{d}{dt}\int_s \mathbf{B}.\mathrm{d}\mathbf{A}$$

$$\int_s \left(curl\ \mathbf{E} + \frac{d\mathbf{B}}{dt}\right).d\mathbf{A}$$

$$\Rightarrow curl\ \mathbf{E} + \frac{d\mathbf{B}}{dt} = 0$$

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$$\Longrightarrow curl\ \mathbf{E} = -\frac{d\mathbf{B}}{dt}$$

Or $\quad \boldsymbol{\nabla} \times \mathbf{E} = -\frac{d\mathbf{B}}{dt}$

This is differential form of Faraday's law of electromagnetism induction.

36.3 Lenz law

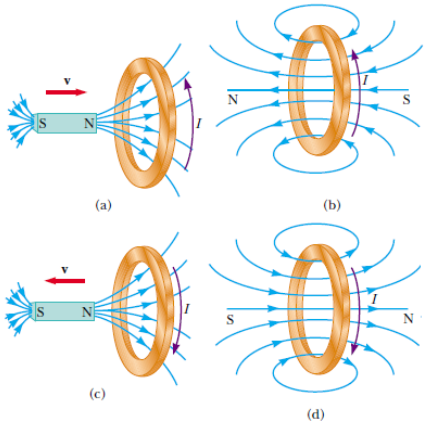
Statement:

The direction of induced current is such that it opposes it own cause.

Explanation:

Consider a bar magnet moves toward a stationary metal loop, as in figure (a). As the magnet moves to the right toward the loop, the external magnetic flux through the loop increases with time. As the result, the induced current set up in the loop which produces magnetic field, as illustrated in figure (b). Knowing that like magnetic poles repel each other, we conclude that the left face of the current loop acts like a north pole and that the right face acts like a south pole.

If the magnet moves to the left, as in figure (c), its flux through the area enclosed by the loop decreases in time. Now the induced current in the loop produces the magnetic field as shown in figure (d). In this case, the left face of the loop is a south pole and the right face is a north pole.



36.3.1 Lenz Law and Conservation of Energy

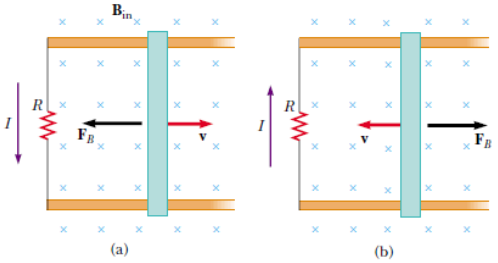
"*Lenz law is the statement of law of conservation of energy for the circuit involving induced current*"

To understand this statement, consider a conducting bar moving to the right on two parallel rails in the presence of a uniform magnetic field as shown in the figure below. As the bar moves to the right, the magnetic flux through the area enclosed by the circuit increases with time because the area increases. As the result, the induced current must be directed counterclockwise when the bar moves to the right. Since the current carrying bar is moving in the magnetic field, it will experience a magnetic force $\mathbf{F_B}$. By using right hand rule, the direction of $\mathbf{F_B}$ is opposite to that of $\mathbf{v}$, that tends to stop the rod. An external dragging force must be applied to keep the rod moving in the magnetic field.

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This dragging force provides the energy for the induced currents to flow. This energy is the source of induced current. Thus the electromagnetic induction is exactly according to the law of conservation of energy.



If the bar is moving to the left, as in figure (b), the external magnetic flux through the area enclosed by the loop decreases with time. Because the field is directed into the page, the direction of the induced current must be clockwise.

36.4 Motional Induction and Motional emf

The emf induced in a loop by moving it a magnetic field is called motional emf.

Consider a straight conductor of length l is placed in a magnetic field which is directed into the plane of paper. For simplicity, we assume that the conductor is moving in a direction perpendicular to the field with constant velocity under the influence of some external agent.

As the bar is pulled to the right with a velocity **v** under the influence of an applied force $\mathbf{F_{app}}$, free charges in the bar experience a magnetic force directed along the length of the bar. This force sets up an induced current because the charges are free to move in the closed conducting path. The magnetic flux passing through the circuit is:

$$\Phi_B = B\,(Area) = B\,(l)\,(x)$$

The induced emf can be determined using Faraday's law of electromagnetic induction:

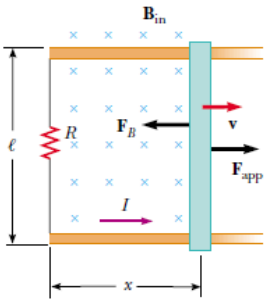
$$\varepsilon = -\frac{d\Phi_B}{dt} = -\frac{d\,(B\,l\,x)}{dt} = -B\,l\left(\frac{dx}{dt}\right)$$

$$\varepsilon = -B\,l\,v$$

This is the expression of induced emf in moving conductor. Due to this induced emf, the current induce in the conductor and is given by:

$$I = \frac{\varepsilon}{R} = \frac{-B\,l\,v}{R} \qquad \text{---------- (1)}$$

Where R is the resistance of the loop.



This induced current gives rise to the magnetic forces $\mathbf{F_B} = I\,(\boldsymbol{l} \times \mathbf{B})$ acting on the conductor. The magnitude of magnetic force will be:

$$F_B = I\,l\,B$$

Putting the values of I from equation (1):

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$$F_B = \left(\frac{-B\ l\ v}{R}\right) l\ B = \frac{-B^2\ l^2\ v}{R}$$

The negative sign is due to the fact that this magnetic force tends to stop the motion of conductor. In order to move the conductor in this uniform magnetic field with constant velocity, the applied force F_{app} and F_B must be equal and opposite. Therefore,

$$F_{app} = \frac{B^2\ l^2\ v}{R}$$

The power expended in moving the conductor can be find out by the expression:

$$P = F_{app}\ v$$

$$P = \left(\frac{B^2\ l^2\ v}{R}\right) v = \frac{B^2\ l^2\ v^2}{R}$$

This is the expression of power delivered to the conductor t move in magnetic field.

Power Dissipation

The power dissipated due to joule heating can be find out using expression:

$$P_{dissipated} = I^2 R$$

Putting value of I, we get:

$$P_{dissipated} = \left(\frac{-B\ l\ v}{R}\right)^2 R = \left(\frac{B^2\ l^2\ v^2}{R^2}\right) R$$

$$P_{dissipated} = \frac{B^2\ l^2\ v^2}{R}$$

So in case of motional induction, the power delivered and power dissipation is equal. Therefore the work done by the external agent is dissipated as joule heating.

36.5 Induced Electric Field

A straight current carrying conductor surrounded by a magnetic field. This magnetic field is surrounded by magnetic lines of force which are found to be concentric circles having their centers on the wire.

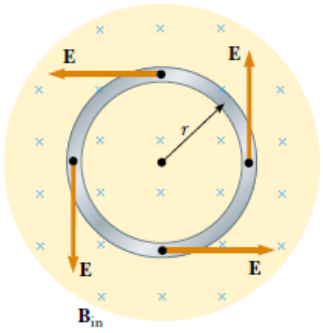
Similarly a changing magnetic field is surrounded by an electric field called induced electric field. This induced electric field is represented by electric lines of force which are found to be concentric circles.

We want to find out the expression of induced electric field due to changing magnetic field. For this, we consider a loop of conducting wire placed in a uniform magnetic field. This magnetic field may be applied by an electromagnet. By varying the current in the electromagnet, we can change the strength of magnetic field.

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When **B** is changed, the magnetic flux through the loop changes. So by using the Faraday's law and Lenz law we can find the magnitude and direction of induced emf and induced current.



Consider a circular path of radius r as shown in the figure below.

Since emf is defined as the work done in moving a unit positive charge from one point to another point. For a close path, the work done per unit charge is expressed as:

$$\varepsilon = \oint \mathbf{E} \cdot d\mathbf{r}$$

If $\mathbf{E} \parallel d\mathbf{r}$,

$$\varepsilon = \oint E\, dr \cos 0°$$

$$\Longrightarrow \varepsilon = \oint E\, dr$$

If the rate of change of magnetic flux is constant, then the induced electric field will also be constant. Therefore,

$$\varepsilon = E \oint dr$$

$$\Longrightarrow \varepsilon = E\ (\ length\ of\ close\ path\ )$$

$$\Longrightarrow \varepsilon = E\ (\ 2\pi r\ ) \qquad \text{----------} \quad (1)$$

By Faraday's law of electromagnetic induction, the induced emf in a coil of single loop will be:

$$\varepsilon = \frac{d\Phi_B}{dt} \qquad \text{----------} \quad (2)$$

Comparing equation (1) and (2), we have

$$E\ (\ 2\pi r\ ) = \frac{d\Phi_B}{dt}$$

$$E\ = \frac{1}{2\pi r} \frac{d\Phi_B}{dt}$$

This is the expression of magnitude of induced electric field.

36.6 Eddy Currents

When the magnetic flux through a piece of conductor changes, induced currents appears in it. These currents are called eddy currents.

In some cases, the eddy currents produce desirable effects and in some cases they may produce undesirable effects. For example, they increase the internal energy and thus can increase the temperature of material. On the other hand, the principle of eddy currents heating is used in induction furnace. In induction furnace, a sample of material can be heated using the repeatedly changing magnetic field.

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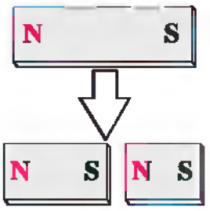
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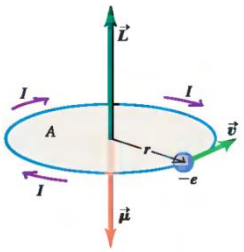
MAGNETIC PROPERTIES OF MATTER

37.1 Gauss's Law for Magnetism

An electric dipole consists of two equal but opposite point charges which are separated by a small distance. On the other hand, the magnetic dipole is considered as combination of a north pole and a south pole which are separated by small distance.

The important difference between an electric and magnetic dipole is that: an electric dipole can be separated into its constituent single charges, but a magnetic dipole cannot. Each time we try to divide a magnetic dipole into separate north and south poles, we create a new pair of poles. This effect occurs microscopically, down to the level of individual atoms. Each atom behaves like a magnetic dipole having a north and a south pole.





The gauss's law can be applied on electric and magnetic fields to find out flux through a close surface. If a close Gaussian surface contains net charge q, then the electric flux Φ_E passing through the Gaussian surface can be find out by using formula:

$$\Phi_E = \oint \mathbf{E}.\,\mathbf{dA} = \frac{q}{\varepsilon_0}$$

If a surface encloses no net charge (e.g., an electric dipole), the electric flux through the surface will be zero. Similarly, the magnetic flux through any close surface can be find out by Gauss's law.

The magnets are found in nature in the form of a magnetic dipole, because every cut through the magnet gives a piece having both a north and a south pole. We can describe this situation for magnetic materials as: "*There is no isolated magnetic charge*" or "*The magnetic monopole does not exist in nature"*.

The Gauss's law can be applied to find magnetic flux Φ_B through a close surface. A Gaussian surface containing a magnetic dipole has no net magnetic charge. Therefore,

$$\Phi_B = \oint \mathbf{B}.\,\mathbf{dA} = 0$$

The net flux of a magnetic field through any close surface is zero.

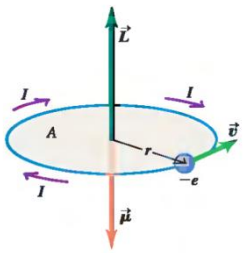
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Magnetic Dipole Moment of Orbiting Electron in an Atom

The magnetic field produced by a current in a coil of wire gives us a hint as to what causes certain materials to exhibit strong magnetic properties. The atoms that make up all matter contain moving electrons, and these electrons form microscopic current loops.

Consider an electron having charge $-e$ in an atom is moving in a circular orbit with radius r and speed v.



This moving charge is equivalent to a current loop. We know that a current loop with area A and current I has a magnetic dipole moment μ_l is described as:

$$\mu_l = IA \qquad \text{----------} \qquad \textbf{(1)}$$

For the orbiting electron

- Current through the loop $I = \frac{-e}{T}$
- Area of the loop is $A = \pi r^2$
- Orbital period $T = \frac{2\pi}{\omega} = \frac{2\pi r}{v}$

Therefore, the equation (1) will become:

$$\mu_l = \left(\frac{-e}{T}\right)(\pi r^2)$$

$$\mu_l = -\frac{e}{(2\pi r/v)}(\pi r^2) = -\frac{ev}{(2\pi r)}(\pi r^2) = -\frac{evr}{2}$$

Multiplying the numerator and denominator of above equation with mass of electron m_e:

$$\mu_l = -\frac{e\,(m_e vr)}{2m_e}$$

Because the magnitude of the orbital angular momentum of the electron is $l = m_e vr$, the magnetic moment can be written as:

$$\mu_l = -\left(\frac{e}{2m_e}\right) l \qquad \text{----------} \qquad \textbf{(1)}$$

This result demonstrates that the magnetic moment of the electron is proportional to its orbital angular momentum, but the vectors μ and l point in opposite directions.

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As the circulation all the electrons in an atom contribute to the magnetic dipole moment, thus, magnetic dipole moment of all the electrons in an atom is described as:

$$\mu_L = -\left(\frac{e}{2m_e}\right) L \qquad \text{----------} \qquad \textbf{(2)}$$

where μ_L is the total magnetic dipole moment of the atom and $L = \sum l$, is the total orbital angular momentum of all the electrons in the atom. In vector form, the equation (2) is described as:

$$\boldsymbol{\mu_L} = -\left(\frac{e}{2m_e}\right) \mathbf{L}$$

Bohr Magneton

From quantum mechanics, we learn that any particular component of angular momentum L is quantized in the integral multiples of $^{h}/_{2\pi}$. The unit of atomic dipole moment, called Bohr magneton μ_B, when the L have the magnitude of $^{h}/_{2\pi}$ in equation (2). Therefore, one Bohr magneton will be:

$$\mu_B = \frac{e}{2m_e}\,\frac{h}{2\pi} = \frac{eh}{4\pi m_e}$$

Substituting the appropriate value for charge and mass of electron, we find:

$$\mu_B = 9.27 \times 10^{-24}\ {}^{\text{J}}/_{\text{T}}$$

Magnetic moments associated with the orbital motion of electrons in atoms typically have the magnitude on the order of μ_B.

Intrinsic Magnetic Moments

The experiments performed in the 1920s, by passing beams of atoms through magnetic fields, showed that only the model of orbital magnetic moment was not sufficient to explain the observed properties. It was necessary to introduce another kind of magnetic moment call intrinsic (spin) magnetic moment.

By analogy with the orbital motion, the relationship between intrinsic angular momentum **s** and intrinsic dipole moment $\boldsymbol{\mu_s}$, for the case of a single electron, is described as:

$$\boldsymbol{\mu_s} = -\left(\frac{e}{m_e}\right) \mathbf{s}$$

The apparent difference of a factor 2 for the case of orbital magnetic dipole moment and intrinsic magnetic dipole moment comes because the basic unit of s is $\frac{1}{2}\left(\frac{h}{2\pi}\right)$, while the basic unit of l is $\frac{h}{2\pi}$.

The total intrinsic magnetic dipole moment for the case of an atom will be:

$$\boldsymbol{\mu_S} = -\left(\frac{e}{m_e}\right) \mathbf{S}$$

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Where $\mathbf{S} = \sum \mathbf{s}$ is the total spin angular momentum of electrons in an atom and $\boldsymbol{\mu}_S$ is the total intrinsic magnetic moment of atom.

The orbital angular momentum and the orbital magnetic dipole moments of a single electron are properties of its particular state of motion. The intrinsic (spin) angular momentum and intrinsic magnetic dipole moments of a single electron are fundamental characteristics of electron itself, along with its mass and electronic charge.

Total Magnetic Moment of an Atom

The magnetic properties of an atom of a particular material are determined by the total magnetic moment. The total magnetic moment $\boldsymbol{\mu}$ is the vector sum of orbital magnetic moment $\boldsymbol{\mu}_L$ and intrinsic (spin) magnetic moment $\boldsymbol{\mu}_S$. Mathematically,

$$\boldsymbol{\mu} = \boldsymbol{\mu}_L + \boldsymbol{\mu}_S$$

$$\boldsymbol{\mu} = -\left(\frac{e}{2m_e}\right)\mathbf{L} - \left(\frac{e}{m_e}\right)\mathbf{S}$$

Nuclear Magnetism

The nucleus of an atom consists of protons and neutrons in orbital motion under the influence of their mutual forces. Moreover, each nucleon in a nucleus has intrinsic angular momentum. Thus, the nuclear magnetic moment has two parts:

- Orbital part, which is due to the motion of protons. The neutrons, being uncharged, don't contribute to the orbital magnetic moment even they have orbital angular momentum.
- Intrinsic part, due to intrinsic magnetic moment of protons and neutrons. The uncharged neutrons have a nonzero intrinsic magnetic moment. If the neutrons were truly an elementary particle with no electric charge, it will indeed have no magnetic dipole moment. The nonzero magnetic dipole moment of a neutron is a clue to its internal structure and can be fairly well accounted for by considering the neutron the neutron to be composed of three charged quarks.

The orbital magnetic dipole moment $\boldsymbol{\mu}_L$ and intrinsic (spin) magnetic dipole moment $\boldsymbol{\mu}_S$ can be found by following equations:

$$\boldsymbol{\mu}_L = \frac{e}{2m_p}\mathbf{L}$$

$$\boldsymbol{\mu}_S = \left(\frac{e}{m_p}\right)\mathbf{S}$$

Typically, the nuclear magnetic dipole moments are smaller than atomic magnetic dipole moments by the factor of the order of 10^{-3}. It is because of the reason that the mass of proton and a neutron is about 1800 times the electron's mass.

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The contribution of nuclear magnetic dipole moment to the magnetic properties of materials is negligible. But the effects of nuclear magnetism become important in case of nuclear magnetic resonance.

Magnetization

Net magnetic dipole moment per unit volume is called magnetization. If $\boldsymbol{\mu} = \sum \boldsymbol{\mu}_i$ is the vector sum of magnetic dipole moment in a volume element V, then we can describe magnetization mathematically **M** as:

$$\mathbf{M} = \frac{\boldsymbol{\mu}}{V} = \frac{\sum \boldsymbol{\mu}_i}{V}$$

Suppose that a material is placed in a uniform magnetic field $\mathbf{B_0}$. This field "*magnetizes*" the material and aligns the magnetic dipoles of the material. The aligned magnetic dipoles produce a magnetic field $\mathbf{B_M}$ of their own. The net magnetic field **B** is the vector sum of the applied field $\mathbf{B_0}$ and the magnetic field produced by the dipoles $\mathbf{B_M}$. Therefore we can write:

$$\mathbf{B} = \mathbf{B_0} + \mathbf{B_M} \qquad \text{----------} \qquad \textbf{(1)}$$

When any magnetic material is placed in a uniform magnetic field $\mathbf{B_0}$, then the relation between the field $\mathbf{B_M}$ and magnetization **M** is expressed as:

$$\mathbf{B_M} = \mu_0 \mathbf{M}$$

Where μ_0 is the permeability of free space. Thus the equation (1) will become:

$$\mathbf{B} = \mathbf{B_0} + \mu_0 \mathbf{M} \qquad \text{----------} \qquad \textbf{(2)}$$

In weak fields, **M** is directly proportional to the applied fields $\mathbf{B_0}$, so **B** must be proportional to $\mathbf{B_0}$. Thus we can write:

$$\mathbf{B} = \kappa_m \mathbf{B_0} \qquad \text{----------} \qquad \textbf{(3)}$$

Where κ_m is called permeability constant of a materials, which is defined relative of vacuum. Combining equation (2) and (3), we get:

$$\kappa_m \mathbf{B_0} = \mathbf{B_0} + \mu_0 \mathbf{M}$$

$$\Rightarrow \mu_0 \mathbf{M} = \mathbf{B_0}(\kappa_m - 1)$$

This expression gives the expression of induced magnetization for a specific material due to the effect of applied magnetic field.

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Magnetic Materials

Substances can be classified as belonging to one of three categories, depending on their magnetic properties.

Paramagnetism

Paramagnetism occurs in those materials whose atoms have permanent magnetic dipole moments. In the absence of external magnetic field, the magnetic dipole moments of paramagnetic materials are randomly oriented. Thus, the magnetization of paramagnetic materials becomes zero, because the random directions of μ cause the vector sum to vanish.

However, when placed in a magnetic field, the magnetic dipoles align parallel to the applied magnetic field, the vector sum of the individual dipole moment is no longer zero and a significant bulk magnetization is observed. The field inside the material now has the two components, the applied field $\mathbf{B_0}$ and the induced field $\mu_0\mathbf{M}$ from the magnetization of dipoles. These two fields are parallel and the alignment of magnetic dipoles enhances the net internal field.

The ratio between the induced magnetization $\mu_0\mathbf{M}$ and the applied field $\mathbf{B_0}$ is determined by $(\kappa_m - 1)$ which is small and positive for paramagnetic materials.

The thermal motion of the atom tends to disturb the alignment of the dipoles, and consequently the magnetization decreases with increasing temperature. The relationship between $\mathbf{M}$ and the temperature was discovered by Pierre Curie in 1895 and is written:

$$\mathbf{M} = C\,\frac{\mathbf{B_0}}{T}$$

This is known as Curie law, the constant C being known as Curie constant.

When the magnetization reaches its saturation value, increase in $\mathbf{B_0}$ have no further effect on the magnetization. Curie law, which requires that $\mathbf{M}$ increases linearly with $\mathbf{B_0}$, is valid only when the magnetization is far from saturation.

When the external magnetic field is removed from a paramagnetic sample, the thermal motion of causes the directions of the magnetic dipole moments to become random again. It is because of this reason that the magnetic force between the atoms is too small to hold the magnetization and prevent magnetization.

Diamagnetism

The diamagnetism occurs in those materials which have no net magnetic dipole moment, as the atoms have completely filled energy sublevels. The orbital and spin magnetic dipole moments of the atoms of diamagnetic materials are added vectorially to zero. However, if a material is positioned in a magnetic field, the temporary dipoles form due to very slight asymmetric alteration in

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the orbital motion of their electrons. The formation of temporary dipole moments results in a small net magnetization. This induced magnetization has the direction opposite to applied magnetic field and the material show a weak repulsion to a magnetic field. This behavior is necessarily present in all materials.

Diamagnetism is much weaker effect than paramagnetism, and therefore it can be most easily observed on in the materials that are not paramagnetic.

The orbiting electrons in an atom behave like current loops. When an external magnetic field $\mathbf{B_0}$ is applied, the flux through the loop changes. By Lenz law, the motion of electrons must change in such a way that the induced field opposes this increase in flux. The change in motion of electron in orbit is accompanied by a slight speeding up or slowing down of orbital motion, such that circular frequency $\Delta\omega$ associated by orbital motion changes by:

$$\Delta\omega = \pm \frac{e\mathbf{B_0}}{2m}$$

Where e and m are the charge and mass of an electron. According to the Lenz law, the induced magnetic dipoles in diamagnetic materials will be oriented in such that they experience a repulsive force in a magnetic field. The ratio between the induced magnetization $\mu_0\mathbf{M}$ and the applied field $\mathbf{B_0}$ is determined by $(\kappa_m - 1)$ have the values in the range -10^{-6} to -10^{-5}.

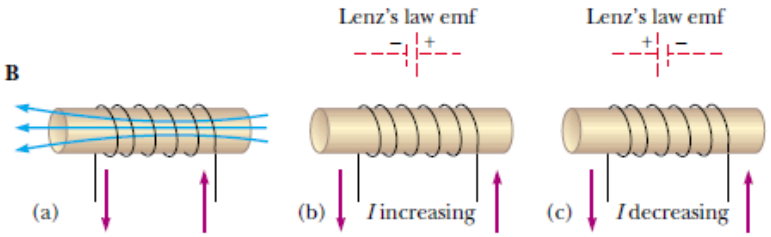
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INDUCTANCE

Inductance

Consider a coil wound on a cylindrical core. Assume that the current in the coil either increases or decreases with time. When the current flow through the coil, a magnetic field will be set up inside the coil as shown in figure below.



As the current changes with time, the magnetic flux through the coil also changes and induces an emf in the coil. This induced emf ε_L is directly proportional to the rate of change of current $\frac{dI}{dt}$.

$$\varepsilon_L \propto \frac{dI}{dt}$$

$$\varepsilon_L = -L\frac{dI}{dt}$$

The negative sign is due to the fact that induced emf oppose the cause which produce it. The symbol L is the inductance or self inductance of the coil which is described as:

"*The ratio between the induced emf and the rate of change of current*"

$$L = -\frac{\varepsilon_L}{\left(\frac{dI}{dt}\right)}$$

The unit of inductance is Henry. The inductance of a coil is one Henry if an emf of one volt is induced in it, when the current changes at the rate of one ampere per second.

$$1\ Henry = \frac{1\ volt}{\left(\frac{1\ ampere}{second}\right)} = 1\ VA^{-1}s$$

Inductance of a Solenoid

Consider a solenoid of length l, having N number of turns. If the current I flows through solenoid, then magnetic field B produced is given by:

$$B = \frac{\mu_0 N\, I}{l} = \mu_0 n\, I$$

Where $n = \frac{N}{l}$ is the number of turns per unit length.

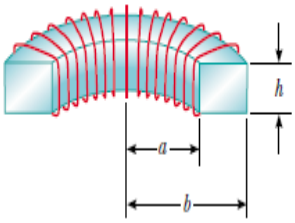
If A is the area of each circular turn of the solenoid, then the flux ϕ_B passing through one turn is:

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$$\phi_B = (B)(A)$$

$$\phi_B = \mu_0 n\, IA$$



The magnetic flux passing through all the N turns is given by:

$$\phi_B = N\mu_0 n\, IA$$

If the current changes through the solenoid, then the magnetic flux also changes with respect to the time, i.e.,

$$\frac{d\phi_B}{dt} = N\mu_0 n\, A\frac{dI}{dt}$$

According to the Faraday's law of electromagnetism, the induced emf ε is described as:

$$\varepsilon = -\frac{d\phi_B}{dt} = -N\mu_0 n\, A\frac{dI}{dt} \qquad \text{----------} \quad \textbf{(1)}$$

If L is the self inductance of the solenoid, then

$$\varepsilon = -L\frac{dI}{dt} \qquad \text{----------} \quad \textbf{(2)}$$

Comparing (1) and (2), we get:

$$L = N\mu_0 n\, A$$

$$\because As\ n = \frac{N}{l} \Longrightarrow N = nl$$

$$\Longrightarrow L = (nl)\mu_0 n\, A$$

$$\Longrightarrow L = \mu_0 n^2 l\, A$$

This expression gives the self inductance of a solenoid.

Inductance of a Toroid

Let N is the number of turns in a toroid through current I is flowing and r is mean radius of the toroid. Let a and b are the inner and outer radii of toroid, respectively, and h is its height.

The magnetic field produced at any point inside toroid can be find out using expression:

$$B = \frac{\mu_0 NI}{2\pi r}$$

Let $h\,dr = dA$ is the small area element, then the magnetic flux passing through the small area element will be:

$$d\phi_B = B\,dA$$

$$d\phi_B = B\,h\,dr$$

Integrating we get:

$$\phi_B = \int_{r=a}^{r=b} B\,h\,dr$$

The magnetic flux passing through all N turns is:

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$$\phi_B = \int_{r=a}^{r=b} N\, B\, h\, dr$$

$$\phi_B = \int_{r=a}^{r=b} N \left(\frac{\mu_0 NI}{2\pi r}\right) h\, dr = \frac{\mu_0 N^2 I\, h}{2\pi} \int_{r=a}^{r=b} \frac{dr}{r}$$

$$\phi_B = \frac{\mu_0 N^2 I\, h}{2\pi} \ln|r|_a^b = \frac{\mu_0 N^2 I\, h}{2\pi} (\ln b - \ln a)$$

$$\phi_B = \frac{\mu_0 N^2 I\, h}{2\pi} \ln\frac{b}{a}$$

$$\frac{\phi_B}{I} = \frac{\mu_0 N^2\, h}{2\pi} \ln\frac{b}{a} \qquad \text{---------- (1)}$$

The self inductance of a toroid is:

$$L = \frac{\phi_B}{I} \qquad \text{---------- (2)}$$

$\because \phi_B \propto I$

$\because \phi_B = L\, I$

By comparing (1) and (2), we get:

$$L = \frac{\mu_0 N^2\, h}{2\pi} \ln\frac{b}{a}$$

So, the inductance of a toroid depends only on geometrical factors.

Inductors with Magnetic Materials

When a magnetic field B_0 acts on a ferromagnetic material, then total magnetic field B is increased which can be expressed as:

$$B = \mu_r B_0$$

Where μ_r is the relative permeability of permeability constant of the magnetic material.

Similarly, the inductance of a coil is increased, if some magnetic material is placed inside the coil. If L_0 and L is the inductance with and without the magnetic material, respectively, then:

$$L = \mu_r L_0 \qquad \text{---------- (1)}$$

Now the inductance of solenoid without magnetic material is:

$$L_0 = \mu_0 n^2 l\, A$$

Putting values of L_0 in equation (1), we have:

$$L = \mu_r \mu_0 n^2 l\, A$$

The effective values of permeability constant μ_r for a ferromagnetic material have the values in range of 10^3 to 10^4 . Therefore the inductance of a coil can be increased by a factor of 10^3 to 10^4 with the core of a ferromagnetic substance.

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Growth of Current in LR Circuit

Consider a series circuit of a resistance R and inductance connected in series with a battery of emf ε as shown in the figure.

When the switch is open, no current will flow through the circuit. So at time $t = 0, I = 0$.

When the switch is closed, the current starts flowing through the circuit. But due to inductance, this current takes some time to get its maximum value. (During this period, the current is called transient current)

Initially, when the current is changing, the back emf $L\frac{dI}{dt}$ appears across the inductance coil which dies away with the passage of time. And its value falls to zero, when the current is maximum.

Let V_R and V_L are the values of potential drop across resistance R and inductance L, respectively. Then, by applying the Kirchhoff's rule:

$$\varepsilon - V_R - V_L = 0$$

$$V_L = \varepsilon - V_R$$

$$L\frac{dI}{dt} = \varepsilon - IR$$

$$\frac{dI}{\varepsilon - IR} = \frac{dt}{L}$$

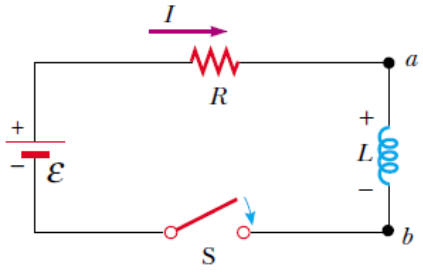
Multiplying both sides by $-R$, we get:

$$\frac{-RdI}{\varepsilon - IR} = -R\frac{dt}{L}$$

Integrating between the limits,

At $t = 0, I = 0$;

At $t = t, I = I$



$$\int_0^I \frac{-RdI}{\varepsilon - IR} = \int_0^t -R\frac{dt}{L}$$

$$|\ln(\varepsilon - IR)|_0^I = -\frac{R}{L}\,|t|_0^t$$

$$\ln(\varepsilon - IR) - \ln(\varepsilon) = -\frac{R}{L}\,t$$

$$\ln\left(\frac{\varepsilon - IR}{\varepsilon}\right) = -\frac{R}{L}\,t$$

$$\frac{\varepsilon - IR}{\varepsilon} = e^{\left(-\frac{R}{L}t\right)}$$

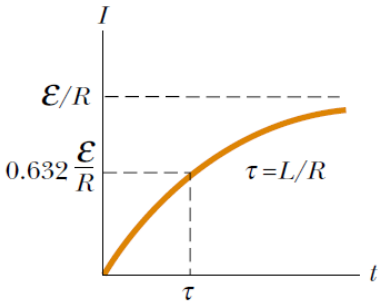
$$IR = \varepsilon - \varepsilon\, e^{\left(-\frac{R}{L}t\right)}$$

$$IR = \varepsilon\left(1 - e^{\left(-\frac{R}{L}t\right)}\right)$$

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$$I = \frac{\varepsilon}{R}\left(1 - e^{\left(-\frac{t}{L/R}\right)}\right)$$



Let $\tau_L = \frac{L}{R}$ = Inductive time constant and $\frac{\varepsilon}{R} = I_0$ = Maximum Current

$$I = I_0\left(1 - e^{\left(-\frac{t}{\tau_L}\right)}\right)$$

This is the equation of growth of current

Special Cases:

At time $t = \tau_L$

$$I = I_0\left(1 - e^{\left(-\frac{\tau_L}{\tau_L}\right)}\right) = I_0(1 - e^{-1}) = I_0(1 - 0.37)$$

$$I = 0.63\ I_0$$

At $t = \infty$

$$I = I_0\left(1 - e^{(-\infty)}\right) = I_0\left(1 - \frac{1}{\infty}\right) = I_0(1 - 0)$$

$$I = I_0$$

Inductive Time Constant

"*It is the interval of time during which the current grows to 63 % of its maximum value*".

The graph shows that current through the circuit increases exponentially.

As

$$I = I_0\left(1 - e^{\left(-\frac{R}{L}t\right)}\right) \quad \text{----------} \quad \textbf{(1)}$$

$$V_R = IR = I_0R\left(1 - e^{\left(-\frac{R}{L}t\right)}\right) = \varepsilon\left(1 - e^{\left(-\frac{R}{L}t\right)}\right) \quad \text{----------} \quad \textbf{(2)}$$

i.e., V_R increases exponentially upto ε.

Differentiating equation (1) with respect to time, we get:

$$\frac{dI}{dt} = -\frac{R}{L}I_0\left(0 - e^{\left(-\frac{R}{L}t\right)}\right)$$

$$\Rightarrow V_L = L\frac{dI}{dt} = I_0R\ e^{\left(-\frac{R}{L}t\right)} = \varepsilon\ e^{\left(-\frac{R}{L}t\right)} \quad \text{----------} \quad \textbf{(3)}$$

i.e., V_L decreases exponentially upto 0.

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Decay of Current in LR Circuit

When the S is connected to the b, as shown in the figure, the current start decaying exponentially with time and becomes zero after some time.

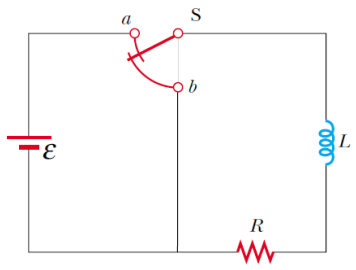
If V_R and V_L are the values of potential difference across the resistance R and inductance L, respectively. Then by applying the Kirchhoff's Rule:

$$V_R + V_L = 0$$

$$IR + L\frac{dI}{dt} = 0$$

$$L\frac{dI}{dt} = -IR$$

$$\frac{dI}{I} = -\frac{R}{L}\,dt$$



Integrating between the limits,

At $t = 0, I = I_0$;

At $t = t, I = I$

$$\int_{I_0}^{I} \frac{dI}{I} = \int_0^t -R\frac{dt}{L}$$

$$|\ln(I)|_{I_0}^{I} = -\frac{R}{L}\,|t|_0^t$$

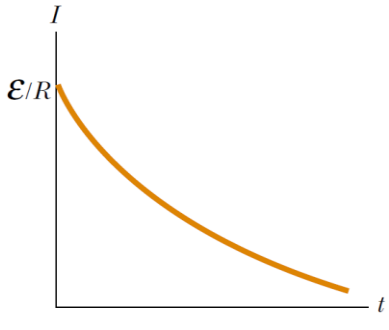
$$\ln(I) - \ln(I_0) = -\frac{R}{L}\,t$$

$$\ln\left(\frac{I}{I_0}\right) = -\frac{R}{L}\,t$$

$$\frac{I}{I_0} = e^{\left(-\frac{R}{L}t\right)}$$

$$I = I_0\, e^{\left(-\frac{R}{L}t\right)}$$

$$I = I_0\, e^{\left(-\frac{t}{L/R}\right)}$$



Let $\tau_L = \frac{L}{R}$ = Inductive time constant

$$I = I_0\, e^{\left(-\frac{t}{\tau_L}\right)}$$

This is the equation of decay of current in an LR circuit.

Special Cases:

At time $t = \tau_L$

$$I = I_0\, e^{\left(-\frac{t}{\tau_L}\right)} = I_0(\,e^{-1}) = I_0(\,0.37)$$

$$I = 0.37\ I_0$$

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At $t = \infty$

$$I = I_0\left(e^{(-\infty)}\right) = I_0\left(\frac{1}{\infty}\right) = I_0(0)$$

$$I = 0$$

Inductive Time Constant

"*It is the interval of time during which the current decays to 63 % of its maximum value*".

As $I = I_0\, e^{\left(-\frac{t}{\tau_L}\right)}$ ---------- **(1)**

The potential difference across R is

$$V_R = IR = I_0 R\ e^{\left(-\frac{R}{L}t\right)}$$

$$V_R = \varepsilon\ \ e^{\left(-\frac{R}{L}t\right)}$$ ---------- **(2)**

Differentiating equation (1) with respect to time, we get:

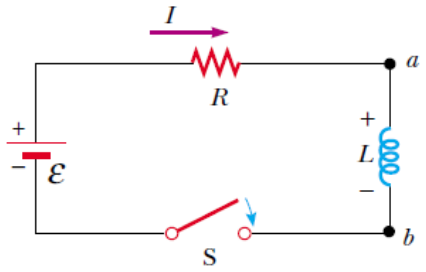
$$\frac{dI}{dt} = -\frac{R}{L} I_0\ e^{\left(-\frac{R}{L}t\right)}$$

$$\Longrightarrow V_L = L\frac{dI}{dt} = -I_0 R\ \ e^{\left(-\frac{R}{L}t\right)}$$

$$\Longrightarrow V_L = -\varepsilon\ \ e^{\left(-\frac{R}{L}t\right)}$$ ---------- **(3)**

Energy Stored in the Magnetic Field

Consider a resistor and an inductor is connected in series with a battery of emf ε. According to the Kirchhoff's 2nd rule;



$$\varepsilon - V_R - V_L = 0$$

$$\varepsilon = V_R + V_L$$

$$\varepsilon = IR + L\ \frac{dI}{dt}$$

Multiplying both sides by 'I', we have:

$$\varepsilon I = I^2 R + LI\ \frac{dI}{dt}$$

Here εI is the power supplied by the source, $I^2 R$ is the power dissipated in resistance and $LI\ \frac{dI}{dt}$ is the energy supplied per time in the inductance coil, where the magnetic field also exists.

If U_B is the total energy stored in the magnetic field of inductance coil, then the energy stored per unit time is:

$$\frac{dU_B}{dt} = LI\ \frac{dI}{dt}$$

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$$dU_B = LI\ dI$$

Integrating both sides, we get:

$$\int_0^{U_B} dU_B = \int_0^I LI\ dI$$

$$U_B = \frac{1}{2}LI^2$$

This is the expression of energy stored in the magnetic field of a current carrying inductor. This equation is similar to the expression of energy stored in the electric field of a capacitor.

$$U_E = \frac{1}{2}\ \frac{q^2}{C}$$

The energy stored in the inductor can be dissipated through joule heating in the resistor, in the similar way the energy stored in the capacitor is dissipated through the joule heating in the resistor during the discharging of a capacitor.

Energy Density

The energy density u_B in case of an inductor can be find out by using expression:

$$Energy\ Density = \frac{Energy\ Stored}{Volume}$$

If U_B is the energy stored in the magnetic field of an inductor of length l having cross-sectional area A, then the energy density in the expression will be:

$$u_B = \frac{U_B}{A\ l}$$

As $U_B = \frac{1}{2}LI^2$, therefore

$$\Longrightarrow u_B = \frac{\left(\frac{1}{2}LI^2\right)}{A\ l} = \frac{LI^2}{2A\ l}$$

In case of a current carrying coil, the induce L can be find out by using expression: $L = \mu_0 n^2 l\ A$

$$\Longrightarrow u_B = \frac{(\mu_0 n^2 l\ A)I^2}{2A\ l} = \frac{\mu_0 n^2 l\ AI^2}{2A\ l}$$

$$\Longrightarrow u_B = \frac{\mu_0 n^2 I^2}{2} = \frac{{\mu_0}^2 n^2 I^2}{2\mu_0}$$

The magnetic field strength inside a current carrying coil is $B = \mu_0 n\ I$

$$\Longrightarrow u_B = \frac{B^2}{2\mu_0}$$

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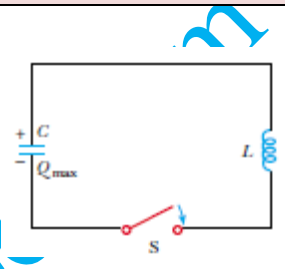
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This expression is similar to the energy density inside the electric field of a capacitor, which is expressed as:

$$u_E = \frac{\varepsilon_0 E^2}{2}$$

where ε_0 is the permittivity of free space and E is the electric field strength inside the capacitor.

Electromagnetic Oscillations (Qualitative discussion)



Consider an oscillating circuit which consists of a capacitor and an inductor. If the capacitor is initially charged and the switch is then closed, we find that both the current in the circuit and the charge on the capacitor oscillate between maximum positive and negative values. If the resistance of the circuit is zero, no energy is transformed to internal energy. In the following analysis, we neglect the resistance in the circuit. We also assume an idealized situation in which energy is not radiated away from the circuit.

When the capacitor is fully charged, the energy U_E in the circuit is stored in the electric field of the capacitor and is equal to

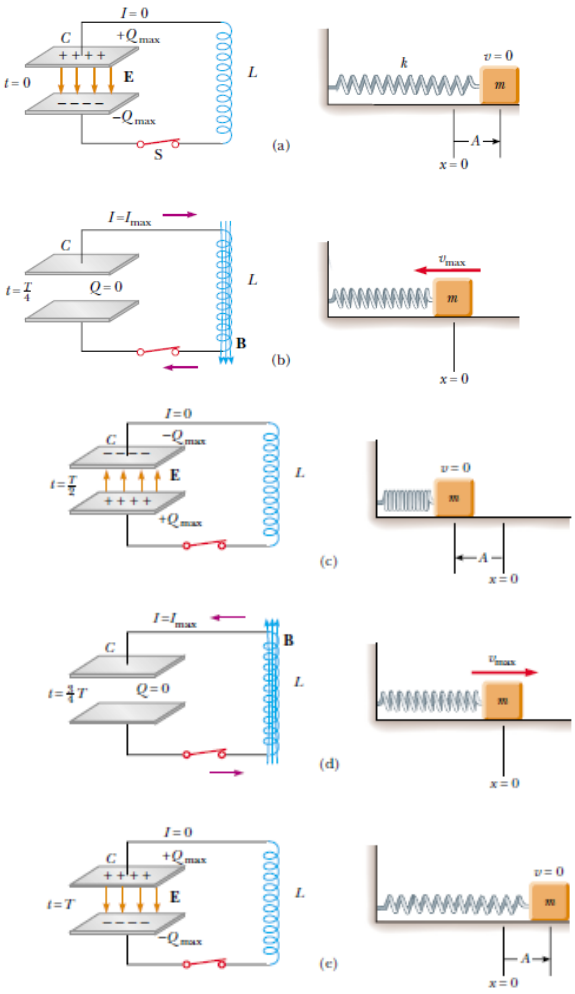
$$U_E = \frac{1}{2}\frac{(q_{max})^2}{C}$$

At this time, the current in the circuit is zero, and therefore no energy is stored in the inductor.

$$U_B = 0$$

As the switch S is closed, the capacitor begins to discharge and the energy stored in its electric field decreases. The discharge of the capacitor represents a current in the circuit, and hence some energy is now stored in the magnetic field of the inductor. Thus, energy is transferred from the electric field of the capacitor to the magnetic field of the inductor. When the capacitor is fully discharged, it stores no energy. At this time, the current reaches its maximum value, and all of the energy is stored in the inductor.

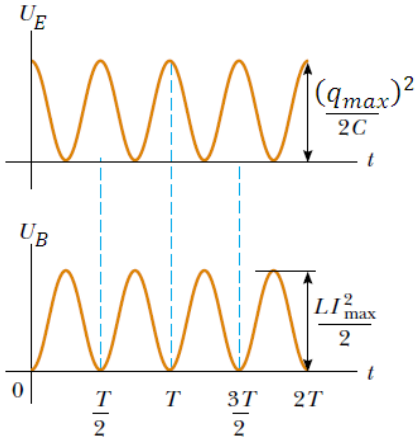
$$(U_E)_{min} = 0, (U_B)_{max} = \frac{1}{2}L(I_{max})^2$$



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The current continues in the same direction, decreasing in magnitude, with the capacitor eventually becoming fully charged again but with the polarity of its plates now opposite the initial polarity. This is followed by another discharge until the circuit returns to its original state of maximum charge q_{max}. The energy continues to oscillate between inductor and capacitor. When the energy stored in the capacitor is maximum, the energy stored in inductor is minimum and vice versa. But the total energy $U = U_E + U_B$ remains constant.



As the energy in such circuit is used to oscillate between capacitor and inductor, therefore, it is also called 'Oscillating Circuit'.

LC Circuit: Electromagnetic Oscillations (Quantitative analysis)

Let U_E and U_B are the values of energy stored in the electric field of capacitor and magnetic field of an inductor, respectively.

So the total energy U stored in the circuit is:

$$U = U_E + U_B$$

$$U = \frac{1}{2}\frac{q^2}{C} + \frac{1}{2}LI^2$$

Differentiating the above equation, we get

$$\frac{dU}{dt} = \frac{d}{dt}\left[\frac{1}{2}\frac{q^2}{C} + \frac{1}{2}LI^2\right]$$

$$\because U = cosntant$$

$$0 = \frac{q}{C}\frac{dq}{dt} + LI\frac{dI}{dt}$$

$$\Rightarrow \frac{q}{C}\frac{dq}{dt} + LI\frac{dI}{dt} = 0$$

$$\because I = \frac{dq}{dt}$$

$$\Rightarrow \frac{q}{C}\frac{dq}{dt} + L.\frac{dq}{dt}.\frac{d}{dt}\left(\frac{dq}{dt}\right) = 0$$

$$\Rightarrow \frac{dq}{dt}\left[\frac{q}{C} + L.\frac{d}{dt}\left(\frac{dq}{dt}\right)\right] = 0$$

$$\Rightarrow \frac{q}{C} + L\frac{d^2q}{dt^2} = 0$$

$$\Rightarrow \frac{d^2q}{dt^2} + \frac{q}{LC} = 0 \qquad \text{---------- (1)}$$

For a block spring system, the differential equation of a mass spring system is:

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$$\frac{d^2x}{dt^2} + \frac{k}{m}x = 0$$

The solution of this equation is:

$$x = x_m \cos(wt + \Phi)$$

Where x_m is the amplitude of oscillating mass spring system and Φ is its initial phase. Similarly, the solution of equation (1) will be:

$$q = q_m \cos(wt + \Phi)$$

And

$$\frac{dq}{dt} = -\,w\,q_m\ \sin(wt + \Phi)$$

$$\frac{d^2q}{dt^2} = -\,w^2\,q_m \cos(wt + \Phi)$$

Putting the values of q and $\frac{d^2q}{dt^2}$ in equation (1), we get:

$$-\,q_m \cos(wt + \Phi)\left[w^2 - \frac{1}{LC}\right] = 0$$

$$\Longrightarrow w^2 - \frac{1}{LC} = 0$$

$$\Longrightarrow w = \frac{1}{\sqrt{LC}}$$

$$\Longrightarrow f = \frac{1}{2\pi}\frac{1}{\sqrt{LC}}$$

This is the expression of frequency of the LC circuit of electromagnetic oscillations.

The energy stored in the electric field of a capacitor is:

$$U_E = \frac{1}{2}\frac{q^2}{C} = \frac{1}{2}\,[q_m \cos(wt + \Phi)]^2$$

$$U_E = \frac{1}{2C}\,[q_m^2 \cos^2(wt + \Phi)]$$

The energy stored in the magnetic field of an inductor will be:

$$U_B = \frac{1}{2}LI^2$$

$$\Longrightarrow U_B = \frac{1}{2}L\left[\frac{dq}{dt}\right]^2 = \frac{1}{2}L\left[\frac{d}{dt}\{q_m \cos(wt + \Phi)\}\right]^2$$

$$\Longrightarrow U_B = \frac{1}{2}L[w^2\,q_m^2\ \sin^2(wt + \Phi)]$$

$$\Longrightarrow U_B = \frac{1}{2}L\left[\frac{1}{LC}\,q_m^2\ \sin^2(wt + \Phi)\right]$$

$$\Longrightarrow U_B = \frac{1}{2C}\,q_m^2\ \sin^2(wt + \Phi)$$

The total energy stored in the LC circuit is:

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$$U = U_E + U_B = \frac{1}{2}\, q_m^2 \cos^2(wt + \Phi) + \frac{1}{2C}\, q_m^2 \ \sin^2(wt + \Phi)$$

$$U = \frac{1}{2}\, q_m^2[\cos^2(wt + \Phi) + \sin^2(wt + \Phi)]$$

$$U = \frac{1}{2}\, q_m^2$$

As the q_m is the maximum value of charged stored in the capacitor, so the total energy stored in LC circuit is constant.

Damped and Forced Oscillations

A resistance is always present in every real LC circuit. When we take this resistance into account, we find that the total electromagnetic energy is not constant but decreases with time as it is dissipated as internal energy in the resistor. For a LC circuit, the total energy U of the circuit is described as:

$$U = U_E + U_B$$

Where U_E is the energy stored in the electric field of a charged capacitor and U_B is the energy stored in the magnetic field of inductor.

$$U = \frac{1}{2}\ \frac{q^2}{C} + \frac{1}{2}LI^2$$

Differentiating the above equation, we get

$$\frac{dU}{dt} = \frac{d}{dt}\left[\frac{1}{2}\ \frac{q^2}{C} + \frac{1}{2}LI^2\right] \qquad \text{---------- (1)}$$

The total energy of LC circuit is no longer constant but rather

$$\frac{dU}{dt} = -I^2R$$

Thus the equation (1) will become:

$$\frac{d}{dt}\left[\frac{1}{2}\ \frac{q^2}{C} + \frac{1}{2}LI^2\right] = -I^2R$$

$$\Rightarrow \frac{q}{C}\ \frac{dq}{dt} + LI.\frac{dI}{dt} = -I^2R$$

$$\because I = \frac{dq}{dt}$$

$$\Rightarrow \frac{q}{C}\ \frac{dq}{dt} + L.\frac{dq}{dt}\ .\frac{d}{dt}\left(\frac{dq}{dt}\right) = -\left(\frac{dq}{dt}\right)^2 R$$

$$\Rightarrow \frac{dq}{dt}\ \left[\frac{q}{C} + L.\frac{d}{dt}\left(\frac{dq}{dt}\right)\right] = -\left(\frac{dq}{dt}\right)^2 R$$

$$\Rightarrow \frac{q}{C} + L.\frac{d}{dt}\left(\frac{dq}{dt}\right) = -\frac{dq}{dt}R$$

$$\Rightarrow L\frac{d^2q}{dt^2} + \frac{dq}{dt}R + \frac{q}{C} = 0$$

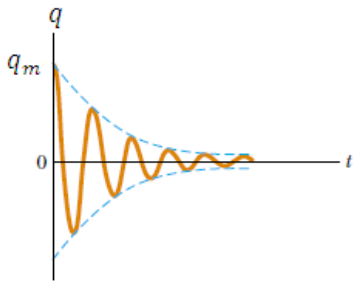
The general solution of the above equation will be:

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$$q = q_m e^{\left(-\frac{Rt}{2L}\right)} \cos(\omega' t + \Phi)$$

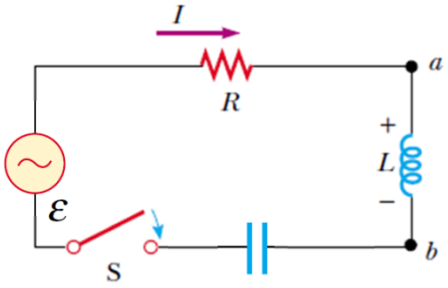
Here $\omega' = \sqrt{\omega^2 - \left(\frac{R}{2L}\right)^2}$

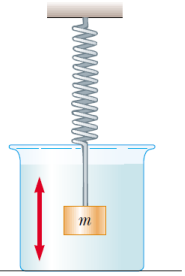


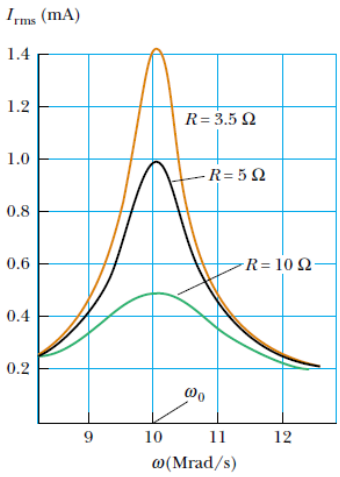
The figure shows the charged stored on the capacitor in a damped LC circuit as the function of time. The current decreases exponentially with time. Moreover the frequency ω' is strictly less than the frequency ω of undammed oscillations.

Forced Oscillations and Resonance

The RLC series circuit is similar to a mechanical system consisting of a simple harmonic oscillator in a damping medium. The oscillating system consist of mass m attached to a spring of spring constant k.







If the frequency of the oscillator is equal to the natural frequency of system $\frac{k}{m}$, the there will be resonance and the amplitude of the oscillation will be maximum.

In the similar way, if ω' is the frequency of sinusoidal voltage source, then the sinusoidal voltage is expressed as:

$$\varepsilon = \varepsilon_m \cos \omega' t$$

Where ε_m is the peak value of sinusoidal voltage. The current flowing through the circuit will be:

$$I = I_m \cos(\omega' t - \Phi)$$

Where I_m is the maximum value of sinusoidal current.

If the angular frequency of external voltage source is equal to the natural frequency $\omega = \frac{1}{\sqrt{LC}}$ of the circuit, then there will be resonance and the maximum current will flow through the circuit.

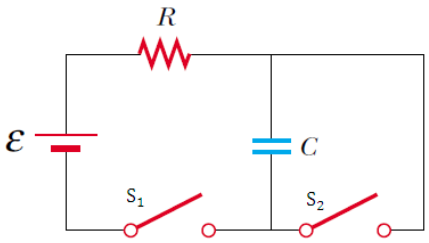
The resistance R provides the damping in the circuit. Greater the resistance, smaller will the current flow through the circuit. Such oscillations which are produced in oscillating system are called forced oscillations.

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Growth of Current in RC Series Circuit

Consider a resistance R and a capacitance C is connected in series with a battery of emf ε as shown in the figure. The S_1 and S_2 are the two switches.



When the switch S_1 is closed, keeping the switch S_2, the capacitor begins to charge and the voltage across the capacitor V_C begins to increase. Let V_R is the potential drop across the resistor, then by applying the Kirchhoff's law:

$$\varepsilon - V_R - V_C = 0$$

$$\varepsilon - IR - \frac{q}{C} = 0$$

$$\varepsilon - \frac{q}{C} = IR$$

$$\varepsilon - \frac{q}{C} = R\,\frac{dq}{dt} \qquad \because I = \frac{dq}{dt}$$

$$\frac{1}{R}dt = \frac{dq}{\left(\varepsilon - \frac{q}{C}\right)}$$

Multiplying equation by '$-\frac{1}{C}$' on both sides:

$$\frac{-\frac{1}{C}dq}{\left(\varepsilon - \frac{q}{C}\right)} = -\frac{1}{RC}dt$$

Integrating the above equation, we get:

$$\int_0^q \frac{-\frac{1}{C}dq}{\left(\varepsilon - \frac{q}{C}\right)} = -\frac{1}{RC}\int_0^t dt$$

$$\Rightarrow \left|\ln\left(\varepsilon - \frac{q}{C}\right)\right|_0^q = -\frac{1}{RC}\,|t|_0^t$$

$$\Rightarrow \ln\left(\varepsilon - \frac{q}{C}\right) - \ln(\varepsilon) = -\frac{1}{RC}\,(t-0)$$

$$\Rightarrow \ln\left(\frac{\varepsilon - \frac{q}{C}}{\varepsilon}\right) = -\frac{t}{RC}$$

$$\Rightarrow \ln\left(1 - \frac{q}{\varepsilon C}\right) = -\frac{t}{RC}$$

$$\Rightarrow 1 - \frac{q}{\varepsilon C} = e^{\left(-\frac{t}{RC}\right)}$$

$$\Rightarrow \frac{q}{\varepsilon C} = 1 - e^{\left(-\frac{t}{RC}\right)}$$

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$$\Longrightarrow q = \varepsilon C\left(1 - e^{\left(-\frac{t}{RC}\right)}\right)$$

Where $\tau_c = RC$ is the capacitive time constant, therefore:

$$\Longrightarrow q = \varepsilon C\left(1 - e^{\left(-\frac{t}{\tau_c}\right)}\right) \quad \text{---------- (1)}$$

When $t = \infty$, the capacitor will fully charged to its saturation value q_0

$$q_0 = \varepsilon C\left(1 - e^{\left(-\frac{\infty}{\tau_c}\right)}\right) = \varepsilon C\left(1 - e^{(-\infty)}\right) = \varepsilon C\left(1 - \frac{1}{e^{\infty}}\right) = \varepsilon C\left(1 - \frac{1}{\infty}\right) = \varepsilon C$$

The equation (1) will become:

$$q = q_0\left(1 - e^{\left(-\frac{t}{\tau_c}\right)}\right) \quad \text{---------- (2)}$$

Growth of Current

Differentiating equation (2) with respect to time, we get:

$$\frac{dq}{dt} = \frac{d}{dt}\left[q_0\left(1 - e^{\left(-\frac{t}{\tau_c}\right)}\right)\right] = \frac{d}{dt}\left[q_0 - q_0\, e^{\left(-\frac{t}{RC}\right)}\right]$$

$$I = 0 - q_0\left(-\frac{1}{RC}\right)\, e^{\left(-\frac{t}{RC}\right)}$$

$$I = \frac{1}{R}\frac{q_0}{C}\, e^{\left(-\frac{t}{RC}\right)}$$

$$\because \frac{q_0}{C} = \varepsilon$$

$$I = \frac{\varepsilon}{R}\, e^{\left(-\frac{t}{RC}\right)}$$

$$\because \frac{\varepsilon}{R} = I_0$$

$$I = I_0\, e^{\left(-\frac{t}{RC}\right)}$$

Special Cases

At $t = 0$

$$q = \varepsilon C\left(1 - e^{\left(-\frac{0}{\tau_c}\right)}\right) = \varepsilon C(1 - e^0) = \varepsilon C(1 - 1)$$

$$q = 0$$

At $t = \tau_c$

$$q = \varepsilon C\left(1 - e^{\left(-\frac{\tau_c}{\tau_c}\right)}\right) = \varepsilon C\left(1 - e^{(-1)}\right) = \varepsilon C\left(1 - \frac{1}{e}\right)$$

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UNIVERSITY OF SARGODHA

B.A. / B.Sc. 1st Annual Exam 2017

Physics **Paper: B**

Time Allowed: 3 Hours **Maximum Marks: 50**

Note: Attempt any five questions in all, selecting one question from section I and four questions from section-II. All questions carry equal marks.

SECTION-I

Q # 1. (a) List at least five properties of ideal gas and then determine the rms speed of ideal gas molecules.

(b) Calculate the most probable speed and average speed for oxygen molecules atT = 300K .

(c) How the speed of sound related to the gas variables in the kinetic theory model?

Q # 2. (a) What is meant by thermal Isolation? Then calculate the work done during adiabatic expansion.

(b) A Carnot engine whose high-temperature reservoir is at 400K has an efficiency of 30%. By how much should the temperature of the low temperature reservoir be changed to increase the efficiency to 40%.

(c) What is the difference between heat of fusion and heat of vaporization?

SECTION-II

Q # 3. (a) Determine the electric field due to a charged disk of radius R.

(b) How much is required to turn an electric dipole 180^0 in a uniform electric field of magnitude $E = 46\,N/C$ if $p = 3.02 \times 10^{-25}\,Cm$ and the initial angle of 64°.

(c) State two Shell theorems of electrostatics.

Q # 4. (a) Define Capacitance and its unit, also determine capacitance due to cylindrical capacitor.

(b) Four 18 Ω are connected in parallel across 25 V ideal battery. What is the current through the battery?

(c) State macroscopic and microscopic forms of Ohm‘s law.

Q # 5. (a) Define electric potential and its unit. Calculate the electric potential due to electric dipole.

(b) An infinite line of charge produces the field of magnitude $4.5 \times 10^4\,N/C$ at a distance of 2 m. Calculate the linear charge density.

(c) What is an analogy between electrical and mechanical oscillator.

Q # 6. (a) Discuss the growth of charge of a capacitor in RC circuit and also find the current.

(b) A particle undergoes uniform circular motion of radius 26.1 μm in a uniform magnetic field. The magnetic force on the particle has the magnitude of $1.60 \times 10^{-17}\,N$. What is the kinetic energy of the particle?

(c) How does doubling the frequency alter the capacitive and inductive reactance?

Q # 7. (a) State and explain Ampere’s law and write its integral and differential form.

(b) A solenoid has length 1.23 m and inner diameter 3.55 cm and it carries the current 5.57A. It consists of five close-packed layers, each with 450 tums along length. What is magnetic field at its center?

(c) What is meant by capacitive reactance? Also write its unit.

Q # 8. (a) Define Inductance and its unit. Derive the expression for Inductance for a Solenoid.

(b) What is magnetic energy per unit length of a coaxial cable of inner radius l.2 mm and outer radius 3.5 mm? If 2.7 current flows.

(c) What is the rms value of secondary voltage if its peak value is 100 V?

Q # 9. (a) An ac emf is applied to RLC series circuit. Calculate the expression for average power dissipated.

(b) A series RLC circuit has inductance L = 12 mH, capacitance C = 1.6 μF and resistance R = 1.5Ω. At what time t will the amplitude of the charge oscillations in the circuit be 50% of its initial value’?

What is difference between acceptor and rejecter circuit?

Q # 10. Write the note on any two of the following

a) Carnot heat Engine
b) Dielectrics and Gauss Law
c) Magnetic Materials
d) Maxwell Equations

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UNIVERSITY OF SARGODHA

B.A. / B.Sc. 1st Annual Exam 2016

Physics **Paper: B**

Time Allowed: 3 Hours **Maximum Marks: 50**

Note: Attempt any five questions in all, selecting one question from section I and four questions from section-II. All questions carry equal marks.

SECTION-I

Q # 1. (a) What is Carnot Heat Engine? Calculate the work done and efficiency of Carnot Heat Engine.

(b) In an interstellar gas cloud. at 50K, the pressure is 1×10^{-8} Pa. Assuming that the molecular diameter of the gases in the cloud are all 20 nm, what is their mean free path?

Q # 2. (a) Discuss entropy and calculate the entropy change for irreversible free expansion of a gas.

(b) A Carnot engine whose high-temperature reservoir is at 400K has an efficiency of 30%. By how much should the temperature of the low temperature reservoir be changed to increase the efficiency to 40%.

(c) What are the two statements of Second Law of Thermodynamic?

SECTION-II

Q # 3: (a) Determine the electric field due to a ring of uniform positive charge distribution.

(b) An electron is in the vacuum near Earth's surface and located at y=0 on the vertical y-axis. At what value of y should the second electron is placed such that its electrostatic force on the first electron balance the gravitational force on the first electron?

(c) Electric charge is Conserved. Comment

Q # 4: (a) Define Capacitance of a capacitor and determine capacitance due to cylindrical capacitor.

(b) A neutral water molecule in its vapor state has an electric dipole moment of magnitude $6.2 \times 10^{-3}\ C\ m$. How far apart are the molecules centers of positive and negative charges?

(c) How Coulomb's Law can be derived from Gauss's Law?

Q # 5: (a) Define electric potential and its unit. Calculate the electric potential due to electric dipole.

(b) A point charge causes an electric flux of 750 $N\text{-}m^2/C$ to pass through a spherical Gaussian surface of l0 cm radius centered on the charge. If the radius of the Gaussian's surface is doubled, how much flux is passed through the surface?

(c) How the energy density and electric field are related.

Q # 6: (a) Discuss the discharging of capacitor in RC circuit. Also calculate the current.

(b) A capacitor discharge through a resistor. After how many time constant does the stored energy fall to one half of its initial value?

(c) Define Ampere's Law.

Q # 7: (a) Define magnetic force and calculate the magnetic force between two parallel current carrying wires

(b) A coil has an inductance of 53mH and a resistance of 0.35 ohms. If a l2V emf is applied across the coil, how much energy is stored in the magnetic field after the current has built up to its equilibrium value?

(c) What is meant be capacitive reactance? Also describe its unit.

Q # 8: (a) What is solenoid? Determine magnetic field due to solenoid by applying Ampere's law.

(b) What is intensity of a travelling electromagnetic wave if B_m is 1×10^{-4} T?

(c) What is the maximum value of an ac voltage whose rms value is 100 V?

Q # 9: (a) An ac emf is applied to RLC series circuit. Calculate the expression for average power dissipated.

(b) A 45.2 mH inductor has the resistance of l.2 kΩ. What is the frequency and find the capacitance of a capacitor with the same reactance.

(c) Define Curie's Law.

Q # 10: Write the note on any two of the following:

(i) Mean free path

(ii) Dielectrics and Gauss Law

(iii) Transformer

(iv) Semiconductors and Superconductors

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UNIVERSITY OF SARGODHA

B.A. / B.Sc. 1st Annual Exam 2015

Physics **Paper: B**

Time Allowed: 3 Hours **Maximum Marks: 50**

Note: Attempt any five questions in all, selecting one question from section I and four questions from section-II. All questions carry equal marks.

SECTION-I

Q # 1. (a) The Maxwell Speed Distribution function is given as:

$$P(v) = \left(\frac{m}{2\pi kT}\right)^{3/2} v^2 e^{-mv^2/2kT}.\ Show\ that\ v_{rms} = \sqrt{\frac{3kT}{m}}$$

(b) One mole of oxygen gas expands isothermally at 310 K from initial volume of 12 L to the final volume of 19 L. What would be the final temperature if the gas had expanded adiabatically to the same final volume?

(c) There are five number 5, 11, 32, 67 and 89. What is the rms value of these numbers.

Q # 2. (a) What is Carnot engine? Calculate the work done and efficiency of Carnot Heat Engine.

(b) The temperature of one mole of mono-atomic gas is raised reversibly 300 K to 400 K with the volume kept constant. What is the entropy change of the gas?

(c) What are the two statements of Second Law of Thermodynamics?

SECTION-II

Q # 3: (a) What is electric dipole? Calculate the electric field at the bisector point at the distance x due to dipole.

(b) The nucleus of iron atom has a radius of about 4×10^{-15} m and contains 26 protons. Calculate the magnitude repulsive force between two of the protons.

(c) What will be the value of flux if the surface encloses negative charge?

Q # 4: (a) Define and prove Gauss's law and write its integral and differential form.

(b) A Gaussian surface in the form of cylinder of radius R immersed in a uniform electric field E, with the cylinder axis parallel to the field. What is the flux of the electric field through this closed surface?

(c) How does the gravitational and electric force resemble each other? How do they differ?

Q # 5: (a) Define electric potential and its unit. Calculate the electric potential from electric field.

(b) A wire 4 m long and 6 mm in diameter has a resistance of 15 $m\Omega$. A potential difference of 23 V is applied between the ends. What is current density?

Q # 6: (a) Discuss the growth of charge in RC circuit also calculate the current.

(b) A capacitor discharge through resistor. After how many time constant does the stored energy fall to one half of its initial value?

(c) Show that the product RC has the dimension of time.

Q # 7: (a) Calculate the magnetic force acting on a moving charge in a magnetic field.

(b) A proton traveling at 23^0 with respect to magnetic field

2.3 mT experiences a magnetic force of $6.48 \times 10^{-17} N$.

Calculate the speed and KE of proton in eV.-

(c) What do you mean by strength of magnetic field and what is its unit?

Q # 8: (a) State Biot-Savart law? And calculate magnetic field due to long straight wire using Biot-Savart law.

(b) At a certain location, the magnetic field has the magnitude 42 μT and point downward at 57° to the vertical. Calculate the magnetic flux through a horizontal surface of area 2.5 m^2.

(c) Does the induced emf depend upon the resistance of the circuit?

Q # 9: (a) An AC emf is applied to RLC series circuit. Calculate the expression for average power dissipated.

(b) A 45.2 mH inductor has the resistance of 1.2 kΩ. What is the frequency and find the capacitance of capacitor with the same resistance?

(c) Why the electromagnetic waves are not deflected by magnetic field.

Q # 10: Write the note on any two of the following:

i. Van Der Waal equation of state
ii. Faraday's law of electromagnetic induction
iii. Transformer
iv. Electromagnetic wave spectrum

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UNIVERSITY OF SARGODHA

B.A. / B.Sc. 1st Annual Exam 2014

Physics **Paper: B**

Time Allowed: 3 Hours **Maximum Marks: 50**

Note: Attempt any five questions in all, selecting one question from section I and four questions from section-II. All questions carry equal marks.

SECTION-I

Q # 1. (a) Discuss the distribution of molecular energies. Hence derive the Maxwell Boltzmann energy distribution formula.

(b) A cylindrical container of length 56 cm and diameter 12.5 cm holds 0.35 moles of nitrogen gas at the pressure of 2.05 atm. Calculate rms speed of the nitrogen molecules.

(c) Show that for adiabatic process $PV^{\gamma} = constant$

Q # 2. (a) What are the two statements of second law of thermodynamics? Show that the statements are equivalent.

(b) A refregerator whose COP is 4.7, extract heat from the food chamber at the rate of 250 J/cycle. How much work per cycle is required to operate the refrigerator?

(c) What is the connection between entropy and the second law of thermodynamics.

SECTION-II

Q # 3: (a) What is electric dipole? Find the electric field at a point on the bisector plane due to dipole.

(b) An infinite line of charge produces the field at a distance of 1.92 m. Calculate the linear charge density.

(c) Do electrons tend to go to region of high potential or low potential?

Q # 4: (a) Derive the formula to calculate the electric potential due to a point charge and collection of point charges.

(b) What is the electric potential at the surface of the gold nucleus? The radius of the surface of the gold nucleus is 7 fm and atomic number is 79.

(c) Can two equipotential surfaces intersect each other?

Q # 5: (a) Define capacitance. Show that capacitance of a cylindrical capacitor depends upon the radii of inner and outer spheres.

(b) What is the capacitance of the earth viewed as an isolating conducting sphere of radius 6370 km?

(c) What type of energy is stored in the capacitor?

Q # 6: (a) Discuss the growth of charge in RC series circuit and the expression of current which is used to charge the capacitor.

(b) The capacitor C discharges through the resistance R. After how many time constants does the charge drop to half of its initial value?

(c) Under what circumstances can the terminal potential difference of a battery exceed its emf?

Q # 7: (a) State and explain Ampere's law and write its integral and differential form.

(b) The earth's magnetic field has a value of 39 μT and is horizontal and due north. The net field is zero 8.13 cm above the long straight horizontal wire that carries a steady current. Calculate current and find its direction of flow.

(c) What type of magnetic field exists around a current carrying straight conductor?

Q # 8: (a) What is atomic magnetism? Find the relation between total orbital magnetic dipole moment and total angular momentum.

(b) What must be the magnitude of uniform electric field if it is to have the same energy density as possessed by a 0.5 T magnetic field?

(c) How does doubling the frequency alter the capacitive and inductive reactance?

Q # 9: (a) Explain the phase lag or lead between current and voltage in a purely inductive AC circuit. Also draw phasor diagram.

(b) In RLC series circuit R= 160 Ω, C=15 μF, L= 230 mH and f=60 Hz. Find the phase constant.

Q # 10: Write the note on any two of the following:

i. Brownian motion
ii. Lenz's law
iii. Maxwell's equations
iv. Electromagnetic wave spectrum

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UNIVERSITY OF SARGODHA

B.A. / B.Sc. 1st Annual Exam 2013

Physics **Paper: B**

Time Allowed: 3 Hours **Maximum Marks: 50**

Note: Attempt any five questions in all, selecting one question from section I and four questions from section-II. All questions carry equal marks.

SECTION-I

Q # 1. (a) State and explain the first law of thermodynamics and discuss its applications for constant temperature.

(b) Apparatus that liquefies helium is in a room at 300 K. If the helium is in the apparatus is at 4.0 K, what is the minimum ratio of heat delivered to the room to the heat removed from the helium?

(c) What factors reduce the efficiency of a heat engine from its ideal value.

Q # 2. (a) What is ideal gas? Describe the work done on ideal gas at constant pressure and temperature.

(b) An ideal gas undergoes a reversible isothermal expansion at 132℃. The entropy of the gas increases by 46 J/K. How much heat was absorbed?

(c) Explain why temperature of a gas drops in an adiabatic expansion.

SECTION-II

Q # 3: (a) Define dipole and dipole moment. Calculate the relation for torque and potential energy of an electric dipole when it is placed in electric field.

(b) An electric dipole consists of charges +2e and -2e separated be 0.78 nm. It is in electric field strength $3.4 \times 10^6\ Vm^{-1}$. Calculate the magnitude of Torque on the dipole, when the dipole moment is (i) parallel to (ii) antiparallel to the electric field.

(c) A point charge is moving in an electric field at right angle to the lines of force. Does any force act on it.

Q # 4: (a) What is electric potential? Find the electric potential at a point P on the axis of charge disc.

(b) The potential at the center of a uniformly charged circular disc of radius $R = 3.5\ cm$ is $V_0 = 550\ V$. What is the total charge on the disc?

(c) Define equipotential surfaces. What is the potential difference between two points on an equipotential surface.

Q # 5: (a) Derive the relation for energy stored in an electric field of charged capacitor, also calculate energy density.

(b) An isolated conducting sphere whose radius is 685 cm has a charge 1.25 nC. What is energy density at the surface of sphere?

(c) What is dielectric and what is the effect of dielectric on capacitance of capacitor?

Q # 6: (a) Discuss decay and growth of charge in RC circuit.

(b) A capacitor of capacitance C is discharging through a resistance R. In terms of time constant $t = RC$, when will its charge be one half of its initial value?

(c) What do you mean by a node in DC circuit and state Kirchhoff's node rule?

Q # 7: (a) State Ampere's law and derive the relation for magnetic field due to a toroid.

(b) A 200 turns solenoid having a length of 25cm carries a current of 0.30A. Find magnetic field B inside the solenoid.

(c) Discuss analogies and differences between Coulomb's law and Biot-savrat law.

Q # 8: (a) Discuss RLC series circuit and derive the relation for impedance and resonance frequency.

(b) At what frequency would a 6.0 mH inductor and 10 μF capacitor have the same resonance?

(c) Define rms value and power factor in AC circuit.

Q # 9: (a) What is inductor? Derive the relation for energy stored in magnetic field due to solenoid?

(b) A coil has an inductance of 53mH and resistance of 0.35 Ω. If a 12 V emf is applied across coil, how much energy stored in magnetic field after current has been built up to its equilibrium value.

(c) Does the induced emf always act to decrease the magnetic flux through a circuit?

Q # 10: Write a note on any two of the following:

i. Hysteresis Loop
ii. Gauss Law in Dielectric
iii. Carnot cycle

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UNIVERSITY OF SARGODHA

B.A. / B.Sc. 1st Annual Exam 2012

Physics **Paper: B**

Time Allowed: 3 Hours **Maximum Marks: 50**

Note: Attempt any five questions in all, selecting one question from section I and four questions from section-II. All questions carry equal marks.

SECTION-I

Q # 1. (a) What is an ideal gas? Discuss work done on an ideal gas during thermal isolation.

(b) Calculate work done by the external agent in compressing 1.12 mole of oxygen from a volume of 22.4 L at 1.53 atm pressure to 15.3 L at the same temperature.

(c) How is the speed of sound related to gas variables in kinetic theory model?

Q # 2. (a) Define mean free path of a molecule travelling through a gas, also calculate mean free path on microscopic level.

(b) Calculate the mean free path for 35 spherical jelly beans in a jar that is vigorously shaken, the volume of jar is 1.0 L and the diameter of jelly beans is 1.0 cm.

(c) List three examples of Brownian motion in physical phenomenon.

SECTION-II

Q # 3: (a). What is an electric dipole? Discuss the motion of dipole in the external electric field. Also calculate work done and energy stored in the dipole.

(b) A proton is placed in a uniform electric field D. What must be the magnitude and direction of the field if the electrostatic force acting on the proton is just to balance its weight?

(c) A point charge is moving in an electric field at right angle to the lines of force. Does any force act on it.

Q # 4: (a). Define electric potential and calculate electric potential due to a dipole.

(b) What must be the magnitude of an isolated positive charge for the electric potential at 15 cm from the charge to be 120 V?

(c) How can you ensure that the electric potential in a given region of space has constant value?

Q # 5: (a) Define capacitance? Calculate the capacitance of a cylindrical capacitor.

(b) A parallel plate capacitor has circular plate of 8.22 cm radius and 1.31 mm separation. Calculate the capacitance and what charge will appear on each plate if a potential difference of 1116 V is applied.

(c) Water has high value of dielectric constant, why is it not used as a dielectric material in capacitor?

Q # 6: (a) Discuss decay of charge in RC series circuit. Calculate current and define time constant of the circuit.

(b) When 115 V is applied across 9.66 m long wire, the current density is 1.44 $A\ cm^{-2}$. Calculate the conductivity of the wire material.

(c) Under what circumstances would you want to connect batteries in parallel and in series?

Q # 7: (a) State Biot Savart Law. Determine the magnetic field at point, distance R away from the long straight wire carrying current.

(b) In the Bohr Model of hydrogen atom the electron circulates around the nucleus in a path of radius $5.29 \times 10^{-9} m$ at a frequency of 6.63×10^{15} Hertz. What value of B is set up at the center of the orbit?

(c) Is B uniform for all points within a circular loop of wire carrying current?

Q # 8: (a) Derive an expression for the average power dissipated in RLC series circuit containing an AC source.

(b) In an RLC series circuit R = 160 Ω, C = 15 μF, = 230 mH, f = 60 Hz and ε=36V (AC supply). Find: (a) the rms emf (b) the rms current (c) the power factor and (d) average power dissipated.

(c) Why it is useful lo use rms notation for alternating current and voltage. '

Q # 9: (a) Discuss the process of generation of electromagnetic waves. How Maxwell equations lead towards the existence of electromagnetic waves.

(b) A beam of light with an intensity I of l2 W cm^{-1} falls perpendicularly on a perfectly reflecting plane mirror of l.5 cm^2 area. What force acts on the mirror.

(c) How does a microwave oven cook food‘?

Q # 10. Write note on any two of the following:

i. Faradays law of electromagnetic induction.
ii. Maxwell Equations
iii. Second Law of thermodynamics and entropy.

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